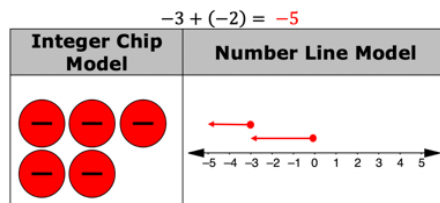


## Unit 6 Overview: Operations With Integers

### Unit Narrative



A concrete-pictorial-abstract learning progression is used in Unit 6 to help students develop both a conceptual understanding of and procedural fluency with integer operations. Students initially use concrete manipulatives and pictorial representations, such as number lines and integer chips, to model integer operations. Students practice evaluating integer operation expressions. Throughout the unit, students develop an appreciation of the utility of integer operations as they solve real-world problems. The unit also includes evaluating exponents at the beginning and the end of the unit. The unit begins with introducing exponents and evaluating exponential expressions with positive bases. After students have learned all four operations with integers and developed procedural fluency, evaluating exponential expressions is revisited, this time including integer bases.



The unit is separated into four sections. Section 1 focuses on evaluating exponential expressions and finding the prime factorization of composite numbers. In Sections 2 and 3, students develop a conceptual understanding of and procedural fluency with adding and subtracting integers. In Section 4, students connect integer addition and subtraction to explore how to multiply and divide integers. The unit culminates with students evaluating exponents with positive rational number or integer bases and natural number exponents.

Students will extend this learning to performing operations with rational numbers with procedural fluency and solving real-world problems with rational numbers in Grade 6 Accelerated Unit 11.

### Unit At-A-Glance

|                                              |                                            |           |                                                           |
|----------------------------------------------|--------------------------------------------|-----------|-----------------------------------------------------------|
| Section 1: Exponents and Factoring           | Benchmarks Addressed                       | Lesson 1  | Exponents                                                 |
|                                              | <b>MA.6.NSO.3.3</b><br><b>MA.6.NSO.3.4</b> | Lesson 2  | Prime Factorization                                       |
| Section 2: Adding Integers                   | Benchmark Addressed                        | Lesson 3  | Adding Integers Using Integer Chips                       |
|                                              | <b>MA.6.NSO.4.1</b>                        | Lesson 4  | Adding Integers Using Number Lines                        |
|                                              |                                            | Lesson 5  | Adding Integers                                           |
| Section 3: Subtracting Integers              | Benchmark Addressed                        | Lesson 6  | Subtracting Integers Using Integer Chips                  |
|                                              | <b>MA.6.NSO.4.1</b>                        | Lesson 7  | Subtracting Integers Using Number Lines                   |
|                                              |                                            | Lesson 8  | Integer Subtraction                                       |
| Section 4: Multiplying and Dividing Integers | Benchmarks Addressed                       | Lesson 9  | Multiplying Integers Using Models                         |
|                                              | <b>MA.6.NSO.4.2</b><br><b>MA.6.NSO.3.3</b> | Lesson 10 | Dividing Integers Using Integer Chips                     |
|                                              |                                            | Lesson 11 | Multiplying and Dividing Integers                         |
|                                              |                                            | Lesson 12 | Multiplying and Dividing Integers With Procedural Fluency |
|                                              |                                            | Lesson 13 | Evaluating Exponents                                      |



# Differentiation

## Universal Design for Learning™ © CAST, 2008

| <p><b>Engagement</b><br/>The “WHY” of learning</p>                                                                                                                            | <ul style="list-style-type: none"> <li>The real-world problems throughout the unit (for example the phone tree in the Lesson 1, 6.1.2 Guided Instruction) help students gain an appreciation for the utility of the mathematics they are learning. The contexts presented will also engage students if they are related to their interests. If needed, you can modify the contexts in the problems based on the interests of your students.</li> <li>The unit includes lots of collaboration and exploration opportunities throughout each lesson so that students can work with partners as they develop a stronger understanding of this new content. This will ensure students are engaged in productive struggle and sustained effort.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                |                 |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <p><b>Representation</b><br/>The “WHAT” of learning</p>                                                                                                                       | <ul style="list-style-type: none"> <li>Allow students to represent their work using integer chips or number lines throughout this unit. This will provide necessary scaffolding and a kinesthetic or visual entry point to build conceptual understanding of operations with integers.</li> <li>Consider creating and displaying an anchor chart or word wall with instructions, examples, and visual models of integer operations.</li> </ul> <div data-bbox="929 781 1415 920"> <p>exponent (or power) <span style="color: red;">4</span><br/>base <span style="color: blue;">3</span> <math>= 3 \cdot 3 \cdot 3 \cdot 3 = 81</math> value (of the expression)<br/>factors</p> </div> <div data-bbox="1869 633 2023 781"> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                |                 |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |
| <p><b>Action &amp; Expression</b><br/>The “HOW” of learning</p>                                                                                                               | <ul style="list-style-type: none"> <li>Several components include constructive-response problems that require students to explain why or how they solved the problem. The teacher guides include alternative methods for students who struggle expressing themselves in writing, such as using an audio recorder or speech-to-text software.</li> <li>The Wrap-Ups throughout the unit, allow students to demonstrate their mastery of the learning targets using different methods, such as writing a reflective self-assessment, solving example problems, or drawing a visual model.</li> </ul> <div data-bbox="1780 963 2023 1223"> <table border="1"> <thead> <tr> <th>Visual Model of Multiplication</th><th>Sign of Product</th></tr> </thead> <tbody> <tr> <td> × </td><td></td></tr> <tr> <td> × </td><td></td></tr> <tr> <td> × </td><td></td></tr> <tr> <td> × </td><td></td></tr> </tbody> </table> </div> | Visual Model of Multiplication | Sign of Product |  ×  |  |  ×  |  |  ×  |  |  ×  |  |
| Visual Model of Multiplication                                                                                                                                                | Sign of Product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |                 |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |
|  ×  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |                 |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |
|  ×  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |                 |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |
|  ×  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |                 |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |
|  ×  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |                 |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |                                                                                                                                                                               |                                                                                       |

## ELL Information

### Partner Reading Strategy

- Prior to each lesson, identify any text-heavy word problems.
- Working in pairs, ask one student to read the word problem aloud while the other student assists in pronunciation or assistance when needed.
- When both students have practiced reading and received feedback, they will begin solving the word problem together to find the solution.

### Whenever possible, act out or provide illustrations of word problems to help students visualize the problem they are going to solve.

- Read problems out loud and gesture or point to the illustration to reinforce the meaning.
- For some problems, students can draw their own pictures. Try having them do this after you have shown them some examples.

## Additional Differentiated Support

| Scaffolding Resources                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | On-Ramp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support.”</p> <ul style="list-style-type: none"> <li>• Base (of an exponent)</li> <li>• Additive property of zero</li> <li>• Commutative property of multiplication</li> <li>• Existence of additive inverses</li> <li>• Exponent (exponential form)</li> <li>• Integers</li> <li>• Prime factorization</li> <li>• Prime number</li> </ul> | <p>On-Ramp to 6<sup>th</sup> Grade Math provides video and practice tools for students to scaffold the following skills:</p> <ul style="list-style-type: none"> <li>• Addition Fluency</li> <li>• Subtraction Fluency</li> <li>• Commutative Property of Multiplication</li> <li>• Associative Property of Multiplication</li> <li>• Distributive Property</li> <li>• Adding Two-Digit and One-Digit Numbers</li> <li>• Prime and Composite Numbers</li> <li>• Number Lines: Whole Number Sums and Differences</li> <li>• Evaluate Expressions</li> <li>• A Fraction is the Quantity Formed by Its Parts</li> <li>• Multiply a Fraction by a Whole Number</li> </ul> |



## Section Overview

### Section 1: Exponents and Factoring (Lessons 1-2)

| Benchmarks Addressed                                                                                                                                                                                                                                                                                                                                                                            | Learning Targets |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>MA.6.NSO.3.3</b> Evaluate positive rational numbers and integers with natural number exponents.</p> <ul style="list-style-type: none"> <li><i>Clarification 1: Within this benchmark, expectations include using natural number exponents up to 5.</i></li> </ul> <p><b>MA.6.NSO.3.4</b> Express composite whole numbers as a product of prime factors with natural number exponents.</p> | Lesson 1         | <ul style="list-style-type: none"> <li>I understand the role of a base and an exponent in the exponential form of a numeric expression.</li> <li>I can write a numeric expression using repeated multiplication in exponential form.</li> <li>I can write an expression in exponential form as a numeric expression using repeated multiplication.</li> <li>I can evaluate a positive rational number raised to a power.</li> </ul> |
|                                                                                                                                                                                                                                                                                                                                                                                                 | Lesson 2         | <ul style="list-style-type: none"> <li>I can express composite whole numbers as a product of prime factors.</li> </ul>                                                                                                                                                                                                                                                                                                              |

### Section 2: Adding Integers (Lessons 3-5)

| Benchmarks Addressed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Learning Targets |                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>MA.6.NSO.4.1</b> Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.</p> <ul style="list-style-type: none"> <li><i>Clarification 1: Instruction begins with the use of manipulatives, models and number lines working towards becoming procedurally fluent by the end of grade 6.</i></li> <li><i>Clarification 2: Instruction focuses on the inverse relationship between the operations of addition and subtraction. If <math>p</math> and <math>q</math> are integers, then <math>p - q = p + (-q)</math> and <math>p + q = p - (-q)</math>.</i></li> </ul> | Lesson 3         | <ul style="list-style-type: none"> <li>I can express integer addition using integer chips.</li> <li>I can add integers using integer chips.</li> </ul> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Lesson 4         | <ul style="list-style-type: none"> <li>I can add integers using a number line.</li> </ul>                                                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Lesson 5         | <ul style="list-style-type: none"> <li>I can add integers with procedural fluency.</li> </ul>                                                          |

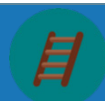


### Section 3: Subtracting Integers (Lessons 6-8)

| Benchmarks Addressed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Learning Targets |                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MA.6.NSO.4.1</b> Apply and extend previous understandings of operations with whole numbers <i>to add and subtract integers with procedural fluency.</i> <ul style="list-style-type: none"> <li>Clarification 1: Instruction begins with the use of manipulatives, models and number lines working towards becoming procedurally fluent by the end of grade 6.</li> <li>Clarification 2: Instruction focuses on the inverse relationship between the operations of addition and subtraction. If <math>p</math> and <math>q</math> are integers, then <math>p - q = p + (-q)</math> and <math>p + q = p - (-q)</math>.</li> </ul> | Lesson 6         | <ul style="list-style-type: none"> <li>I can use integer chips to subtract integers.</li> </ul>                                                                                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Lesson 7         | <ul style="list-style-type: none"> <li>I can subtract integers using a number line.</li> <li>I can use number lines to understand how the rules for adding integers are related to the rules for subtracting integers.</li> </ul> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Lesson 8         | <ul style="list-style-type: none"> <li>I can subtract integers with values greater than 100.</li> </ul>                                                                                                                           |

### Section 4: Multiplying and Dividing Integers (Lessons 9-13)

| Benchmarks Addressed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Learning Targets |                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MA.6.NSO.3.3</b> Evaluate positive rational numbers and integers with natural number exponents. <ul style="list-style-type: none"> <li>Clarification 1: Within this benchmark, expectations include using natural number exponents up to 5.</li> </ul><br><b>MA.6.NSO.4.2</b> Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency. <ul style="list-style-type: none"> <li>Clarification 1: Instruction includes the use of models and number lines and the inverse relationship between multiplication and division, working toward becoming procedurally fluent by the end of grade 6.</li> <li>Clarification 2: Instruction focuses on the understanding that integers can be divided, provided that the divisor is not zero, and every quotient of integers (with non-zero divisor) is a rational number. If <math>p</math> and <math>q</math> are integers where <math>q \neq 0</math>, then <math>-\left(\frac{p}{q}\right) = \frac{-p}{q}</math>, <math>-\left(\frac{p}{q}\right) = \frac{p}{-q}</math> and <math>\left(\frac{p}{q}\right) = \frac{-p}{-q}</math>.</li> </ul> | Lesson 9         | <ul style="list-style-type: none"> <li>I can multiply integers using integer chips, number lines, and patterns.</li> </ul>                                                     |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Lesson 10        | <ul style="list-style-type: none"> <li>I can use integer chips and patterns to discover rules for dividing integers.</li> </ul>                                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Lesson 11        | <ul style="list-style-type: none"> <li>I can multiply and divide integers with procedural fluency.</li> </ul>                                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Lesson 12        | <ul style="list-style-type: none"> <li>I can multiply and divide integers with procedural fluency.</li> </ul>                                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Lesson 13        | <ul style="list-style-type: none"> <li>I can evaluate integers with natural number exponents.</li> <li>I can evaluate a positive rational number raised to a power.</li> </ul> |



## Vertical Alignment

Grade 6 Accelerated Unit 6  
Operations with Integers

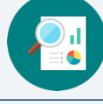


Grade 7 Accelerated Unit 3 and Unit 4  
Exponents and Scientific Notation  
Solving Problems with Rational Numbers

| Prior Course(s)                                                                                                                                                                                                                                     | Course Level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Future Course(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>MA.4.AR.3.1</b> Determine factor pairs for a whole number from 0 to 144. Determine whether a whole number from 0 to 144 is prime, composite, or neither.</p> <p><b>MA.5.NSO.2</b> Add, subtract, multiply, and divide multidigit numbers.</p> | <p><b>MA.6.NSO.3.3</b> Evaluate positive rational numbers and integers with natural number exponents.</p> <p><b>MA.6.NSO.3.4</b> Express composite whole numbers as a product of prime factors with natural number exponents.</p> <p><b>MA.6.NSO.4.1</b> Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.</p> <p><b>MA.6.NSO.4.2</b> Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.</p> | <p><b>MA.8.NSO.1.3</b> Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency</p> <p><b>MA.8.NSO.1.5</b> Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.</p> <p><b>MA.8.NSO.1.7</b> Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.</p> |

## 6.1 Exponents

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                   |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.3.3 Evaluate positive rational numbers and integers with natural number exponents.                                                                                                                                                                                                                                                                               |  Learning Targets        | <ul style="list-style-type: none"><li>• I understand the role of a base and an exponent in the exponential form of a numeric expression.</li><li>• I can write a numeric expression using repeated multiplication in exponential form.</li><li>• I can write an expression in exponential form as a numeric expression using repeated multiplication.</li><li>• I can evaluate a positive rational number raised to a power.</li></ul> |                                                                                                   |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                                                                                               |                                                                                                            | Working Toward                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                   |
|                                                                                                       | N/A                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                            | MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.                                                                                                                                                                                                                                        |                                                                                                   |
|  Essential Questions | <ul style="list-style-type: none"><li>• How do you write the exponential form of a numeric expression using repeated multiplication?</li><li>• How do you write repeated multiplication as the exponential form of a numeric expression?</li><li>• How do you evaluate positive rational numbers raised to powers?</li></ul>                                              |                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                   |
|  Lesson Narrative    | Students learn how to rewrite exponential expressions as repeated multiplication. They learn that the exponent indicates how many times to multiply the base by itself. Students also rewrite repeated multiplication of the same positive rational numbers as numeric exponential expressions. Finally, students evaluate positive rational numbers raised to exponents. |                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                   |
|  Component Summary | 6.1.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                   | Relate graphic designing to studying mathematics ( <i>SE p. 233</i> )                                      | 6.1.4 Collaboration (10 min.)                                                                                                                                                                                                                                                                                                                                                                                                          | Collaboratively evaluate exponential expressions and recognize common errors ( <i>SE p. 235</i> ) |
|                                                                                                       | 6.1.2 Guided Instruction (10 min.)                                                                                                                                                                                                                                                                                                                                        | Represent repeated multiplication as an exponential expression and evaluate ( <i>SE p. 234</i> )           | 6.1.5 Try It (5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                  | Practice evaluating exponential expressions ( <i>SE p. 236</i> )                                  |
|                                                                                                       | 6.1.3 Guided Practice (5 min.)                                                                                                                                                                                                                                                                                                                                            | Practice writing exponential expressions using repeated multiplication and vice versa ( <i>SE p. 235</i> ) | 6.1.6 Your Turn! (5-10 min.)                                                                                                                                                                                                                                                                                                                                                                                                           | Identify and write equivalent expressions ( <i>SE p. 237</i> )                                    |
|                                                                                                       | 6.1.7 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                   | Error analysis of student work evaluating exponential expressions ( <i>SE p. 237</i> )                     |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                   |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                            | Additional Preparation                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                   |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>• Base</li><li>• Exponent</li><li>• Evaluate</li></ul> <u>Additional Terms:</u> N/A                                                                                                                                                                                                                               | <ul style="list-style-type: none"><li>• (Optional) Large chart paper for anchor chart</li></ul>            | <ul style="list-style-type: none"><li>• 6.1.1 Warm-Up contains a video to accompany the question prompts.</li></ul>                                                                                                                                                                                                                                                                                                                    |                                                                                                   |

| <br>Teacher<br>Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                    | Things to Consider                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                   | <ul style="list-style-type: none"><li>Students may use terms like “solve” when referring to evaluating expressions.</li><li>Students may confuse the directions “evaluate” and “rewrite” (in an equivalent form).</li><li>Students may multiply the base by the exponent when asked to evaluate an exponential expression or rewrite an equivalent expression.</li></ul> | <ul style="list-style-type: none"><li>Students are expected to multiply positive rational numbers with procedural fluency by the end of Grade 6 (MA.6.NSO.2.2). Consider ways to support students in developing this fluency with multiplying positive whole numbers, decimals, and fractions when addressing this benchmark instead of giving them a calculator.</li></ul> |
|                                                                                                                   | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                           | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                          |
| <br>Differentiate<br>Instruction | <b>Notice and Wonder</b><br>In 6.1.6 Your Turn!, ask students what they notice and wonder about the different forms of the equivalent expressions. These discussions will help students make connections and build their understanding.                                                                                                                                  | <b>MA.K12.MTR 4.1: Discussion and Reflection</b><br>6.1.4 Collaboration includes error analysis problems. In question 2, students determine if the provided student work is correct, and if it is incorrect, they describe and correct the error(s). In question 3, they determine if the expressions are equivalent and evaluate each expression to justify their claim.   |
|                                                                                                                   | Consider creating an anchor chart to help students learn key terms in 6.1.2 Guided Instruction. Include the definitions and examples to provide a visual entry point.                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                                                                   | Use the Think Aloud strategy to debrief 6.1.3 Guided Practice to provide an auditory entry point.                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                             |
| Consider allowing students who struggle writing explanations to use an audio recorder in the 6.1.7 Wrap-Up.       |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                             |
| Lesson Notes:                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                             |

## 6.1.1 Warm-Up: Graphic Designer

**Component Overview:** Play the career video featuring a graphic designer for the class. Then, the students will answer three questions relating the messages from the video to their own life experiences and goals.



### 6.1.1 Warm-Up: Graphic Designer

1. Maurice Woods is a graphic designer and founder of the Inneract Project. He says, "Design is really about a process of thinking creatively and critically about how to solve something in a way that not only makes sense to people, but produces a solution to a problem." How does that process relate to your study of mathematics?

*Answers will vary. In math class I use my understanding of both math and situations I have experienced to come up with a logical solution to the problem.*

2. Maurice states that one of the skills he brought with him from his basketball career to his design career was his experience with teamwork and what it means to be flexible and compromise. Describe an experience when teamwork was needed for you to reach a goal.

*Answers will vary. As a soccer player I could run down the field and try to score a goal on my own. If I work with my teammates to pass the ball moving down the field, that increases the chances of making a goal.*

3. Maurice said, "I want to be remembered for giving back to the community and giving back to young people ... I get more joy out of that than any project I work on." What do you want to be remembered for in life?

*Answers will vary. I want to be remembered for helping others in need. I want to be remembered for inventing something that helps people live healthier lives.*



*Consider an extension project in which students explore career(s) they are interested in pursuing and research how their desired careers use mathematical content and/or practices.*



*Prompt students to share their responses with a partner or monitor for strong responses and have students share whole group.*

## 6.1.2 Guided Instruction: Expressions with Exponents

**Component Overview:** Students represent repeated multiplication as exponential expressions and evaluate exponential expressions. Students develop an understanding of the key terms “base,” “exponent,” and “evaluate.”



### 6.1.2 Guided Instruction: Expressions with Exponents

1. The principal at Fitzgerald Middle School has been tasked with contacting everyone at the school regarding an upcoming change. The principal uses a phone tree to contact the staff. The principal calls 2 people, then those 2 people will call 2 other people. This process repeats until everyone has been contacted.

- a. Fill in the blanks for the phone tree.

| Phone Tree             |                                                                                     |
|------------------------|-------------------------------------------------------------------------------------|
| First round of calls:  | 2                                                                                   |
| Second round of calls: | $2 \cdot 2$                                                                         |
| Third round of calls:  | $2 \cdot 2 \cdot 2$                                                                 |
| Fourth round of calls: | $2 \cdot \underline{2} \cdot \underline{2} \cdot \underline{2}$                     |
| Fifth round of calls:  | $2 \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2}$ |

- b. What do you notice about the phone tree?  
*Answers will vary. Each round increases by a multiple of two.*

- c. How many people were contacted on the fifth round?  
*32*

The fifth round of calls can be rewritten in exponential form as  $2^5$ .

- d. How does the exponential expression for the fifth round of calls relate to the repeated multiplication written in the phone tree?  
*In the exponential expression, the 5 indicates the number of times that 2 is multiplied in the phone tree.*



Remind students to use talk moves, such as revoicing and thinking aloud, when working with their partners and during the whole-class debrief discussion.



How would the expression using repeated multiplication change if the base was 5 and the exponent was 2, as in  $5^2$ ?



## 6.1.2 Guided Instruction: Expressions with Exponents (continued)

The numeric expression  $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$  can be rewritten in exponential form as  $2^5$ , where 2 is the **base** and 5 is the **exponent**.



The **base** of an exponent is the number used as a factor in exponential form.



The **exponent** is the number of times the base occurs as a factor.

Exponents are also commonly referred to as powers. For example,  $2^5$  is read as "2 to the 5<sup>th</sup> power."

2. Complete the statements using the definitions of base and exponent.

a. In the exponential expression  $9^3$ , the 9 is the

- ☐ base
- ☐ exponent

of the expression and the 3 is the

- ☐ base.
- ☐ exponent.

b. The exponential expression  $9^3$  has the same value

as

- ☐  $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$ .
- ☐  $9(3)$ .
- ☐  $(9)(9)(9)$ .



In mathematics, **evaluate** is an instruction that means "find the value of."

3. Evaluate  $9^3$ .

729



Consider creating an anchor chart to help students learn key terms. Include the definitions and examples to provide a visual entry point. Also, consider using different colors to support students in differentiating between the base and the exponent. Pay attention to precision in how students are writing the exponent as a superscript. This visual representation can also be found later in this unit.

$$\begin{array}{c} \text{exponent (or power)} \\ 3^4 \\ \text{base} \end{array} = \underbrace{3 \cdot 3 \cdot 3 \cdot 3}_{\text{factors}} = 81 \quad \begin{array}{c} \text{value} \\ \text{(of the} \\ \text{expression)} \end{array}$$



Consider asking students the following:

- How have you used the word "evaluate" in the past? How is this different from the definition we use in math?
- In your own words, if the question asks you to evaluate, what would you do?

## 6.1.3 Guided Practice: Equivalent Expressions

**Component Overview:** Students practice rewriting exponential expressions using repeated multiplication and vice versa. Students can complete this activity independently and peer review their responses with their partner.



### 6.1.3 Guided Practice: Equivalent Expressions

1. Write each numeric expression in exponential form.

a.  $5 \cdot 5 \cdot 5 \cdot 5$

$5^4$

b.  $\left(\frac{6}{7}\right) \times \left(\frac{6}{7}\right) \times \left(\frac{6}{7}\right)$

$\left(\frac{6}{7}\right)^3$

c.  $(3.25)(3.25)$

$(3.25)^2$

2. Rewrite each expression given in exponential form as repeated multiplication.

a.  $7^5$

$7 \cdot 7 \cdot 7 \cdot 7 \cdot 7$

b.  $(1.5)^2$

$(1.5)(1.5)$

c.  $\left(\frac{3}{4}\right)^3$

$\left(\frac{3}{4}\right) \cdot \left(\frac{3}{4}\right) \cdot \left(\frac{3}{4}\right)$



Consider asking students the following:

- When rewriting an expression with the same number being multiplied by itself repeatedly, how do you know what exponent to use?
- When expanding an expression given in exponential form, how do you know what the factors are and how many factors to include?



Challenge learners to evaluate the expressions and write an explanation of the reasoning behind their answers if they finish early.

This will be addressed in the next components.



## 6.1.4 Collaboration: Evaluating Exponential Expressions

**Component Overview:** In this activity, students will transition from rewriting to evaluating expressions. This component includes error analysis problems to highlight and address common misconceptions. Students will also explore equivalent expressions written in different forms.



### 6.1.4 Collaboration: Evaluating Exponential Expressions

- With your partner, discuss the four expressions in the table.

|              |                |
|--------------|----------------|
| Expression A | $(2)(2)(2)(2)$ |
| Expression B | $(2^4)$        |
| Expression C | 16             |
| Expression D | $2 \times 4$   |

- Which expression does not represent the same value as the others?  
*Expression D,  $2 \times 4$*
- Explain your answer or show the work that led to your conclusion.

| Expression A                                            | Expression B                                                         | Expression C             | Expression D          |
|---------------------------------------------------------|----------------------------------------------------------------------|--------------------------|-----------------------|
| $(2)(2)(2)(2)$<br>$= (4)(2)(2)$<br>$= (8)(2)$<br>$= 16$ | $(2^4)$<br>$= (2)(2)(2)(2)$<br>$= (4)(2)(2)$<br>$= (8)(2)$<br>$= 16$ | <i>The expression 16</i> | $2 \times 4$<br>$= 8$ |

- Explain what mathematics process should be done if the directions state to **evaluate** an expression.  
*Answers will vary. Evaluate an expression means "find the value of" the expression. Use the order of operations to find the value of the expression.*
- Marta's work for evaluating the expression  $3^2$  is shown:

$$3^2 = 3 \times 2 = 6$$

- Discuss Marta's work with your partner.  
*Partner discussion.*



Have students practice interpreting these mathematical statements, particularly reading the same statement in multiple ways.



Prompt students to discuss the difference between  $a^b$  and  $a \times b$ , or have them provide examples to prove their explanation.

## 6.1.4 Collaboration: Evaluating Exponential Expressions (continued)

- b. Explain whether her work is correct or incorrect. If it is correct, describe how you know. If it is incorrect, describe her error and provide the correct evaluation of the expression.

*Answers will vary. Martha is incorrect. She multiplied the base by the power. The 2 in the exponential expression is the power and the 3 is the base. This means that 2 factors of 3 are to be multiplied. The correct answer is:  $3 \cdot 3 = 9$ .*



Ask student how they might explain to Marta the correct way to complete the math problem.

3. Determine if each equation in the table is true or false. Show your work to support your answer.

| Equation                                                                                  | True or False? | Work                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $16^1 = 8^2$                                                                              | <i>False</i>   | $16^1 = 16$<br>$8^2 = 8 \cdot 8$<br>$= 64$<br>$16 \neq 64$                                                                                                                                                                                    |
| $3 + 3 + 3 = 3^3$                                                                         | <i>False</i>   | $3 + 3 + 3 = 9$<br>$3^3 = 3 \cdot 3 \cdot 3$<br>$= 9 \cdot 3$<br>$= 27$<br>$9 \neq 27$                                                                                                                                                        |
| $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = 4 \cdot \frac{1}{2}$ | <i>False</i>   | $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{1}{2}$<br>$= \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$<br>$4 \cdot \frac{1}{2} = \frac{4}{2} = 2$<br>$\frac{1}{16} \neq 2$ |
| $(5)^3 = 5 \cdot 5 \cdot 5$                                                               | <i>True</i>    | $(5)^3 = 5 \cdot 5 \cdot 5$<br>$= 25 \cdot 5$<br>$= 125$                                                                                                                                                                                      |

## 6.1.5 Try It: Evaluating Exponential Expressions

**Component Overview:** Students practice evaluating exponential expressions. The bases represent a variety of rational numbers (a fraction, whole number, and decimal).



### 6.1.5 Try It: Evaluating Exponential Expressions

1. Evaluate each expression written in exponential form.

a.  $\left(\frac{2}{5}\right)^2$   
 $\frac{4}{25}$

b.  $4^5$   
 $1,024$

c.  $(6.2)^3$   
 $238.328$



Students who struggle may benefit from writing each expression out using repeated multiplication before evaluating the problem.

## 6.1.6 Your Turn!

**Component Overview:** Students will practice identifying equivalent expressions, rewriting expressions, and evaluating them. Circulate while students work to monitor progress, assist struggling students, and determine which students you will call on to share during the whole-class debrief.



### 6.1.6 Your Turn!

1. For each set of expressions, circle the equivalent expressions.

a.

$$(4)^3$$

$$(3)^4$$

$$(4)(4)(4)$$

$$4 + 4 + 4$$

b.

$$\left(\frac{5}{3}\right)^2$$

$$\left(\frac{3}{5}\right)^2$$

$$\frac{10}{6}$$

$$\frac{25}{9}$$

2. Complete the table by rewriting the numeric expression in the form indicated. Then, find the value of each expression.

| Exponential Form              | Expanded Form                                                                               | Value               |
|-------------------------------|---------------------------------------------------------------------------------------------|---------------------|
| $11^3$                        | $(11)(11)(11)$                                                                              | 1,331               |
| $\left(\frac{7}{15}\right)^3$ | $\left(\frac{7}{15}\right) \cdot \left(\frac{7}{15}\right) \cdot \left(\frac{7}{15}\right)$ | $\frac{343}{3,375}$ |
| $(5.2)^2$                     | $(5.2) \cdot (5.2)$                                                                         | 27.04               |
| $\left(\frac{4}{5}\right)^4$  | $\frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5}$                         | $\frac{256}{625}$   |



Ask students what they notice and wonder about the different forms of the equivalent expressions. These discussions will help students make connections and build their understanding.

## 6.1.7 Wrap-Up: Ticket Out the Door

**Component Overview:** Students demonstrate their understanding of the lesson learning targets by analyzing samples of student work for errors. Consider using data from this formative assessment to determine which students need support before the next lesson.



### 6.1.7 Wrap-Up: Ticket Out the Door

Jose and Georgette were asked to each roll a number cube five times and record the number from each roll onto a chart. The number that Jose rolls represents the base of an exponential expression, and Georgette's roll represents the exponent of the expression. Their table below shows the results of rolling the number cubes five times.

| Roll | Jose's Number Cube | Georgette's Number Cube | Exponential Expression |
|------|--------------------|-------------------------|------------------------|
| 1    | 5                  | 3                       | $5^3$                  |
| 2    | 5                  | 4                       | $5^4$                  |
| 3    | 6                  | 2                       | $6^2$                  |
| 4    | 4                  | 5                       | $4^5$                  |
| 5    | 1                  | 4                       | $1^4$                  |

Jose conjectures that the value of the exponential expression from Roll 3 is the greatest because the exponential expression that has the largest base is always greater.

Georgette conjectures that the value of the exponential expression from Roll 4 is the greatest because the exponential expression that has the largest exponent is always greater.

1. Determine if either Jose or Georgette were right. If neither are correct, then which exponential expression has the largest value?

*In this case, Georgette is correct because  $4^5 = 1,024$ , which is greater than  $5^4 = 625$ .*

2. Determine if it is possible that Jose's conjecture or Georgette's conjecture is always true using the exponential expressions  $2^5$  and  $5^2$ . Explain your findings.

*Answers will vary.  $2^5 = 32$  and  $5^2 = 25$ . Based on the answers, Jose's conjecture is not always true because although 5 is the larger base, the value of the exponential expression  $5^2$  is less. Georgette's conjecture is also not always true. For example, if the base is 1, the overall value will always be 1, even if raised to a higher power.*



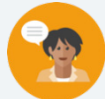
Provide students with scaffolding prompts if they are unsure where to begin.

- Read the information for context.
- After reading, what is the question asking?
- Compare what you think the answer is to Jose's and Georgette's responses. Then answer questions 1 and 2 based on your findings.

## 6.2 Prime Factorization

### Lesson Introduction

| Lesson Introduction                                                                                   |                                                                                                                                                                                                                                                                                                                  |                                                                                                                 |                                                                                                     |                                                                                                                  |                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.3.4 Express composite whole numbers as a product of prime factors with natural number exponents.                                                                                                                                                                                                        |                                                                                                                 |  Learning Target | • I can express composite whole numbers as a product of prime factors.                                           |                                                                                                                                                                                                                                                      |
|  Benchmark Coherence | Building On<br>MA.4.AR.3.1 Determine factor pairs for a whole number from 0 to 144. Determine whether a whole number from 0 to 144 is prime, composite, or neither.                                                                                                                                              |                                                                                                                 | Working Toward<br>N/A                                                                               |                                                                                                                  |                                                                                                                                                                                                                                                      |
|  Essential Questions | <ul style="list-style-type: none"><li>• What is a factor?</li><li>• What is a prime factor?</li><li>• How do you determine the prime factorization of a composite whole number?</li></ul>                                                                                                                        |                                                                                                                 |                                                                                                     |                                                                                                                  |                                                                                                                                                                                                                                                      |
|  Lesson Narrative    | In this lesson, students use their prior knowledge of factors and prime numbers to explore finding the prime factors of composite numbers using factor trees. Students extend their understanding of prime factorizations by expressing composite whole numbers as the product of prime factors using exponents. |                                                                                                                 |                                                                                                     |                                                                                                                  |                                                                                                                                                                                                                                                      |
|  Component Summary | 6.2.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                          | Select which scenario you prefer and support the choice mathematically ( <i>SE p. 238</i> )                     | 6.2.4 Guided Practice (10 min.)                                                                     | Practice expressing composite numbers as a product of prime factors ( <i>SE p. 241</i> )                         |                                                                                                                                                                                                                                                      |
|                                                                                                       | 6.2.2 Collaboration (5-10 min.)                                                                                                                                                                                                                                                                                  | Represent composite numbers as products of factors and explain how the factor pairs relate ( <i>SE p. 239</i> ) | 6.2.5 Your Turn! (10 min.)                                                                          | Independent practice expressing three-digit composite numbers as a product of prime factors ( <i>SE p. 241</i> ) |                                                                                                                                                                                                                                                      |
|                                                                                                       | 6.2.3 Guided Instruction (10 min.)                                                                                                                                                                                                                                                                               | Express composite numbers as a product of prime factors ( <i>SE p. 239</i> )                                    | 6.2.6 Wrap-Up (~5 min.)                                                                             | Demonstrate understanding of writing the prime factorization of composite numbers ( <i>SE p. 242</i> )           |                                                                                                                                                                                                                                                      |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                         |                                                                                                                 | Materials                                                                                           |                                                                                                                  | Additional Preparation                                                                                                                                                                                                                               |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>• Prime number</li><li>• Prime factor</li><li>• Prime factorization</li></ul> <u>Additional Terms:</u> N/A                                                                                                                                               |                                                                                                                 | <ul style="list-style-type: none"><li>• (Optional) Factor tree graphic organizers</li></ul>         |                                                                                                                  | <ul style="list-style-type: none"><li>• If differentiating for varying writing abilities, consider preparing sentence starters/stems.</li><li>• Consider prepping a list of prime factors for students to reference as they build fluency.</li></ul> |

|                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher<br>Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Things to Consider                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                                                                                   | <ul style="list-style-type: none"><li>Students may struggle to see how the factor pairs relate to each other or may struggle to explain the connection in writing in the 6.2.2 Collaboration.</li><li>Students without a strong grasp of multiplication may struggle to find the prime factors of three- and four-digit numbers.</li></ul>                                                                                                                                                         | <ul style="list-style-type: none"><li>Consider reviewing the definition of prime number before the lesson to activate prior knowledge.</li><li>Consider providing factor tree graphic organizers, particularly in 6.2.3 Guided Instruction, 6.2.4 Guided Practice, 6.2.5 Your Turn!, and 6.2.6 Wrap-Up.</li><li>Consider providing a list of prime factors for students to reference throughout the lesson as an additional scaffold.</li></ul>                                                                                                             |
|                                                                                                                   | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                                                                                   | <b>Think-Pair-Share</b><br>In 6.2.4 Guided Practice and 6.2.5 Your Turn!, have students first complete problems independently and then discuss their work with a partner. These discussions will build student confidence and understanding.                                                                                                                                                                                                                                                       | <b>MA.K12.MTR.5.1: Patterns and Structures</b><br>In 6.2.2 Collaboration, the task lends itself well to students using patterns and structure to help understand and connect mathematical concepts in building toward understanding of prime factorization.<br><br><b>MA.K12.MTR.4.1: Discussion and Reflection</b><br>In question 2 of 6.2.4 Guided Practice, students have an opportunity to analyze the errors in a sample of student work. Students identify the error, write a letter to the student explaining the error, and then correct the error. |
| <br>Differentiate<br>Instruction | Consider providing students with factor tree graphic organizers to provide a visual entry point to help them express composite whole numbers as products of prime factors. Discussions during 6.2.2 Collaboration provide auditory entry points.<br><br>Consider providing a choice of composite numbers for 6.2.5 Your Turn! question 3 for students who may struggle to come up with a number on their own. This will allow students to focus on the process of writing the prime factorization. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Lesson Notes:                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



## 6.2.1 Warm-Up: Would You Rather?

**Component Overview:** Students select which of two scenarios they prefer and support their choice mathematically.



### 6.2.1 Warm-Up: Would you Rather?

- Which of the following would you rather choose? Justify your reasoning using mathematics.

Deposit \$3 in the bank and have it triple each week for 4 weeks.

Deposit \$4 in the bank and have it quadruple each week for 3 weeks.

*Answers will vary.*

*If you deposit \$3 in the bank and have it triple every 4 weeks you will have the following: Week 1 (\$3). Week 2 ( $3 \cdot 3 = \$9$ ). Week 3 ( $9 \cdot 3 = \$27$ ). Week 4 ( $27 \cdot 3 = \$81$ ).*

*If you deposit \$4 in the bank and have it quadruple each week for 3 weeks you will have the following: Week 1 (\$4), Week 2 ( $4 \cdot 4 = \$16$ ). Week 3 ( $16 \cdot 4 = \$64$ ).*

*I would rather have the \$3 tripled for 4 weeks because I would end up with more money in the end.*



Consider asking students the following:

- How can you represent each scenario using multiplication?
- How much money will be in the bank after 4 weeks?
- How much will be in the bank after 3 weeks?

## 6.2.2 Collaboration: Exploring Factors

**Component Overview:** Students work with a partner to write whole numbers as factor pairs. Students then draw connections among the factor pairs.



### 6.2.2 Collaboration: Exploring Factors

Work with your partner on the following.

- Write the number 24 as a factor pair that uses the greatest factor other than itself.  
 $2 \cdot 12$
- Write the number 12 as a factor pair that uses the greatest factor other than itself.  
 $2 \cdot 6$
- Write the number 6 as a factor pair that uses the greatest factor other than itself.  
 $2 \cdot 3$
- Write the number 3 as a factor pair that uses the greatest factor other than itself.  
*There is no way to do this because the only factors of 3 are 3 and 1.*
- How did each of the above tasks, problems 2-4, relate to the factor pair that came before it?  
*Answers will vary. With the exception of task number 4, each factor pair has 2 as a factor. Each factor pair are factors of 24. The greatest factor of each task is also a factor of the greatest factor of the pair of the previous task.*
- What was different about rewriting the number 3 as a factor pair using the greatest factor other than itself?  
*Answers will vary. The number 3's greatest factor other than itself is 1.*



Consider your students when planning this Collaboration to ensure equitable opportunity for contribution. Students may either just work independently side by side instead of collaborating, or one student may dominate the conversation. Prompt discussions during this component using intentional pauses, or give students specified roles and questions to then compare with their partner or a small group.



**MA.K12.MTR.5.1: Patterns and Structure**  
Encourage students to make use of structure in making meaningful connections for question 6.

## 6.2.3 Guided Instruction: Prime Factorization

**Component Overview:** Students review the definition of prime numbers and learn the definitions of prime factorization and prime factors. Students use factor trees to determine the prime factorization of composite numbers, representing repeated multiplication in the prime factorizations using exponents.



### 6.2.3 Guided Instruction: Prime Factorization

Each of the factor pairs from the Collaboration can be used to rewrite 24 using more than one pair of factors.

For example, the first factor pair was  $2 \cdot 12$ , so 24 can be rewritten as  $24 = 2 \cdot 12$ .

Because 12 can be written as  $2 \cdot 6$ , 24 can also be rewritten as  $24 = 2 \cdot 2 \cdot 6$ .

1. Since 6 can be expressed as  $2 \cdot 3$ , then 24 can be rewritten as  $24 = 2 \cdot 2 \cdot 2 \cdot 3$ .

This is called the **prime factorization** of 24 because all of the factors are **prime numbers**.



A **prime number** is a whole number greater than 1 that is not divisible by any whole number other than 1 and itself.

A **prime factor** is a factor that is a prime number.

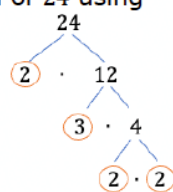


**Prime factorization** is the expression of a number as the product of prime factors.

2. Cory determined the prime factorization of 24 using different factor pairs.

- a. Explain whether Cory's method changed the final result.  
*Cory's method would not change the result. The prime factorization remains:  $2 \cdot 2 \cdot 2 \cdot 3$ .*

- b. Why did Cory circle the values he did?  
*Cory is identifying the prime factors.*



Utilize the Study Expert-led Glossary Videos for additional support with these glossary terms in multiple languages.



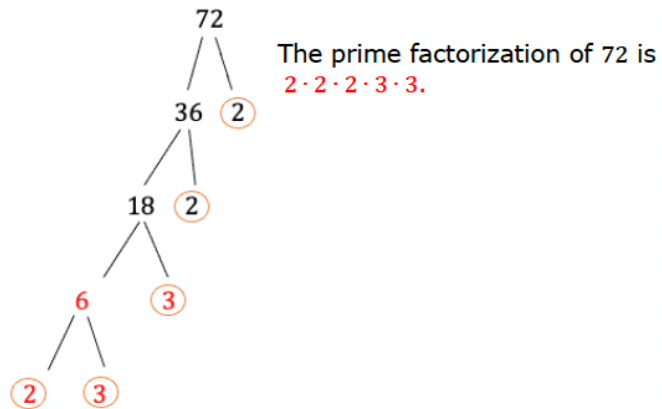
How do you determine if a number is prime?  
If a prime factorization has repeated prime factors, how do you represent them?



Consider providing a list of prime factors for students to reference as they begin the process of prime factorization of whole numbers as additional scaffolded support.

## 6.2.3 Guided Instruction: Prime Factorization (continued)

3. Find the prime factorization of 72 by completing the work below.



4. Consider the expression  $2 \times 2 \times 2 \times 3 \times 3 \times 5$ .
- What is the product of  $2 \times 2 \times 2 \times 3 \times 3 \times 5$ ?  
**360**
  - The prime factorization of **360** is  
 $2 \times 2 \times 2 \times 3 \times 3 \times 5$ .

Numbers with multiple repeating prime factors can be written using exponents. For example, the prime factorization of 360 can be written as  $2^3 \times 3^2 \times 5$ .

5. Why is it helpful to write the prime factorization with exponents?
- Answers will vary. When there are many of the same factor(s), the prime factorization becomes very long to write. Prime factorization with exponents allows for the factors to be seen more easily.*



*For students who may struggle with multiplication, encourage them to start by multiplying two factors at a time, chunking the expression into more manageable parts.*



*Consider using the Think-Pair-Share routine before question 4 to provide students an opportunity to discuss the reasons it is helpful to use exponents when writing prime factorization.*

## 6.2.4 Guided Practice: Prime Factorization

**Component Overview:** Students find the prime factorization of composite numbers and represent them using exponents. Students also analyze the errors in a sample of student work.



### 6.2.4 Guided Practice: Prime Factorization

- Find the prime factorization of each value. Use exponents where appropriate.

a. 132      b. 270      c. 882  
 $2^2 \cdot 3 \cdot 11$        $2 \cdot 3^3 \cdot 5$        $2 \cdot 3^2 \cdot 7^2$

- Sophie said the prime factorization of 210 is  $3 \cdot 7 \cdot 10$ .

- Determine the error Sophie made. Then, write her a brief note explaining her error and how she can learn from it.

**Means**  
**I**  
**Start**  
**To**  
**Acquire**  
**Knowledge**  
**Experience**  
**Skills**

*Answers will vary. Sophie, you are almost there! When factoring a number, each factor must be a prime number. 10 is not a prime number. Recall that a prime number is a whole number greater than 1 that is not divisible by any whole number other than 1 and itself. To complete your steps, 10 will need to be factored using 2 and 5, which are prime numbers.*

*When factoring, remember to ask yourself, "is each factor a prime number?" You got this!*

- What is the prime factorization of 210?  
 $2 \cdot 3 \cdot 5 \cdot 7$



Consider providing sentence starters on question 2 for students who struggle writing an explanation of the error.

- "When finding the prime factors, you should..."
- "The \_\_\_\_ is wrong because..."
- "Next time, it might be helpful if you..."



Have a few students share the error they identified and how to correct it.



Encourage students to finish their note with encouragement for Sophie to foster growth mindset.

## 6.2.5 Your Turn!

**Component Overview:** Students independently practice writing the prime factorization of composite whole numbers.



### 6.2.5 Your Turn!

1. Find the prime factorization of each value. Use exponents where appropriate.

a. 904  
 $2^3 \cdot 113$

b. 1,125  
 $3^2 \cdot 5^3$

c. 80  
 $5 \cdot 2^4$

2. Which number includes the largest exponent when written in prime factorization? Explain your answer.

5,670

64

$2 \cdot 3^4 \cdot 5 \cdot 7$

$2^6$

*Written in prime factorization, 64 has the largest exponent, 6. There are 6 factors of 2 in 64. There are 4 factors of 3 in the prime factorization of 5,670.*

3. Find a number that meets the following conditions.
- The value of the number is less than 150.
  - One of its prime factors has an exponent of 3.
  - When expressed in exponential form, its prime factorization has only two unique bases.

*Answers will vary.  $2^3 \cdot 5 = 40$ ,  $3^3 \cdot 2^2 = 108$*



*If time allows after completing the Your Turn!, have students compare their answers to question 2 with a partner. Elicit any responses of students who changed their answer because of something their partner shared.*



*If students finish early, have them come up with additional examples that meet the requirements in question 3 and challenge a family member at home or another teacher with their number.*



## 6.2.6 Wrap-Up: Ticket Out the Door

**Component Overview:** Students demonstrate their understanding of the lesson learning targets. Use data from this formative assessment to determine which students need support on this benchmark.



### 6.2.6 Wrap-Up: Ticket Out the Door

1. Find the prime factorization of each value. Use exponents where appropriate.

a. 168  
 $2^3 \cdot 3 \cdot 7$

b. 1,040  
 $2^4 \cdot 5 \cdot 13$

c. 144  
 $2^4 \cdot 3^2$



Consider providing factor tree graphic organizers for students to use to show their work for more information when assessing their progress.



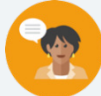
If students need additional scaffolding, consider having them make two attempts. In their first attempt, they should show their work and mark as “without a calculator.” With their second attempt, they should show work marked as “with a calculator.”



## 6.3 Adding Integers Using Integer Chips

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                      |                                                                                                                                  |                                                                                                                                                                                                      |                                                                                                                                                     |                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.                                                                                                                                         |                                                                                                                                  |  Learning Targets                                                                                                 | <ul style="list-style-type: none"><li>I can express integer addition using integer chips.</li><li>I can add integers using integer chips.</li></ul> |                                                                                                                                                   |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                          |                                                                                                                                  | Working Toward                                                                                                                                                                                       |                                                                                                                                                     |                                                                                                                                                   |
|                                                                                                       | MA.5.NSO.2 Add, subtract, multiply, and divide multidigit numbers.                                                                                                                                                                                                                   |                                                                                                                                  | MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.<br><br>MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers. |                                                                                                                                                     |                                                                                                                                                   |
|  Essential Questions | <ul style="list-style-type: none"><li>What is an integer?</li><li>How can you use integer chips to represent integer addition?</li><li>What is a zero pair?</li><li>How do you use integer chips to add integers?</li></ul>                                                          |                                                                                                                                  |                                                                                                                                                                                                      |                                                                                                                                                     |                                                                                                                                                   |
|  Lesson Narrative    | In this lesson, students represent integer addition using integer chips to begin developing an understanding of how to add signed numbers. Students explore zero pairs and begin generalizing about patterns they observe when adding integers with the same sign or opposite signs. |                                                                                                                                  |                                                                                                                                                                                                      |                                                                                                                                                     |                                                                                                                                                   |
|  Component Summary | 6.3.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                              | Make connections between elevations and positive and negative integers ( <i>SE p. 243</i> )                                      | 6.3.4 Bring It Together (5 min.)                                                                                                                                                                     | Summarize how to add integers with the same sign using integer chips ( <i>SE p. 245</i> )                                                           |                                                                                                                                                   |
|                                                                                                       | 6.3.2 Guided Instruction (5-10 min.)                                                                                                                                                                                                                                                 | Represent simple money contexts with positive and negative integers and integer chips and define zero pairs ( <i>SE p. 243</i> ) | 6.3.5 Exploration (15 min.)                                                                                                                                                                          | Explore using zero pairs to add integers with opposite signs using integer chips ( <i>SE p. 246</i> )                                               |                                                                                                                                                   |
|                                                                                                       | 6.3.3 Exploration (10 min.)                                                                                                                                                                                                                                                          | Explore adding integers with the same sign using integer chips ( <i>SE p. 245</i> )                                              | 6.3.6 Bring It Together (5 min.)                                                                                                                                                                     | Summarize how to add integers with opposite signs ( <i>SE p. 248</i> )                                                                              |                                                                                                                                                   |
|                                                                                                       | 6.3.7 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                              | Demonstrate understanding of the lesson learning targets ( <i>SE p. 248</i> )                                                    |                                                                                                                                                                                                      |                                                                                                                                                     |                                                                                                                                                   |
|  Resources         | Glossary                                                                                                                                                                                                                                                                             |                                                                                                                                  | Materials                                                                                                                                                                                            |                                                                                                                                                     | Additional Preparation                                                                                                                            |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Integers</li><li>Zero pair</li><li>Zero pair as a mathematical equation</li></ul> Additional Terms: N/A                                                                                                                      |                                                                                                                                  | <ul style="list-style-type: none"><li>Integer chips</li><li>(Optional) \$1 manipulatives</li></ul>                                                                                                   |                                                                                                                                                     | <ul style="list-style-type: none"><li>Consider creating sentence stems and/or answer choices for question 2 in 6.3.6 Bring It Together.</li></ul> |

| <br>Teacher Tips                                                                                                                                                               | Common Misconceptions                                                                                                                                                                                                  | Things to Consider                                                                                                                                                                                                                                                                                                                                      |      |       |                                                                                     |                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"><li>Students may struggle to come up with real-world scenarios to represent addition expressions, particularly when addends are negative.</li></ul>                                  | <ul style="list-style-type: none"><li>Consider providing integer chips for students in 6.3.5 Exploration or encouraging students to draw a representation if they get stuck.</li><li>Consider using \$1 manipulatives in 6.3.3 Exploration to help students connect the idea of spending and earning money to positive and negative integers.</li></ul> |      |       |                                                                                     |                                                                                     |
|                                                                                                                                                                                                                                                                 | Literacy and Language Routines                                                                                                                                                                                         | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                      |      |       |                                                                                     |                                                                                     |
|                                                                                                                                                                                                                                                                 | <b>Notice and Wonder</b><br>In 6.3.3 Exploration and 6.3.5 Exploration, provide an opportunity for students to notice and wonder about patterns in the sums when adding integers with the same or opposite signs.      | <b>MA.K12.MTR.2.1: Multiple Representations</b><br>In 6.3.3 Exploration and 6.3.5 Exploration, students have opportunities to represent adding integers using multiple representations. Students make connections among written descriptions of real-world scenarios, numeric expressions, and integer chips.                                           |      |       |                                                                                     |                                                                                     |
| <br>Differentiate Instruction                                                                                                                                                  | Lesson 3 incorporates integer chips into instruction to provide meaningful learning strategies for making sense of adding integers. Provide integer chips for students to explore each activity throughout the lesson. |                                                                                                                                                                                                                                                                                                                                                         |      |       |                                                                                     |                                                                                     |
| <div><table><tr><th>Earn</th><th>Spend</th></tr><tr><td></td><td></td></tr></table></div> |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                         | Earn | Spend |  |  |
| Earn                                                                                                                                                                                                                                                            | Spend                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                         |      |       |                                                                                     |                                                                                     |
|                                                                                                                                                                              |                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                         |      |       |                                                                                     |                                                                                     |
| Lesson Notes:                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                         |      |       |                                                                                     |                                                                                     |

## 6.3.1 Warm-Up: Below Sea Level

**Component Overview:** Students connect positive and negative integers to distances above and below sea level and sketch an example.



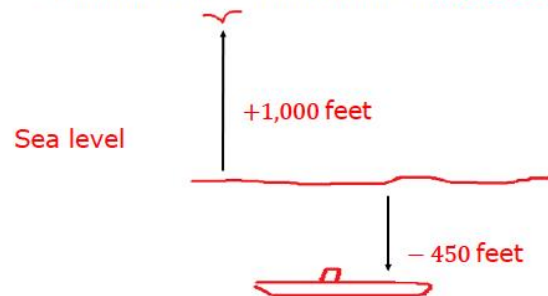
### 6.3.1 Warm-Up: Below Sea Level

Dr. Ernest Everett Just was a pioneering African American biologist who conducted research at Woods Hole Marine Biological Laboratory in Massachusetts. He published two books and more than 70 scientific papers, most of which focused on the cell. Today, marine biologists use submarines to study marine life.



A submarine explorer begins at sea level and descends at a rate of 150 feet per minute.

1. Sketch a picture that shows how far below sea level the submarine will be in 3 minutes. *Answers will vary.*



2. Add a bird that is flying 1,000 feet above sea level to your picture. *See above*
3. What number represents sea level?  
**0**
4. The location of the submarine is represented by a

- ☐ positive value,
- ☒ negative value,

represented by a

- while the location of the bird is
- ☒ positive value.
  - ☐ negative value.



*Dr. Ernest Everett Just accomplished many things in his career that may pique students' interests in the STEM field. Consider an interdisciplinary extension project with a science teacher on one of his many roles.*



*As students work on question 1, consider asking them the following:*

- *Does your sketch more closely resemble a horizontal or vertical number line? Why?*
- *What value does sea level represent in your sketch?*

## 6.3.2 Guided Instruction: Zero Pairs

**Component Overview:** Students learn how positive and negative integers can be represented using integers and integer chips. The concept of zero pairs is introduced in the context of adding integers. Students review the definition and then use zero pairs to add integers.



### 6.3.2 Guided Instruction: Zero Pairs

Positive and negative values are often useful in the contexts of money, banking, and accounting.

- In the context of money, 

○ positive  
○ negative

 numbers can be used to represent money spent or money owed to someone.
- Positive  
○ Negative

 numbers can be used to represent money earned.



Consider providing a few real-world examples of positive and negative numbers representing money. These examples will activate prior knowledge and help students connect spending and earning to positive and negative numbers.

Money can be used to learn how to add integers.

- A value of  $+1$  means that you *earned* \$1.
- A value of  $-1$  means that you *spent* \$1 or *owe* \$1.

The table below shows a positive integer chip, which represents earning \$1, and a negative integer chip, which represents spending \$1.

| Earn | Spend |
|------|-------|
|      |       |

Integer chips can also be used to represent integers.



**Integers** are whole numbers and their opposites.

## 6.3.2 Guided Instruction: Zero Pairs (continued)

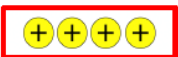
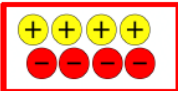
A positive number can be written with a positive sign,  $+1$  or  $(+1)$ , or it can be written without any sign,  $1$ .

A negative number can be written as  $-1$ ,  $(-1)$  or  $\bar{1}$ .

- Complete the table. In the last column, sketch the integer chips that represent the phrase and the integer. Place a  $+$  sign in a circle to represent  $+$  and place a  $-$  sign in a circle to represent  $-$ .

| Phrase     | Integer | Integer Chips                                                                     |
|------------|---------|-----------------------------------------------------------------------------------|
| Earned \$6 | $(+6)$  |  |
| Spent \$3  | $-3$    |  |

- Complete the table by sketching integer chips for each phrase.

| Earned \$4                                                                        | Spent \$4                                                                         | Earned \$4, Then Spent \$4                                                        |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|  |  |  |

- Joleen earned \$4, then spent \$4. How much money does she have left?  
 $0$

The last column in the table and problem above both represent a **zero pair**.

A **zero pair** is a pair of numbers whose sum is zero.



Consider the tools available for instruction during this activity. Modeling each example using inquiry would benefit students as they make connections from the integer-chip representations to their numerical representations.

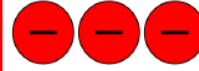
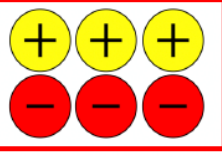


Consider asking students the following:

- Can you explain what a zero pair is in your own words?
- Can you provide an example of a zero pair?

## 6.3.2 Guided Instruction: Zero Pairs (continued)

4. Complete the table by drawing the negative integer chips that create a zero pair. In the **sum** column, draw all of the chips from both columns.

| Positive                                                                          | Negative                                                                          | Sum                                                                               |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|  |  |  |

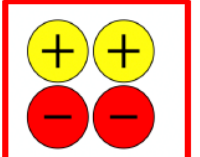
Each zero pair can be represented by an equation. The box shows two mathematical equations that represent the zero pair of  and .



### Zero Pair as a Mathematical Equation

$$+1 + (-1) = 0 \quad \text{or} \quad 1 + ^-1 = 0$$

5. Write an equation that represents the **sum** column in the previous table.  
 $+3 + (-3) = 0$
6. Complete the table by drawing the positive integer chips that create a zero pair. In the **sum** column, draw all of the chips from both columns.

| Positive                                                                            | Negative                                                                            | Sum                                                                                 |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |

7. Write an equation that represents the **sum** column in the previous table.  
 $+2 + (-2) = 0$



Students are only given the visual model for positive integer chips. They complete the visual model for negative integer chips and the sum.



To challenge advanced learners, consider having students write a word problem to go with the equations they have created to represent zero pairs.



Students are only given the visual model for negative integer chips. They complete the visual model for positive integer chips and the sum.



### 6.3.3 Exploration: Adding Integers with the Same Sign

**Component Overview:** Students work with a partner to complete a table that includes multiple representations of adding integers with the same sign. The table includes the numeric expression, money sentence, integer chips, and sum. Next, students draw conclusions about patterns they notice about the signs of the sums and if there are any zero pairs when adding integers with the same sign.



#### 6.3.3 Exploration: Adding Integers with the Same Sign

1. Work with your partner to complete the table.

| Expression    | Money Sentence                                                                                                                                                    | Integer Chips | Sum |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----|
| $5 + 2$       | I earn \$5 babysitting and another \$2 for cleaning the dishes.                                                                                                   |               | 7   |
| $3 + 2$       | <i>Answers will vary. I earn \$3 mowing the lawn and another \$2 for taking out the trash.<br/>I earn \$3 washing dishes and another \$2 for babysitting.</i>     |               | 5   |
| $-4 + -3$     | <i>Answers will vary. I owe my mom \$4 and I spent \$3 at the dollar store.<br/>I loaned my friend \$4 and I spent \$3 on my mom's birthday present.</i>          |               | -7  |
| $(-2) + (-1)$ | <i>Answers will vary. I spent \$2 at the candy store and I paid back my mom the \$1 I owed her.<br/>I bought a \$2 ice cream cone and I loaned my sister \$1.</i> |               | -3  |

2. With your partner, discuss the patterns you notice about the signs of the sums when adding integers with the same sign.

*Answers will vary. I notice that when the signs were the same, we added the total number of chips.*

3. Were any zero pairs created when adding integers with the same sign? Explain.

*Answers will vary. No zero pairs were created when adding integers with the same sign. Instead, the result was a larger quantity of the signed integers.*



Students are only given the visual model of integer chips for the first row.



#### Notice and Wonder

Consider engaging students with a discussion regarding what they notice and wonder about the signs and if there were any zero pairs prior to them writing their responses to questions 2 and 3.



Consider asking students the following:

- What is the sign of the sum of two positive integers?
- What is the sign of the sum of two negative integers?



## 6.3.4 Bring It Together: Adding Integers with the Same Sign

**Component Overview:** Students summarize how to add integers with the same sign using integer chips. Consider creating an anchor chart and displaying it for students to reference.

### 6.3.4 Bring It Together: Adding Integers with the Same Sign

1. Complete the visual model of the rule of adding two positive integers by drawing the integer chip that represents the **sign** of the sum.

| Visual Model of Addition                                                          | Sign of Sum                                                                       |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|  |  |

2. Complete the visual model of the rule of adding two negative integers by drawing the integer chip that represent the **sign** of the sum.

| Visual Model of Addition                                                            | Sign of Sum                                                                         |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |



Prompt students to justify their visual models to a partner. Highlight any students that make changes based on their partner's feedback to encourage growth mindset.



Consider asking students the following:

- Will the sum of two positive integers always be positive?
- Will the sum of two negative integers always be negative?



To add complexity, consider asking students who finish early to write equations to match each visual model of addition.

## 6.3.5 Exploration: Using Zero Pairs to Add Integers with Opposite Signs

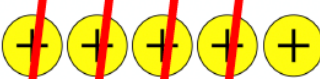
**Component Overview:** Students use integer chips to represent the sum of integers with opposite signs, determining the sum based on the zero pairs. Students then work with a partner to continue adding integers with opposite signs and create money sentences to represent addition expressions. Students work independently to find the sum of integers, check their partner's work, and use the sums to generalize how to add integers with opposite signs.



### 6.3.5 Exploration: Using Zero Pairs to Add Integers with Opposite Signs

It is possible to add a positive number and a negative number. To find the answer using integer chips, combine a  $+$  and  $-$  to create a zero pair. Make as many zero pairs as possible. Then, determine the number of chips that are left over.

- Complete the table to show how the sum  $5 + (-4)$  can be represented using integer chips.

| Step                                                                                                  | Response                                                                          |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Represent the first addend with integer chips.                                                        |  |
| Represent the second addend with integer chips.                                                       |  |
| Determine the number of zero pairs created, if any.                                                   | 4 zero pairs                                                                      |
| Cross out the integer chips that represent zero pairs to indicate they are eliminated from the model. |                                                                                   |
| Determine the <b>sum</b> using the number of integer chips remaining.                                 | $5 + (-4) = 1$                                                                    |

- What do you notice about the remaining integer chips and the sign of the sum?  
*There was 1 positive integer chip remaining. The sum was 1.*



Consider asking students the following:

- How can you represent 5 with integer chips?
- How can you represent  $-4$  using integer chips?
- Why is the sum of 5 and  $-4$  positive?



For both questions 1 and 3, students are given the visual model for the first two steps and must complete the third row in addition to synthesis questions 2 and 4.

### 6.3.5 Exploration: Using Zero Pairs to Add Integers with Opposite Signs (continued)

3. Complete the table independently to show how the sum  $4 + (-7)$  can be represented using integer chips. Check your work with your partner.

| Step                                                                                                  | Response                                                                          |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Represent the first addend with integer chips.                                                        |  |
| Represent the second addend with integer chips.                                                       |  |
| Determine the number of zero pairs created, if any.                                                   | 4 zero pairs                                                                      |
| Cross out the integer chips that represent zero pairs to indicate they are eliminated from the model. |                                                                                   |
| Determine the <b>sum</b> using the number of integer chips remaining.                                 | $4 + (-7) = -3$                                                                   |

4. What do you notice about the remaining integer chips and the sign of the sum?  
*There were 3 negative integer chips remaining. The sum was  $-3$ .*



Consider having students discuss what they think the sign of the sums will be based on what they notice about the addition expressions before having them complete the table for question 5. Guide students to make logical predictions based on what they know about adding signed numbers and absolute value.

### 6.3.5 Exploration: Using Zero Pairs to Add Integers with Opposite Signs (continued)

5. Work with your partner to complete the table. The first row is completed.

| Expression | Money Sentence                   | Integer Chips | Sign of Sum | Sum |
|------------|----------------------------------|---------------|-------------|-----|
| $4 + (-2)$ | I earn \$4 and spend \$2.        |               | +           | +2  |
| $3 + (-6)$ | <i>I earn \$3 and spend \$6.</i> |               | -           | -3  |
| $-7 + 5$   | <i>I spend \$7 and earn \$5.</i> |               | -           | -2  |
| $(-1) + 8$ | <i>I spend \$1 and earn \$8.</i> |               | +           | +7  |



Consider having students who finish early represent each expression in an additional way, either with an equivalent expression, a visual representation, or a written scenario.

6. Complete the table below independently. Check your work with your partner.

| Mathematical Expression | Number of Zero Pairs | Sign of Sum | Sum |
|-------------------------|----------------------|-------------|-----|
| $(-10) + 12$            | 10                   | +           | 2   |
| $12 + -7$               | 7                    | +           | 5   |
| $(-11) + 4$             | 4                    | -           | -7  |
| $8 + -8$                | 8                    | No sign     | 0   |
| $-10 + 3$               | 3                    | -           | -7  |
| $9 + (-2)$              | 2                    | +           | 7   |



#### Turn and Talk

Ask students to verbalize their responses before completing their written explanation to questions 7 and 8. This will solidify student understanding and allow for any misconceptions to be resolved before the 6.3.7 Wrap-Up.

7. What do you notice about expressions where the sum is a positive number?

*Answers will vary. The number with the larger absolute value in the mathematical expression is positive.*

8. What do you notice about expressions where the sum is a negative number?

*Answers will vary. The number with the larger absolute value in the mathematical expression is negative.*



What do you notice about the sum when the addend with the larger absolute value is negative?

## 6.3.6 Bring It Together: Adding Integers with the Opposite Sign

**Component Overview:** Students write expressions to represent a visual model of addition of integers with opposite signs. Then, students explain two things they noticed about adding integers with opposite signs.



### 6.3.6 Bring It Together: Adding Integers with Opposite Signs

The table below shows visual models of rules for adding integers with opposite signs. In each visual model, the larger circle symbolizes the integer represented by the most integer chips.

- Write an example of the rule. The first row has been completed.

| Visual Model of Addition | Sign of Sum | Example                                                 |
|--------------------------|-------------|---------------------------------------------------------|
|                          |             | $2 + (-5) = -3$                                         |
|                          |             | Answers will vary.<br>$-3 + 10 = 7$<br>$-5 + 7 = 2$     |
|                          |             | Answers will vary.<br>$6 + (-4) = 2$<br>$10 + (-6) = 4$ |
|                          |             | Answers will vary.<br>$-9 + 8 = -1$<br>$-12 + 4 = 8$    |

- Give two observations about the rules for adding integers with opposite signs.

Answers will vary.

When adding integers with opposite signs:

- Ignoring the signs of the numbers in the mathematical expression, subtract the numbers.
- The sign of the sum will be the sign of the number with the greater absolute value in the mathematical expression.



Consider asking students the following:

- What do you notice about the size of these integer chips?
- How does the absolute value of each addend impact the sign of the sum?



**Common Misconception**

Monitor for students who may articulate their answers with phrases like “the bigger number will always be larger” or “the answer will take the sign of the bigger number.” The sum will take the sign of the addend with the larger absolute value, which is not the larger number if it is negative.

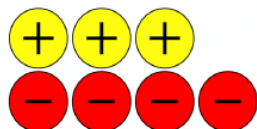
## 6.3.7 Wrap-Up: Ticket Out the Door

**Component Overview:** Students demonstrate their understanding of the lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should not use a calculator to complete the Wrap-Up.



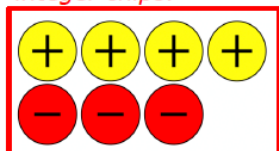
### 6.3.7 Wrap-Up: Ticket Out the Door

1. Simone created the sketch below to represent the expression  $4 + (-3)$ .



Her sketch is incorrect. Explain why and create a new sketch to find the sum.

*There should be 4 positive integer chips and 3 negative integer chips.*



2. Complete the table for the expression below.

| Expression      | $3 + -5$                                                                                                                                       |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Integer Chips   |                                                                                                                                                |
| Money Sentence  | <i>Answers will vary. I earned \$3 mowing the lawn and spend \$5 on pizza.<br/>I earned \$3 taking out the garbage and spend \$5 on candy.</i> |
| Sign of the Sum | -                                                                                                                                              |
| Sum             | -2                                                                                                                                             |



Consider providing integer chips to provide a kinesthetic entry point.

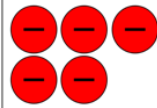
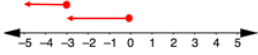
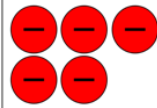
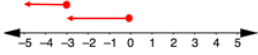
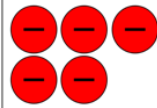
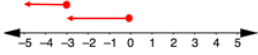
Also, consider providing sentence starters for question 2 to help students who struggle writing explanations.

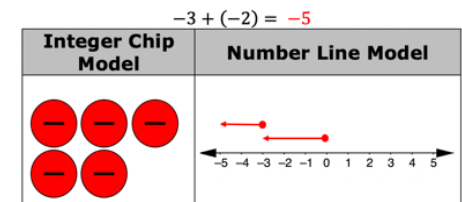


## 6.4 Adding Integers Using Number Lines

### Lesson Introduction

|                                                                                                                  |                                                                                                                                                                                                                                                                      |                                                                                                                                                               |                                                                                                                                                                                                                    |                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <div>Benchmark</div>           | MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.                                                                                                                         |                                                                                                                                                               |  <div>Learning Target</div>                                                                                                     | <ul style="list-style-type: none"><li>I can add integers using a number line.</li></ul>                                                                  |
|  <div>Benchmark Coherence</div> | Building On<br>MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.                                                                                                                                                                                   |                                                                                                                                                               | Working Toward<br>MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.<br>MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers. |                                                                                                                                                          |
|  <div>Essential Questions</div> | <ul style="list-style-type: none"><li>Where do negative integers fall on the number line? What about positive numbers?</li><li>How can you represent adding integers using a number line?</li><li>How do you use a number line to add integers?</li></ul>            |                                                                                                                                                               |                                                                                                                                                                                                                    |                                                                                                                                                          |
|  <div>Lesson Narrative</div>    | In this lesson, students add integers using number line models. Students connect their prior work modeling integer addition using integer chips to represent integer addition on a number line. Students also review and apply the commutative property of addition. |                                                                                                                                                               |                                                                                                                                                                                                                    |                                                                                                                                                          |
|  <div>Component Summary</div> | 6.4.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                              | Watch a video about how speed is not important in math and reflect ( <i>SE p. 249</i> )                                                                       | 6.4.4 Collaboration (10 min.)                                                                                                                                                                                      | Apply the commutative property of addition to integer addition problems represented with integer chips, number lines, and equations ( <i>SE p. 251</i> ) |
|                                                                                                                  | 6.4.2 Refresher (5 min.)                                                                                                                                                                                                                                             | Plot positive and negative integers on a number line ( <i>SE p. 250</i> )                                                                                     | 6.4.5 Try It (5 min.)                                                                                                                                                                                              | Practice adding integers using number lines ( <i>SE p. 252</i> )                                                                                         |
|                                                                                                                  | 6.4.3 Guided Instruction (10 min.)                                                                                                                                                                                                                                   | Model integer addition with integer chips and on number lines ( <i>SE p. 250</i> )                                                                            | 6.4.6 Your Turn! (10 min.)                                                                                                                                                                                         | Additional practice adding integers using number lines ( <i>SE p. 253</i> )                                                                              |
|                                                                                                                  | 6.4.7 Wrap-Up (~5 min.)                                                                                                                                                                                                                                              | Self-assessment of understanding and confidence using number lines to add integers ( <i>SE p. 254</i> )                                                       |                                                                                                                                                                                                                    |                                                                                                                                                          |
|  <div>Resources</div>         | Glossary                                                                                                                                                                                                                                                             | Materials                                                                                                                                                     |                                                                                                                                                                                                                    | Additional Preparation                                                                                                                                   |
|                                                                                                                  | <u>Key Terms:</u> N/A<br><u>Additional Terms:</u> <ul style="list-style-type: none"><li>Integer</li><li>Ray</li><li>Commutative Property</li></ul>                                                                                                                   | <ul style="list-style-type: none"><li>(Optional) Integer chips</li><li>(Optional) Colored pencils</li><li>(Optional) Dry erase boards &amp; markers</li></ul> |                                                                                                                                                                                                                    | <ul style="list-style-type: none"><li>6.4.1 Warm-Up contains a video to accompany the question prompts.</li></ul>                                        |

| <br>Teacher Tips              | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>Students may struggle to represent multiple addends on a number line, particularly where to begin the second addend.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Things to Consider</b> <ul style="list-style-type: none"><li>Consider providing sentence stems for 6.4.1 Warm-Up and question 4 for 6.4.4 Collaboration.</li></ul>                                                                                                                                                                                                                   |                    |                   |                                                                                     |                                                                                     |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                                                                                | <b>Literacy and Language Routines</b><br><b>Take Turns</b> <p>In 6.4.2 Refresher, students can first work independently to complete the problems. Then students can work with a partner and take turns sharing their responses as they review how to represent integers on a number line.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Mathematical Thinking and Reasoning (MTR) Standard</b><br><b>MA.K12.MTR.2.1: Multiple Representations</b> <p>In 6.4.3 Guided Instruction, students represent integer addition using integer chips, number lines, and numerical expressions. Ask students to share the similarities and differences among the representations and how they decide which representation(s) to use.</p> |                    |                   |                                                                                     |                                                                                     |
|                                                                                                                | <p>Multiple representations (MA.K12.MTR.2.1) provide students the opportunity to demonstrate their understanding in a variety of ways, particularly finding models or modes that fit their understanding. Provide opportunities for students to engage with manipulatives, sketch and draft their own representations, and transition their representations from concrete to abstract. 6.4.3 Guided Instruction provides explicit models and samples for using representations to demonstrate understanding of adding integers. Encourage students not to isolate this approach to just this lesson but instead think creatively and strategically in which approach best fits their thinking and reasoning to apply throughout the unit.</p> |                                                                                                                                                                                                                                                                                                                                                                                         |                    |                   |                                                                                     |                                                                                     |
| <br>Differentiate Instruction | <div><div><math>-3 + (-2) = -5</math></div><table><tr><th>Integer Chip Model</th><th>Number Line Model</th></tr><tr><td></td><td></td></tr></table></div>                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                         | Integer Chip Model | Number Line Model |  |  |
| Integer Chip Model                                                                                             | Number Line Model                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                         |                    |                   |                                                                                     |                                                                                     |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                         |                    |                   |                                                                                     |                                                                                     |
| Lesson Notes                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                         |                    |                   |                                                                                     |                                                                                     |



## 6.4.1 Warm-Up: Speed Is Not Important

**Component Overview:** Play the video, “Speed is not Important” to launch the Warm-Up. Students learn that math involves deep thinking and therefore can be completed slowly. This component is designed to debunk the myth that speed is important in math.



### 6.4.1 Warm-Up: Speed Is Not Important

How fast you perform calculations in mathematics is not at all related to how “good” you are at math. Fast calculation is not important. What matters is deep thinking and understanding.

1. Describe a time when you felt rushed to perform calculations on a math assignment or assessment.  
*Answers will vary.*



#### Think-Pair-Share

Have students share their responses with a partner after writing. Monitor for strong responses that highlight growth mindset and elicit those responses whole group. Also, prompt students to share something that their partner said that made them feel more confident or think about math in a different way.

2. If you were given as much time as you needed in the situation you described, how would that have affected your feelings and performance on completing the assignment or assessment?  
*Answers will vary.*

## 6.4.2 Refresher: Plotting Numbers on a Number Line

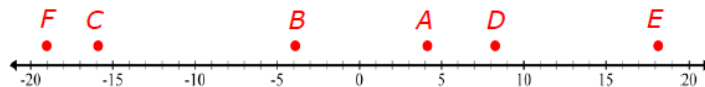
**Component Overview:** Students activate prior knowledge of integers and plotting numbers on number lines by plotting positive and negative integers on a number line. Students fill in statements to review positive and negative directions on a number line.



### 6.4.2 Refresher: Plotting Numbers on a Number Line

1. Plot each of the following integers on the number line below. Label each point with the letter indicated.

$$A = 4 \quad B = -4 \quad C = -16 \quad D = 8 \quad E = 18 \quad F = -19$$



2. Complete the statements.

- a. When moving from left to right ( $\rightarrow$ ) on the number line, the values ☐ increase. ☐ decrease. This can be described as moving in a ☐ positive ☐ negative direction.

- b. When moving from right to left ( $\leftarrow$ ) on the number line, the values ☐ increase. ☐ decrease. This can be described as moving in a ☐ positive ☐ negative direction.



#### Take Turns

Students can work first independently to complete the problems. Then students can work with a partner and take turns sharing their responses as they review how to represent integers on a number line.



Do not spend too much time on this Refresher as it is only meant to activate prior knowledge, not reteach a concept. What do you notice about the numbers to the left of the zero on the number line? What about the numbers to the right? What does this say about values as they go to the right on the number line?

## 6.4.3 Guided Instruction: Rules for Adding Integers on a Number Line

**Component Overview:** Students connect the integer-chip model to the number-line model, representing addition expressions using both. Students reason about how to accurately represent addends using rays and consider when number lines might be preferred representations.

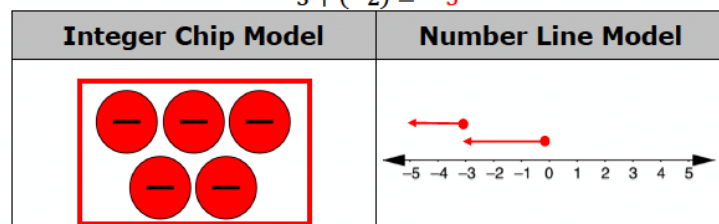


### 6.4.3 Guided Instruction: Rules for Adding Integers on a Number Line

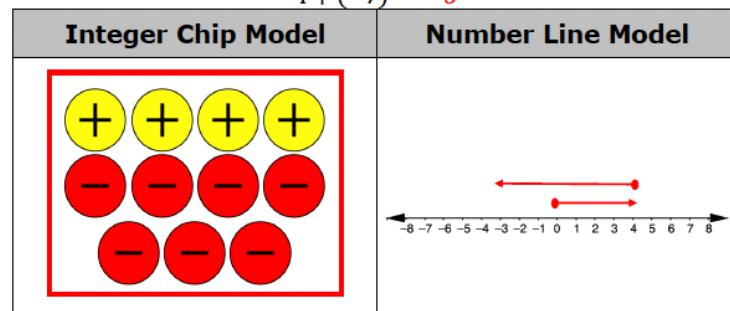
In addition to integer chips, number lines are another helpful tool you can use to model adding integers.

1. Complete the following to model integer addition.

$$-3 + (-2) = -5$$



$$4 + (-7) = -3$$



**MA.K12.MTR.2.1: Multiple Representations**  
Utilize inquiry to position as sense makers, connecting pictorial and concrete models and manipulatives to adding integers.

How can you represent  $-3$  on a number line? How can you represent adding a  $-2$ ? Would you just put a dot on  $-2$ ? Why or why not?

How might you represent two values with different signs on a number line?

Where is the sum located on the number line?

## 6.4.3 Guided Instruction: Rules for Adding Integers on a Number Line (continued)

2. The sum of integers can be represented using rays on a number line. A ray has a single endpoint and extends in a specified direction based on the sign of the integer.

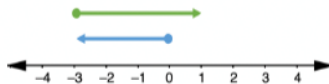


- A ray that represents a positive integer has its endpoint at 0, with the arrow extending to the *right*.
- A ray that represents a negative integer has its endpoint at 0, with the arrow extending to the *left*.
- To represent a sum, draw a ray for the first integer that starts at 0 and extends in the direction indicated by the *sign* of the first integer. It ends at the location of the first integer on the number line.
- Then, draw a ray for the second integer that starts where the first ray's arrow ends and extends in the direction indicated by the *sign* of the second integer. It ends after moving the number of units indicated by the second integer.
- The sum of the two integers is the integer where the second ray ends.



When providing Guided Instruction, make a note for students about how the blue and green paragraphs correspond with the blue and green arrows on the number line in question 3.

3. The diagram shown represents  $-3 + 4 = +1$



Give an example or explain when a number line model would be more useful than using integer chips.

*Answers will vary. The number line model might be more useful when the integers are larger numbers, because there may be too many chips involved.*



## 6.4.4 Collaboration: Commutative Property

**Component Overview:** The commutative property of addition is defined and reviewed. Working with a partner, students practice modeling the commutative property of addition using number lines. Students discuss why the models of the commutative property are equivalent.

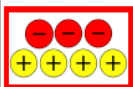
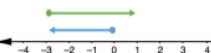
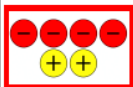
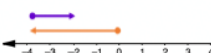


### 6.4.4 Collaboration: Commutative Property

The **commutative property of addition** states that the order in which addition is performed does not change the value of the expression.

With your partner, add integers using the commutative property.

- Complete the table by sketching the integer chip model for the first column, the number line model for the third column, and answering the question in the last column.

| Integer Chip Model                                                                              | Number Line Model                                                                               | Commutative Property Number Line Model                                                            | Yes/No                                                                       |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| $-3 + 4$<br>  | $-3 + 4$<br> | $4 + (-3)$<br> | Is the sum of $-3 + 4$ the same as the sum of $4 + (-3)$ ?<br><br><b>Yes</b> |
| $-4 + 2$<br> | $-4 + 2$<br> | $2 + (-4)$<br> | Is the sum of $-4 + 2$ the same as the sum of $2 + (-4)$ ?<br><br><b>Yes</b> |

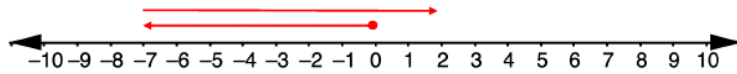


As students work on question 1, prompt them with the following:

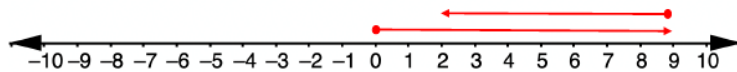
- How can you represent 4 on the number line?
- Where should you start to represent  $-3$ ?
- Which direction does a negative addend's arrow point?
- How do the number-line models show the sums of both addition expressions are the same?

## 6.4.4 Collaboration: Commutative Property (continued)

2. Show the model for  $(-7) + 9$  on the number line below.



3. Show the model for  $9 + (-7)$  on the number line below.



4. Discuss with your partner how the two number line models for  $(-7) + 9$  and  $9 + (-7)$  show that  $(-7) + 9 = 9 + (-7)$ . Write a summary of your reasoning.  
*Answers will vary. In the number line model of  $(-7) + 9$ , a ray begins at 0 and moves 7 units to the left, stopping at  $-7$ . The next ray begins at  $-7$ , and moving 9 units in the opposite direction, to the right, and stops at 2.*  
*In the number line model of  $9 + (-7)$ , a ray begins at 0 and moves 9 units to the right, stopping at 9. The next ray begins at 9, and moving 7 units, in the opposite direction, to the left, stops at 2.*  
*In both models, the final location on the number line is 2.*



As students work on questions 2-4, prompt them with the following:

- Are  $(-7) + 9$  and  $9 + (-7)$  equivalent? How do you know?
- How do the representations demonstrate this relationship? How does this show the commutative property of addition?



Consider providing sentence stems on question 4 for students who struggle to begin writing their responses. Prompt students to use the words “equal to” or “equivalent” in context.

## 6.4.5 Try It: Adding Integers Using Number Lines

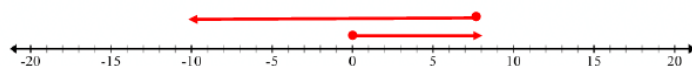
**Component Overview:** Students practice adding integers using number lines. Consider allowing students to work with a partner on this component.



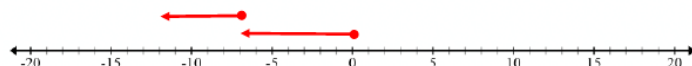
### 6.4.5 Try It: Adding Integers Using Number Lines

- Find each sum. Use the number line to model the addition.

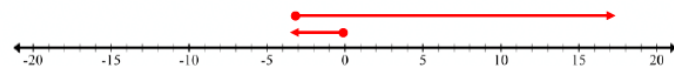
a.  $8 + (-18) = -10$



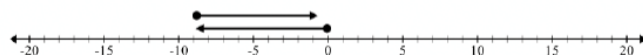
b.  $-7 + (-5) = -12$



c.  $-3 + 20 = 17$



- Write the numerical expression for the model below. Then, find the sum.



| Numerical Expression | Sum  |
|----------------------|------|
| $-9 + 8$             | $-1$ |



Consider providing tools and resources for students to successfully engage with this activity.  
Integer chips to verify their answers or represent their work in a different way  
Colored pencils to sketch integer chips or other creative models for integers

## 6.4.6 Your Turn!

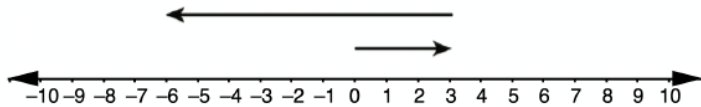
**Component Overview:** Students create addition expressions to match number line models and find sums. Students represent addition of temperatures on a vertical number line in context.



### 6.4.6 Your Turn!

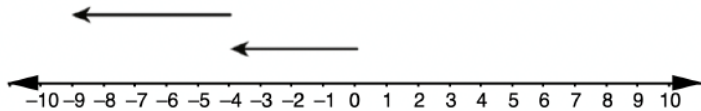
- For each number line model, write the addition expression represented and find the sum.

a.



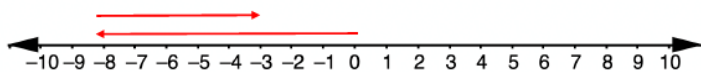
| Addition Expression | Sum  |
|---------------------|------|
| $3 + (-9)$          | $-6$ |

b.



| Addition Expression | Sum  |
|---------------------|------|
| $(-4) + (-5)$       | $-9$ |

- Use the number line to show the sum of  $-8 + 5$ . Then, complete the equation.



$$-8 + 5 = -3$$

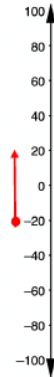


Consider providing tools and resources for students to successfully engage with this activity.  
Integer chips to verify their answers or represent their work in a different way.  
Colored pencils to sketch integer chips or other creative models for integers.

## 6.4.6 Your Turn! (continued)

3. In a single day, the temperature in degrees Fahrenheit ( $^{\circ}\text{F}$ ) of a town in Canada began at a low of  $-20^{\circ}\text{F}$  and increased  $40^{\circ}\text{F}$  by noon.

- Plot a point at the low temperature.
- Draw a ray along the number line to represent the increase of  $40^{\circ}\text{F}$ .
- Which direction does the arrow extend to show the increase in temperature?  
*Up*
- What was the temperature in the town by noon?  
 *$20^{\circ}\text{F}$*



*This is the only vertical number line depicted in this lesson. Consider providing real-world examples of vertical number lines, such as a thermometer, to activate students' prior knowledge and provide an entry point.*



*To add complexity, consider having students who finish early write additional real-world examples of integer addition. Then ask them to represent the addition on a number line and with an equation.*

## 6.4.7 Wrap-Up: Reflection

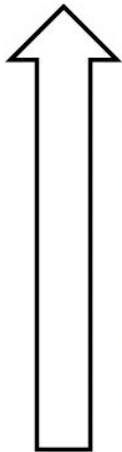
**Component Overview:** Students self-assess their understanding and confidence with using number lines to add integers.



### 6.4.7 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current level of confidence with using a number line to add integers.

*Answers will vary.*



I'm very confident! I can teach anyone how to add integers.

I can help others add integers.

I can add integers on my own.

I can add integers with assistance.

I'm having trouble adding integers.

I'm not sure what integers are.



*For students who color the arrow from "I can add integers with assistance" and below, consider prompting those students to provide an example of where they have trouble or what questions they still have.*

*For students who color the arrow from "I can add integers on my own" and above, consider prompting them to provide an example or write an explanation to a peer or younger family member.*



## 6.5 Adding Integers

### Lesson Introduction

| Lesson Introduction                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                         |                                                                                                                                                                                                  |                                                                                             |                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.                                                                                                                                                                                                                                                                     |                                                                                                         |  Learning Target                                                                                              | <ul style="list-style-type: none"><li>I can add integers with procedural fluency.</li></ul> |                                                                                                                                                                                                                                                      |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                         | Working Toward                                                                                                                                                                                   |                                                                                             |                                                                                                                                                                                                                                                      |
|                                                                                                       | MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.                                                                                                                                                                                                                                                                                                                                              |                                                                                                         | MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.<br>MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers. |                                                                                             |                                                                                                                                                                                                                                                      |
|  Essential Questions | <ul style="list-style-type: none"><li>What are additive inverses?</li><li>How can you use the signs of addends and their absolute values to determine the sign of their sum?</li><li>Describe how to add integers with the same signs and with different signs?</li></ul>                                                                                                                                        |                                                                                                         |                                                                                                                                                                                                  |                                                                                             |                                                                                                                                                                                                                                                      |
|  Lesson Narrative    | In this lesson, students begin by modeling additive inverses on number lines and with integer chips. Students discover why the sign of the sum stays the same when the signs of the addends are the same. They also discover why, when the signs of the addends are not the same, the answer is the amount remaining after canceling the zero pairs with the sign of the addend with the greater absolute value. |                                                                                                         |                                                                                                                                                                                                  |                                                                                             |                                                                                                                                                                                                                                                      |
|  Component Summary | 6.5.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                          | Write true addition equations using integers from $-3$ to $3$ ( <i>SE p. 255</i> )                      | 6.5.4 Collaboration (10 min.)                                                                                                                                                                    | Fill in missing addends and sums in integer addition equations ( <i>SE p. 257</i> )         |                                                                                                                                                                                                                                                      |
|                                                                                                       | 6.5.2 Exploration (10 min.)                                                                                                                                                                                                                                                                                                                                                                                      | Explore the concept of additive inverses ( <i>SE p. 255</i> )                                           | 6.5.5 Your Turn! (10 min.)                                                                                                                                                                       | Practice finding the sum of one-, two-, and three-digit integers ( <i>SE p. 258</i> )       |                                                                                                                                                                                                                                                      |
|                                                                                                       | 6.5.3 Guided Instruction (10 min.)                                                                                                                                                                                                                                                                                                                                                                               | Connect additive inverses to zero pairs for adding with the same or opposite signs ( <i>SE p. 256</i> ) | 6.5.6 Wrap-Up (~5 min.)                                                                                                                                                                          | Reflect on what they are still unsure about when adding integers ( <i>SE p. 258</i> )       |                                                                                                                                                                                                                                                      |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                         | Materials                                                                                                                                                                                        |                                                                                             | Additional Preparation                                                                                                                                                                                                                               |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Existence of additive inverses</li></ul> <u>Additional Terms:</u> <ul style="list-style-type: none"><li>Absolute value</li><li>Integer</li></ul>                                                                                                                                                                                                         |                                                                                                         | <ul style="list-style-type: none"><li>Integer chips</li></ul>                                                                                                                                    |                                                                                             | <ul style="list-style-type: none"><li>Consider creating an anchor chart to illustrate how to add integers with the same and with different signs. Include instructions, visual models (number line and integer chips), and a few examples.</li></ul> |

|  <p>Teacher Tips</p>              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Things to Consider                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                    | <ul style="list-style-type: none"> <li>Students may struggle to explain how many zero pairs are created in 6.5.2 Guided Instruction.</li> <li>Students may have difficulty finding the missing addends in 6.5.4 Collaboration.</li> <li>Students may add integers of opposite signs and give the value a negative sign. For example, <math>-3 + 2 \neq -5</math>.</li> </ul>                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Consider displaying an anchor chart that illustrates how to add integers with the same and different signs. Include instructions, visual models (number line and integer chips), and a few examples.</li> <li>Consider rewriting expressions in 6.5.3 Guided Instruction to show the additive inverses involved as students develop procedural fluency when adding without manipulatives or number lines.</li> <li>Consider allowing students to use the models that make sense to them to add integers if they are not yet at a place to do so without. Encourage the use of number-line models in this case because of the size of the addends in the lesson.</li> </ul> |
|                                                                                                                    | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  <p>Differentiate Instruction</p> | Take Turns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA.K12.MTR.4.1: Discussion and Reflection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                                                                                    | <p>In 6.5.4 Collaboration, consider having students take turns explaining how they added the integers with different signs. These discussions will improve student understanding and confidence.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>6.5.4 Collaboration provides an opportunity for students to engage in mathematical discussion through explaining their mathematical process and justifying their results to a peer. Use this as an opportunity for students to practice articulating their mathematical thinking and comparing their methods to others.</p>                                                                                                                                                                                                                                                                                                                                                                                    |
|  <p>Differentiate Instruction</p> | <p>Each activity in this lesson provides opportunity to differentiate instruction by content, by process, or by product.</p> <ul style="list-style-type: none"> <li>Provide integer chips for 6.5.2 Exploration and 6.5.4 Collaboration for kinesthetic and visual entry points.</li> <li>Offer verbal responses for students to submit via phones or apps for the 6.5.6 Wrap-Up.</li> <li>For 6.5.5 Your Turn!, consider substituting the given integer addition problems with less difficult problems if students are struggling. If students finish early, supplement with more complex problems, such as four-digit integers or sums with three addends.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                    | <p>Lesson Notes:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## 6.5.1 Warm-Up: True Equations

**Component Overview:** Students fill in the blanks to write two addition equations using integers from  $-3$  to  $3$ .



### 6.5.1 Warm-Up: True Equations

- Using integers between  $-3$  and  $3$  one time each, fill in the blanks to make the equation true.

*Answers will vary.*

$$\boxed{-3} + \boxed{2} = \boxed{-1}$$



*For students who may finish early, consider having them create as many true equations as they can.*



**MA.K12.MTR.3.1: Fluency**

*Did you fill in the addends or the sum first?  
Which method made more sense for you? Why?*

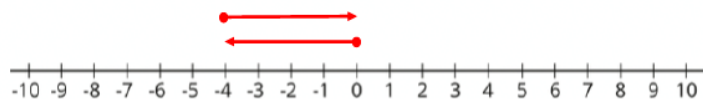
## 6.5.2 Exploration: Additive Inverses

**Component Overview:** Students explore multiple representations of additive inverses and finding their sum. Students see additive inverses represented on number lines and with integer chips. Students connect additive inverses to zero pairs. Finally, students create an example of an additive inverse and share their example with a partner.



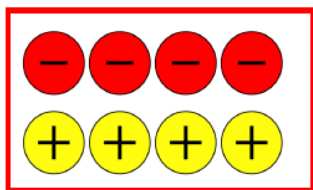
### 6.5.2 Exploration: Additive Inverses

1. Use the number line to show the sum of  $-4 + 4$ . Then, complete the equation.

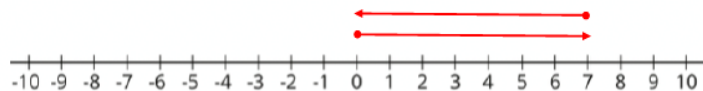


$$-4 + 4 = 0$$

2. Represent the sum of  $-4 + 4$  using integer chips.

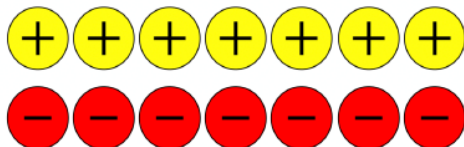


3. Use the number line to show the sum of  $7 + (-7)$ . Then, complete the equation.



$$7 + (-7) = 0$$

4. Represent the sum of  $7 + (-7)$  using integer chips.



Consider using question stems for students to answer on their own or for you to ask while circulating.

- What do you notice about the sum of each question?
- What do you notice about the values being added?
- Do additive inverses remind you of anything you learned about when modeling integers with integer chips?



#### Existence of Additive Inverses

For every  $a$  there exists  $(-a)$  so that  $a + (-a) = 0$  and  $(-a) + a = 0$ .

## 6.5.2 Exploration: Additive Inverses (continued)

Additive inverses are another way to represent the concept of a zero pair.

5. What number is the additive inverse of  $-15$ ?

$15$

6. What number is the additive inverse of  $27$ ?

$-27$

7. Choose any positive integer and write it in the blank:

*Answers will vary. Any positive integer is correct. The example provided here uses 12.*

Use your number to fill in the blanks in the statement.

For the value  $12$ , there exists  $(-12)$  so that

$$12 + (-12) = 0 \text{ and } (-12) + 12 = 0.$$

8. Write your partner's positive integer:

*Answers will vary. The example provided here uses 9.*

Use your partner's number to fill in the blanks in the statement.

For the value  $9$ , there exists  $(-9)$  so that

$$9 + (-9) = 0 \text{ and } (-9) + 9 = 0.$$



*How do you know that 15 is the additive inverse of  $-15$ ?*

## 6.5.3 Guided Instruction: Adding Integers

**Component Overview:** Students summarize concepts explored in previous lessons adding integers and relate these concepts to how to think about adding integers without using manipulatives or number-line models. They formalize this learning as they continue developing procedural fluency.



### 6.5.3 Guided Instruction: Adding Integers

Understanding the existence of additive inverses can be helpful when adding integers with larger absolute values.

1. Complete each statement based on your understanding of adding integers so far.

- a. When adding two positive integers, the sum is

- ☐ positive.
- ☐ negative.

- b. When adding two negative integers, the sum is

- ☐ positive.
- ☐ negative.

When adding a positive and a negative integer, the sum will be either positive or negative, depending on which value has the larger absolute value.

2. Consider the expression  $40 + (-165)$ .

- a. What will be the sign of the sum?

*The sign of the sum will be negative.*

- ☐ positive
- ☐ negative

- b. Explain your reasoning.

*The sum will be negative because  $-165$  has the greater absolute value.*



*How does the number of zero pairs formed relate to additive inverses?*



*Consider rewriting the expression as  $40 + (-40) + (-125)$  and think aloud to compare the two expressions and discuss how this shows how the additive inverse of 40 can be used to consider the zero pairs created and eliminated when adding  $40 + (-165)$ . The chips that would remain after eliminating the zero pairs are 125 negative chips, or  $-125$ .*

## 6.5.3 Guided Instruction: Adding Integers (continued)

When adding these integers, zero pairs are created.

- c. Use additive inverses to explain how many zero pairs are created.

*40 zero pairs are created. There are the 40 positive integers, from the addend 40, and 40 negative integers, from the  $-165$  addend, that can be combined to form zero pairs.*

- d. What remains after combining the additive inverses?

*125 negative integers remain.*

- e. Complete the equation.

$$40 + (-165) = -125$$

3. Complete the statements to describe how to find the sum of  $-1,029 + (-38)$ .

When adding  $-1,029 + (-38)$ , ☐ zero pairs ☒ no zero pairs are

created because the signs of the addends are

- ☒ the same.  
☐ opposites.

The sum can be found by adding the

absolute values of the addends. Since both addends are

- ☐ positive,  
☒ negative,

the ray for the second addend of the sum

will go farther to the

- ☐ right ( $\rightarrow$ )  
☒ left ( $\leftarrow$ )

of zero on a number

line.

The sum of  $-1,029 + (-38)$  is

- ☐ 991.  
☐ -991.  
☐ 1067.  
☒ -1067.



*If students struggle to understand why 40 zero pairs are created, consider modeling a problem with smaller values to model how the zero pairs are created using integer chips.*



*Utilize talk moves to position students for synthesis. "Karma, in your own words, how would you describe the way to use zero pairs for adding integers? Chantoy, restate Karma's approach but in your own words."*



## 6.5.4 Collaboration: Reasoning with Integers

**Component Overview:** Working with a partner, students determine the missing addends in integer addition equations. Students finish the activity by explaining how they determined the missing addends.



### 6.5.4 Collaboration: Reasoning with Integers

1. Work with your partner to fill in the blanks with an integer to make each equation true.

$$\boxed{14} + \boxed{-49} = \boxed{-35}$$

$$\boxed{304} + \boxed{-70} = \boxed{234}$$

$$\boxed{-136} + \boxed{161} = \boxed{25}$$

2. Choose one equation and explain your thinking in determining the missing value.

*Student choice will vary.*

$$\boxed{14} + \boxed{-49} = \boxed{-35}$$

*Since 14 is positive and the sum is negative, the other addend must be negative. The sign of the sum indicates which addend is the integer with the greater absolute value. Therefore, the missing addend will be negative.*

*To determine the missing value, I added the absolute values of both existing numbers in the equation, so that zero pairs can be created. Then, I kept the sign of the sum.  $|14| + |-35| = 49$ . The missing addend is therefore  $-49$ .*



#### Turn and Talk

*This routine provides students an opportunity to share their thoughts, reflect, and finalize their answers. To increase active listening, consider asking partners to explain each other's work.*

## 6.5.5 Your Turn!

**Component Overview:** Students continue practice finding the sum of one-, two-, and three-digit integers.



### 6.5.5 Your Turn!

1. Find each sum.

a.  $10 + (-204) = -194$       b.  $(-99) + (99) = 0$

c.  $904 + (-11) = 893$       d.  $-1 + 76 = 75$

e.  $205 + (-643) = -438$       f.  $-151 + (-70) = -221$

g.  $-399 + (-405) = -804$       h.  $-80 + 606 = 526$

i.  $703 + (-190) = 513$       j.  $-600 + (-78) = -678$



- How can you determine the sign of the sum when the integers have the same sign?
- How can you determine the sign of the sum when the integers have opposite signs?



Consider displaying an anchor chart with instructions, visual models, and examples for adding integers with the same and with opposite signs. Referencing this chart will help students develop procedural fluency with adding integers.

## 6.5.6 Wrap-Up: Growth Mindset

**Component Overview:** Students share what they are still unsure about and develop a plan for how they can fill in the gap(s). Consider using students' perceptions of their understanding, along with their accuracy on the questions in 6.5.5 Your Turn!, to determine which students need support before the next lesson.



### 6.5.6 Wrap-Up: Growth Mindset

1. Today, I am still unsure about \_\_\_\_\_

\_\_\_\_\_.

2. To fill in this gap, I intend to \_\_\_\_\_

\_\_\_\_\_.

*Answers will vary.*



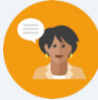
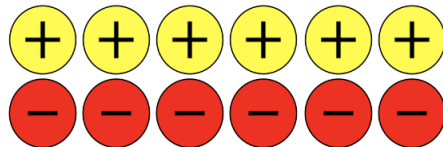
Consider providing answer frames for students to select all that apply.

- The meaning of additive inverse
- The meaning of absolute value
- How to use integer chips to represent zero pairs
- Fluency adding integers with the same sign
- Fluency adding integers with opposite signs

## 6.6 Subtracting Integers Using Integer Chips

### Lesson Introduction

| Lesson Introduction                                                                                   |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                              |                                                                                                                                                                                                  |                                                                                                 |                                                                                                                   |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.                                                                                                                                                                                      |                                                                                                                              |  Learning Target                                                                                              | <ul style="list-style-type: none"><li>I can use integer chips to subtract integers.</li></ul>   |                                                                                                                   |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                                                       |                                                                                                                              | Working Toward                                                                                                                                                                                   |                                                                                                 |                                                                                                                   |
|                                                                                                       | MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.                                                                                                                                                                                                                                                               |                                                                                                                              | MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.<br>MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers. |                                                                                                 |                                                                                                                   |
|  Essential Questions | <ul style="list-style-type: none"><li>How do you represent integer subtraction with integer chips?</li><li>How do you use zero pairs to determine the difference of integers?</li></ul>                                                                                                                                           |                                                                                                                              |                                                                                                                                                                                                  |                                                                                                 |                                                                                                                   |
|  Lesson Narrative    | In this lesson, students begin developing an understanding of subtracting integers using integer chips. Previously students represented and solved integer addition problems using integer chips. Students connect and extend their knowledge and skills to represent and solve integer subtraction problems using integer chips. |                                                                                                                              |                                                                                                                                                                                                  |                                                                                                 |                                                                                                                   |
|  Component Summary | 6.6.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                           | Math content and practices in everyday life ( <i>SE p. 259</i> )                                                             | 6.6.4 Guided Practice (10 min.)                                                                                                                                                                  | Practice representing and solving subtraction problems using integer chips ( <i>SE p. 262</i> ) |                                                                                                                   |
|                                                                                                       | 6.6.2 Exploration (10-15 min.)                                                                                                                                                                                                                                                                                                    | Predict the sign of the difference of two integers. Compare with a partner and check with a calculator. ( <i>SE p. 260</i> ) | 6.6.5 Your Turn! (5-10 min.)                                                                                                                                                                     | Additional practice subtracting integers using integer chips ( <i>SE p. 263</i> )               |                                                                                                                   |
|                                                                                                       | 6.6.3 Guided Instruction (10-15 min.)                                                                                                                                                                                                                                                                                             | Model integer subtraction with integer chips ( <i>SE p. 261</i> )                                                            | 6.6.6 Wrap-Up (~5 min.)                                                                                                                                                                          | Demonstrate understanding of lesson learning targets ( <i>SE p. 264</i> )                       |                                                                                                                   |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                          |                                                                                                                              | Materials                                                                                                                                                                                        |                                                                                                 | Additional Preparation                                                                                            |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Additive inverse property of zero</li></ul><br><u>Additional Terms:</u> N/A                                                                                                                                                                                               |                                                                                                                              | <ul style="list-style-type: none"><li>Integer chips</li><li>Anchor charts</li><li>(Optional) \$1 manipulatives</li><li>calculators</li></ul>                                                     |                                                                                                 | <ul style="list-style-type: none"><li>6.6.1 Warm-Up contains a video to accompany the question prompts.</li></ul> |

|                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Things to Consider                                                                                                                                                                                                                                                                   |
|                                                                                                                | <ul style="list-style-type: none"><li>Students may struggle determining how many zero pairs to add to the model to be able to subtract.</li></ul>                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"><li>Consider guiding students through the process of how to complete the first row in the table in 6.6.2 Exploration.</li><li>Consider creating an anchor chart during 6.6.3 Guided Instruction with images, instructions, and examples.</li></ul> |
|                                                                                                                | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                   |
|                                                                                                                | <b>Contemplate, then Calculate</b><br>Creating a plan in problem-solving before jumping in to calculating builds fluency (MA.K12.MTR.3.1) and mitigates errors. Notice the purposeful uses of “guessing” in questions 2 and 3 of 6.6.2 Exploration so students are negotiating and discussing the relationship between positive and negative integers.                                                                                                                                                           | <b>MA.K12.MTR.5.1: Patterns and Structure</b><br>6.6.4 Guided Practice provides an opportunity for students to make connections among several subtraction problems. Students develop an understanding of when, why, and how to create zero pairs and use them to subtract integers.  |
| <br>Differentiate Instruction | <p>Integer Chips</p> <p>Providing concrete manipulatives offers different entry points for students to engage in mathematical discussion. Integer chips can also be printed, cut, and laminated for use in class. Consider displaying an anchor chart complete with the notes from class with images, instructions, and examples for students to reference during 6.6.4 Guided Practice and 6.6.5 Your Turn!.</p> <div></div> |                                                                                                                                                                                                                                                                                      |
| Lesson Notes:                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                      |

## 6.6.1 Warm-Up: When Will I Use This?

**Component Overview:** Students reflect on a video about the application of mathematical content and practices in everyday life. This component may help students see the utility of math. Students also reflect on the practices or thinking skills that they are developing in math.



### 6.6.1 Warm-Up: When Will I Use This?

1. The speaker states, "It's not about what you learn, it's about what methods, tools, and tactics you have to develop in order to solve the problem that you may never see again for the rest of your life ..." What does he mean by "methods, tools, and tactics"?



*Answers will vary. When the speaker says "methods, tools, and tactics" he is referring to the actions that we may take to solve a particular problem. He is stating that we may not ever have to solve a particular problem again, but the ways in which we solve that problem (the methods, tools, tactics we use) will be used over and over again in our lives.*

2. The quote continues to say, "... but you will see other problems where these methods and tools will become immensely valuable to you." Where might these methods, tools, and tactics help you in everyday life?

*Answers will vary. These methods might help me in everyday life when I encounter a problem that requires me to think critically or persevere.*



#### Think-Pair-Share

Give students the opportunity to share their reflections with a peer after initially reflecting independently. This will expose students to more applications of mathematics content and practices.



Explain if you agree with the statement from the video that the math practices are more important than the math content. Provide justification for your response.

## 6.6.2 Exploration: Subtracting Integers

**Component Overview:** Students predict the sign of the difference of two integers and compare their predictions with a partner. Next, students check their predictions with a calculator.



### 6.6.2 Exploration: Subtracting Integers

1. In the first column, a larger circle represents a number that has a larger absolute value than that of the smaller circle. Without discussing it with your partner, enter a numeric example for each visual model. Then, in the "My Guess: Answer Will be + or -" column, enter whether you think the result will be positive (+) or negative (-).

After completing all of your own examples and guesses, record your partner's examples and guesses.

*Answers will vary. Students should have examples that match the model. For example, in line 1 the expression should have a larger positive integer subtracting a smaller positive integer.*

| Visual Model of Subtraction | My Example  | My Guess: Answer Will be + or - | Partner's Example | Partner's Guess: Answer Will be + or - |
|-----------------------------|-------------|---------------------------------|-------------------|----------------------------------------|
|                             | $8 - 4$     | +                               | $7 - 3$           | +                                      |
|                             | $2 - 5$     | -                               | $4 - 9$           | -                                      |
|                             | $5 - (-5)$  | +                               | $6 - (-6)$        | +                                      |
|                             | $-5 - 5$    | -                               | $-6 - 6$          | -                                      |
|                             | $-1 - (-7)$ | +                               | $-2 - (-4)$       | +                                      |
|                             | $-9 - (-1)$ | -                               | $-8 - (-2)$       | -                                      |



Precision in size difference is less important than the discussion about the relationship between positive and negative integers, as the chips are meant to be a model or representation rather than a rote formula. Encourage students to both agree and disagree.



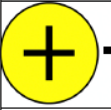
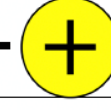
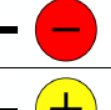
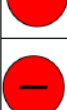
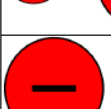
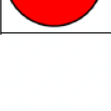
What does a larger integer chip represent in terms of absolute value?  
What happens when you subtract a larger integer from a smaller integer? A smaller integer from a larger integer?  
What happens when you subtract a negative integer?



## 6.6.2 Exploration: Subtracting Integers (continued)

2. Now, form a group of four by joining another pair of classmates. Discuss your guesses. Do all four of you agree on all of the problems? If not, list which problems you disagree on.  
*Answers will vary.*

3. Tell your teacher when you are ready to test out the guesses with a calculator. Use the calculator to find the answer for one problem from each visual model. When each of you are in agreement with the rule, check the box in the column that represents the answer.

| Visual Model of Subtraction                                                                                                                                               |  |  |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|  -      |                                                                                   | ✓                                                                                 |                                                                                   |                                                                                   |
|  -      |                                                                                   |                                                                                   |                                                                                   | ✓                                                                                 |
|  -    | ✓                                                                                 |                                                                                   |                                                                                   |                                                                                   |
|  -  |                                                                                   |                                                                                   | ✓                                                                                 |                                                                                   |
|  -  |                                                                                   | ✓                                                                                 |                                                                                   |                                                                                   |
|  -  |                                                                                   |                                                                                   |                                                                                   | ✓                                                                                 |



### Contemplate, then Calculate

Notice the intentional scaffolding that provides students an opportunity to critically analyze and formulate a rationale (contemplate) before finding the answer (calculate). Monitor during Exploration to ensure students are actively participating in this routine for using patterns and structure to help understand and connect mathematical concepts (MA.K12.MTR.5.1).



Students are provided the visual model in the left column and can mark the “guess/answer” columns with dots, checks, or their preference. It is not the expectation that all students answer these “correctly” at this point. They will continue formalizing their understanding throughout the lesson.



What do you notice about the sign of the difference when you subtracted an integer with a larger absolute value? What did you notice about the sign of the differences for each of the examples?

## 6.6.3 Guided Instruction: Using Integer Chips to Show Subtraction

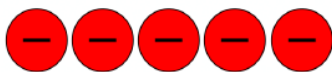
**Component Overview:** Students model integer subtraction with integer chips. Students explore how they can use zero pairs to determine the difference.



### 6.6.3 Guided Instruction: Using Integer Chips to Show Subtraction

Integer chips can also be used to model subtraction.

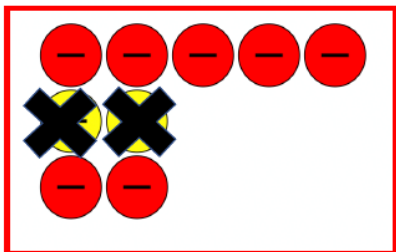
To find the difference of  $(-5) - 2$  using integer chips, begin by first arranging 5 negative integer chips on your desk, as shown below.



1. Draw a picture of how you think integer chips can be used to represent subtracting positive 2 from the 5 negative integer chips that are already on your desk.

Consider using zero pairs. It is okay if you do not know; just try your best.

*Answers will vary and it is not the expectation students have correct models at this point.*



#### Additive Identity Property of Zero

$$a + 0 = a \quad \text{or} \quad 0 + a = a$$

When adding 0 to a value, the value of the expression does not change. In a model, adding zero pairs represents adding 0. The value of the model does not change, but the integer chips that are available do.



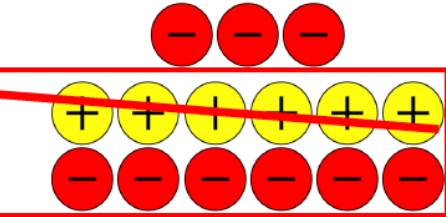
*Consider providing students with integer chips (concrete manipulatives) to provide kinesthetic and visual entry points.*



*How many zero pairs do you need to add to be able to take away 2 positive integers? Explain if adding zero pairs changes the value of the expression.*

## 6.6.3 Guided Instruction: Using Integer Chips to Show Subtraction (continued)

2. Complete the table to show how  $(-3) - 6$  can be represented using integer chips.

|                                                                                                                                 |                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Begin by representing the first integer, $-3$ , with integer chips.                                                             |                                          |
| Determine what should be subtracted from the model.                                                                             | 6 <div> <div> ○ positive<br/>○ negative </div> </div> integer chips need to be subtracted from the model                  |
| If the chips that need to be removed from the model are not available, add the number of zero pairs needed to be able to do so. |  <p><i>Need to add 6 Zero Pairs.</i></p> |
| Cross out the integer chips from the model drawn above to represent the subtraction. <i>Shown above.</i>                        |                                                                                                                           |
| Find the value of the expression based on the integer chips remaining.                                                          | $-3 - 6 = -9$                                                                                                             |

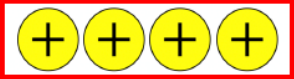
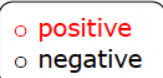
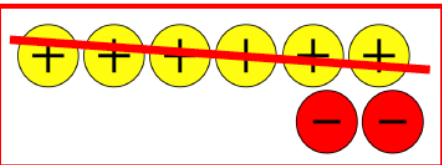
3. Look at your guess for representing  $(-5) - 2$ . Using the directions in the table for  $(-3) - 6$ , determine if your guess was correct. If your guess was correct, explain how you decided you should add two zero pairs to the model.  
If your guess was not correct, explain what confused you about the zero pairs.  
*Answers will vary.*



How many zero pairs are needed to subtract 6 positive integers? How can you represent this using integer chips?

## 6.6.3 Guided Instruction: Using Integer Chips to Show Subtraction (continued)

4. Complete the table to show how  $4 - 6$  can be represented using integer chips.

|                                                                                                                                 |                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Begin by representing the first integer with integer chips.                                                                     |                                                             |
| Determine what should be subtracted from the model.                                                                             | $-6$  integer chips<br>need to be subtracted from the model |
| If the chips that need to be removed from the model are not available, add the number of zero pairs needed to be able to do so. | <br><i>Need to add 2 Zero Pairs.</i>                        |
| Cross out the integer chips from the model drawn above to represent the subtraction. <i>Shown above.</i>                        |                                                                                                                                              |
| Find the value of the expression based on the integer chips remaining.                                                          | $4 - 6 = -2$                                                                                                                                 |



Consider providing sentence stems for students who struggle to begin writing their responses.



Can you take away 6 positive integers from 4 positive integers?

How many zero pairs need to be added to the model so that there are 6 positive integer chips to subtract from it?



*Notice and Wonder*

Ask students to share what they notice and wonder about the process of representing and subtracting integers using integer chips.

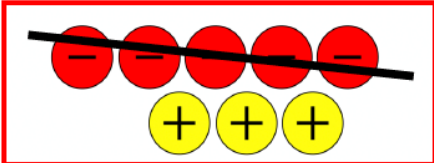
## 6.6.4 Guided Practice: Subtraction with Integer Chips

**Component Overview:** Students practice representing and solving integer subtraction problems using integer chips. Students then apply the skills they have developed to a real-world scenario involving money.



### 6.6.4 Guided Practice: Subtraction with Integer Chips

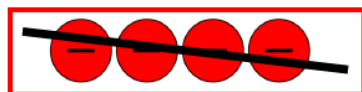
- Complete the table to show how  $(-2) - (-5)$  can be represented using integer chips.

|                                                                                                                                 |                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Begin by representing the first integer with integer chips.                                                                     |                                                                                                   |
| Determine what should be subtracted from the model.                                                                             | <div> <div>5</div> <div> <input type="radio"/> positive<br/> <input checked="" type="radio"/> negative </div> </div> <div>integer chips need to be subtracted from the model</div> |
| If the chips that need to be removed from the model are not available, add the number of zero pairs needed to be able to do so. |  <div>Need to add 3 Zero Pairs</div>                                                             |
| Cross out the integer chips from the model drawn above to represent the subtraction. <i>Shown above</i>                         |                                                                                                                                                                                    |
| Find the value of the expression based on the integer chips remaining.                                                          | $-2 - (-5) = 3$                                                                                                                                                                    |



Do you have enough negative integer chips to subtract  $-5$ ? How many zero pairs need to be added to the model to have  $-5$  available to subtract? Does adding zero pairs to the model change the value of the model? Does it still represent a value of  $-2$ ? What do you think the sign of the difference will be? Why?

- Use integer chips to represent how  $-4 - -4 = 0$ .

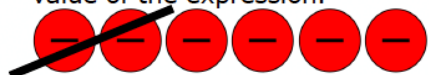


4 negative integer chips will need to be subtracted. No zero pairs need to be added to the model because 4 negative integer chips are already available to be subtracted. After subtracting, there are no integer chips remaining, so  $-4 - (-4) = 0$ .

## 6.6.4 Guided Practice: Subtraction with Integer Chips (continued)

3. Jennifer borrowed money from Amber twice. The first time, it was \$1 for a bottle of water. The second time, it was \$5 for ice cream. Jennifer owes Amber a total of \$6. Later, Amber told Jennifer not to worry about paying her back \$2 of the \$6. The amount that Jennifer still owes Amber can be represented by the expression  $-6 - (-2)$ .

- a. Draw integer chips to represent  $-6 - (-2)$  and the value of the expression.



*2 negative integer chips will need to be subtracted. No zero pairs need to be added to the model because 2 negative integer chips are already available to be subtracted. After subtracting, 4 negative integer chips remain. Therefore, the answer is  $-4$ .*

- b. Determine the amount that Jennifer still owes Amber.

*Jennifer still owes Amber \$4.*



*Read the problem aloud and model with \$1 manipulatives to provide auditory, visual, and kinesthetic entry points.*



*How can you represent the original amount Jennifer borrows using integer chips? Will you need to use positive or negative integers? How do you know? Explain if you need to add zero pairs to complete the subtraction.*



## 6.6.5 Your Turn!

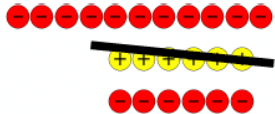
**Component Overview:** Students continue to practice subtracting signed integers using integer chips. Give students the choice of either using physical integer chip manipulatives or drawing pictorial representations of integer chips.



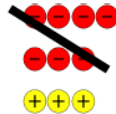
### 6.6.5 Your Turn!

1. Use integer chips to find the value of each expression.

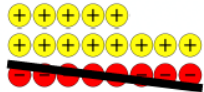
a.  $-11 - 6 = -17$



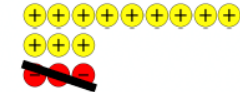
b.  $-4 - (-7) = 3$



c.  $5 - (-8) = 13$



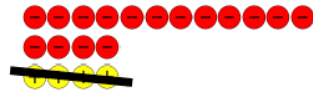
d.  $9 - (-3) = 12$



e.  $-1 - 2 = -3$



f.  $-12 - 4 = -16$



Consider displaying an anchor chart that displays work from 6.6.2 Exploration with images, instructions, and examples for students to reference.



## 6.6.6 Wrap-Up: Cool Down

**Component Overview:** Students demonstrate their understanding of adding integers with the same and opposite signs. Use data from this formative assessment to determine which students need support before the next lesson.



### 6.6.6 Wrap-Up: Cool Down

1. Choose two of the visual models of subtraction that you are sure will result in a positive answer.

- a. Draw an asterisk (\*) next to both in the table.

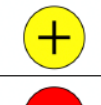
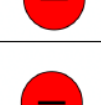
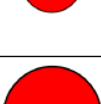
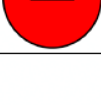
*Answers will vary. All the correct combinations have been starred.*

- b. Give an example of them below.

*Answers will vary. Sample examples are below in order from top to bottom.*

$$5 - 3 = 2$$

$$2 - (-5) = 7$$

| Visual Model of Subtraction                                                         |                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|    |    |
|    |    |
|    |    |
|    |    |
|   |   |
|  |  |

2. Choose one visual model that is a challenge for you.

- a. Draw an exclamation point (!) next to it in the table.

*Answers will vary.*

- b. Explain using an example why it is a challenge.

*Answers will vary.*



Possible student responses:

- The ones with different colors because I'm not sure what to do when they are different.
- The ones that are all red because the negative and minus signs confuse me.

Consider using 6.7.3 Collaboration, in the next lesson, to connect the student responses.

## 6.7 Subtracting Integers Using Number Lines

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                             |                                                                                           |                                                                                                                                                                                                  |                                                                                                                                                                                                                                |                                                                                                                        |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.                                                                                                                                                                                                                |                                                                                           |  Learning Targets                                                                                             | <ul style="list-style-type: none"><li>I can subtract integers using a number line.</li><li>I can use number lines to understand how the rules for adding integers are related to the rules for subtracting integers.</li></ul> |                                                                                                                        |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                                                                                 |                                                                                           | Working Toward                                                                                                                                                                                   |                                                                                                                                                                                                                                |                                                                                                                        |
|                                                                                                       | MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.                                                                                                                                                                                                                                                                                         |                                                                                           | MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.<br>MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers. |                                                                                                                                                                                                                                |                                                                                                                        |
|  Essential Questions | <ul style="list-style-type: none"><li>How do you subtract integers using a number line?</li><li>How are the integer chips and number-line models of integer subtraction related?</li><li>How are the patterns of adding and subtracting integers connected?</li></ul>                                                                                       |                                                                                           |                                                                                                                                                                                                  |                                                                                                                                                                                                                                |                                                                                                                        |
|  Lesson Narrative    | In this lesson, students extend their knowledge and skills of subtracting integers using integer chips to now also using number lines. Students also discover subtraction can be rewritten as adding the additive inverse. Students make connections between the patterns of integer subtraction and addition as they solve problems throughout the lesson. |                                                                                           |                                                                                                                                                                                                  |                                                                                                                                                                                                                                |                                                                                                                        |
|  Component Summary | 6.7.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                     | Draw integer chips to represent and evaluate expressions ( <i>SE p. 265</i> )             | 6.7.3 Collaboration (20-25 min.)                                                                                                                                                                 | Relate integer addition and subtraction ( <i>SE p. 266</i> )                                                                                                                                                                   |                                                                                                                        |
|                                                                                                       | 6.7.2 Guided Instruction (10 min.)                                                                                                                                                                                                                                                                                                                          | Subtract integers representing temperatures using a number line ( <i>SE p. 265</i> )      | 6.7.4 Your Turn! (5 min.)                                                                                                                                                                        | Practice subtracting integers using a number line ( <i>SE p. 268</i> )                                                                                                                                                         |                                                                                                                        |
|                                                                                                       | 6.7.5 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                     | Self-assess understanding of using number lines to subtract integers ( <i>SE p. 269</i> ) |                                                                                                                                                                                                  |                                                                                                                                                                                                                                |                                                                                                                        |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                                    |                                                                                           | Materials                                                                                                                                                                                        |                                                                                                                                                                                                                                | Additional Preparation                                                                                                 |
|                                                                                                       | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> <ul style="list-style-type: none"><li>Additive inverse</li><li>Zero pair</li></ul>                                                                                                                                                                                                                    |                                                                                           | <ul style="list-style-type: none"><li>Integer chips</li><li>Chart paper and markers</li></ul>                                                                                                    |                                                                                                                                                                                                                                | <ul style="list-style-type: none"><li>Create an anchor chart with instructions, examples, and visual models.</li></ul> |

|                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Things to Consider                                                                                                                                                                                                                                                                                    |
|                                                                                                                | <ul style="list-style-type: none"><li>Students may believe any negative value means a decreasing sum or that any negative value moves left on a horizontal number line. For example, students may believe <math>-9 - (-30) = -39</math>.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"><li>Consider creating an anchor chart with instructions, examples, and visual models. Explain the chart in 6.7.2 Guided Instruction.</li><li>Consider advising students to highlight the integers being added/subtracted in 6.7.3 Collaboration.</li></ul>          |
|                                                                                                                | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                    |
|                                                                                                                | <b>Think-Pair-Share</b><br>In 6.7.3 Collaboration, have students initially complete the problems. Then have students discuss their work with a partner. The discussions will help students gain confidence and understanding.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>MA.K12.MTR.6.1: Reasonableness</b><br>6.7.3 Collaboration provides students multiple opportunities to assess the reasonableness of their solutions. Students discuss the problems with a partner. These discussions will help students clear up any misconceptions and deepen their understanding. |
| <br>Differentiate Instruction | Consider your students when planning for 6.7.3 Collaboration. Use data or informal observations from the previous lesson to inform strategic pairings for the activity to maximize engagement and flexibility. <ul style="list-style-type: none"><li>Partner two students with different communication styles. For example, pair one student who performed well with modeling integer chips in the previous lesson with another student who effectively explains what they are seeing and excels with verbal processing.</li><li>Alternatively, students could each choose a different context before comparing their work with a partner as an expert in that context. This jigsaw approach may fit classrooms with students who work best by trying first before comparing with a partner.</li></ul> |                                                                                                                                                                                                                                                                                                       |
| Lesson Notes:                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                       |

## 6.7.1 Warm-Up: Bell Work

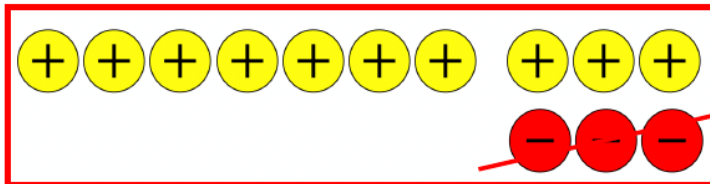
**Component Overview:** Students draw integer chips to represent and solve a subtraction expression. Give students the choice of either using physical integer chip manipulatives or drawing pictorial representations of integer chips.



### 6.7.1 Warm-Up: Bell Work

1. Draw integer chips that represent the expression below and evaluate.

$$7 - (-3)$$



$$7 - (-3) = 10.$$



Can you take away  $-3$  from the integer chips you have represented?  
How many zero pairs do you need to add to the model?

## 6.7.2 Guided Instruction: Subtracting Integers Using a Number Line

**Component Overview:** Students represent a temperature scenario using a vertical number line and integer chips. Students fill in guided notes that explain the process of subtracting integers using a number line.



### 6.7.2 Guided Instruction: Subtracting Integers Using a Number Line

1. The temperature in degrees Fahrenheit ( $^{\circ}\text{F}$ ) of a city in Alaska was  $-4^{\circ}\text{F}$ . After sunset, the temperature dropped  $6^{\circ}\text{F}$ .

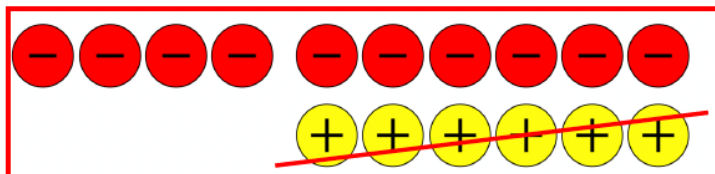
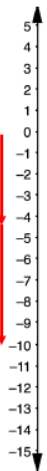
- a. Represent the change in temperature on the number line.
- b. Write the subtraction number sentence that is represented from the context and on the number line.

$$-4 - 6$$

- c. After the decrease in temperature, what was the temperature?

$$-4 - 6 = -10$$

- d. Represent the temperature change using integer chips to verify the solution.



$$-4 - 6 = -10$$



#### MA.K12.MTR 2.1: Multiple Representations

Prompt students to compare the number line model to the integer chips.

- How can you show  $-4$  degrees using a number line? How is a negative temperature represented?
- If the temperature is decreasing by 6 degrees, is it going up or down the number line?
- How can you take 6 positive integers away from  $-4$ ? Will you need to add zero pairs to the model? Why or why not?
- What does that look like with integer chips?

## 6.7.2 Guided Instruction: Subtracting Integers Using a Number Line (continued)

2. Complete the statements to explain how to represent subtraction on a horizontal number line.

- a. To find the difference of  $-9 - (-30)$ , start by plotting

- ☐ 9
- ☒ -9
- ☐ 30
- ☐ -30

on the number line first.

- b. The subtraction sign indicates moving to the

- ☐ right
- ☒ left

on the number line, but it is followed by a negative sign with 30, which requires moving in the opposite direction.

- c. Therefore, the answer will be found by moving

- ☐ 9
- ☒ 30

units to the ☐ right ☒ left of -9 on the number line.



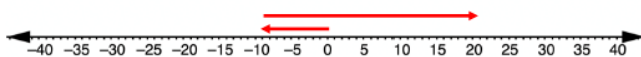
### Turn and Talk

Have students consider why the arrow for  $-30$  will move to the right and discuss with a partner. Have some students share their thoughts with the whole class. Consider also asking students to compare the integer chip and number line representations.



What happens when you subtract a negative number? How does this relate to the integer-chip model of representing subtraction?

3. Show the subtraction on the number line below.



The difference of  $-9 - (-30)$  is 21.

## 6.7.3 Collaboration: Relating Integer Addition and Subtraction

**Component Overview:** With a partner, students choose a context to represent with a number-line model, a subtraction expression, and a related addition equation. Students practice rewriting subtraction equations with the additive inverse of the second addend and represent this on number lines. Partners form groups of four and check each other's work. Finally, students practice representing additive inverses with integer chips.



### 6.7.3 Collaboration: Relating Integer Addition and Subtraction

Three contexts are given below. Choose one context to work on with your partner. Circle your choice.

|                  |                                                                                                                                                                                                                                                                     |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Context A</b> | Tamara didn't realize she had a balance of \$19 in her bank account when she used her bank card to purchase a \$50 purse. When the purchase is processed by the bank, what will be her new account balance?                                                         |
| <b>Context B</b> | The elevation of the top floor of a parking garage in Death Valley, California, is 282 feet (ft) below sea level. A car that leaves the parking garage from the top floor loses 72 ft in elevation before reaching the street. What is the elevation of the street? |
| <b>Context C</b> | Ginny mows lawns for her summer job. Before she started mowing, she spent \$32 on gas for the lawn mower. Then, she spent another \$17 on other supplies she bought at the hardware store. What were Ginny's total expenses?                                        |

For the context selected, represent the mathematics in the context in each of the following ways.

*Answers will vary. Example is based on Context A.*

1. Draw a number line to represent the given context.



2. Write two equations that can be used to find the solution to the question in the context. One equation should use subtraction and the other should use addition.

$$19 - 50 = -31$$

$$19 + (-50) = -31$$



#### Think-Pair-Share

Have students initially complete the problems. Then have students discuss their work with a partner. The discussions will help students gain confidence and understanding.



- How do you represent elevations below sea level on a number line?
- If there's a starting balance of \$19 and Tamara spends more than that, what will her account balance be? How do you represent this on a number line?
- How can you use either subtraction or addition to represent the same scenario?



## 6.7.3 Collaboration: Relating Integer Addition and Subtraction (continued)

- Create a group of four by joining with another pair of students. Verify your work by exchanging with the other pair of students.
- Continue working with the group of four to complete the table.

| Example Subtraction Problem | Example Rewritten as an Addition Problem | Number Lines: Show the subtraction on the top number line and the addition on the bottom number line. |
|-----------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------|
| $10 - 7 = 3$                | $10 + (-7) = 3$                          |                                                                                                       |
| $7 - 10 = -3$               | $7 + (-10) = -3$                         |                                                                                                       |
| $8 - -3 = 11$               | $8 + 3 = 11$                             |                                                                                                       |
| $-5 - 3 = -8$               | $-5 + (-3) = -8$                         |                                                                                                       |
| $-7 - -15 = 8$              | $-7 + 15 = 8$                            |                                                                                                       |
| $-9 - -7 = -2$              | $-9 + 7 = -2$                            |                                                                                                       |



What do you notice about the models for each subtraction problem and the expression rewritten with addition?

### 6.7.3 Collaboration: Relating Integer Addition and Subtraction (continued)

When subtracting integers, problems can be rewritten as an addition problem using the additive inverse of the second integer.

5. Describe one advantage of changing a subtraction problem to an addition problem.

*It is easy to forget to go the opposite direction when subtracting. It's easier to add or subtract the absolute values of the addends.*

6. Complete the table by rewriting the second integer in the subtraction model as its additive inverse and indicating the sign of the value of either expression. The first row has been completed. Once your group is done, split back into pairs and check your table with a different pair of students.

| Visual Model of Subtraction | Visual Model of Addition | Sign of the Answer |
|-----------------------------|--------------------------|--------------------|
|                             |                          | +                  |
|                             |                          | -                  |
|                             |                          | +                  |
|                             |                          | -                  |
|                             |                          | +                  |
|                             |                          | -                  |



Answers will vary for question 5. Consider monitoring for and eliciting specific or targeted responses.

For example, students may share, "When the addition sign is used, I can more easily compare the 'size' or absolute value of each integer to determine the sign."



Students are provided with the first visual model for addition in their workbook. Highlighted in light red, students complete the rest of the second column by rewriting the second integer from the subtraction model as its additive inverse.

## 6.7.4 Your Turn!

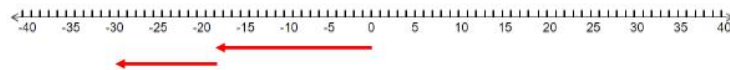
**Component Overview:** Students are provided with additional practice independently subtracting integers using number lines.



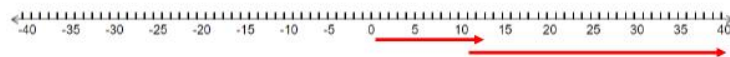
### 6.7.4 Your Turn!

- Represent each subtraction statement on a number line and find the value of the expression.

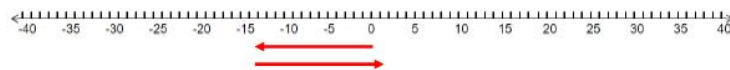
a.  $-19 - 11 = -30$



b.  $12 - (-28) = 40$



c.  $(-14) - (-15) = 1$



Have students rewrite the subtraction expressions as an addition problem. Ask them which expression they would prefer to solve and why.

## 6.7.5 Wrap-Up: Reflection

**Component Overview:** Students complete a self-assessment of their understanding of using number lines to subtract integers. Use student reflections to determine which students may need support before the next lesson.



### 6.7.5 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I can subtract integers using a number line.

|                                      |                                                 |                                             |
|--------------------------------------|-------------------------------------------------|---------------------------------------------|
| I need help.<br>I can't get started. | I'm getting there, but<br>I need more practice. | I understand this,<br>and I feel confident. |
|--------------------------------------|-------------------------------------------------|---------------------------------------------|

← *Answers will vary.* →

2. I can use number lines to understand how the rules for adding integers are related to the rules for subtracting integers.

|                                      |                                                 |                                             |
|--------------------------------------|-------------------------------------------------|---------------------------------------------|
| I need help.<br>I can't get started. | I'm getting there, but<br>I need more practice. | I understand this,<br>and I feel confident. |
|--------------------------------------|-------------------------------------------------|---------------------------------------------|

← *Answers will vary.* →



For additional information or to provide alternative means of reflection, prompt students to select one of the following:

- Provide an example based on your location on the number line.
- Provide a mathematical sentence that describes your journey through the lesson. For example,  $3 - (-2)$  could represent initially understanding, being unsure, but ending up positive.
- Write question numbers or activities next to where you marked to indicate specific moments in the lesson.

## 6.8 Integer Subtraction

### Lesson Introduction

| Lesson Introduction                                                                                   |                                                                                                                                                                                                                                                                                      |                                                                                             |                                                                                                                                                                                                  |                                                                                                       |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.                                                                                                                                         |                                                                                             |  Learning Target                                                                                              | <ul style="list-style-type: none"><li>I can subtract integers with values greater than 100.</li></ul> |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                          |                                                                                             | Working Toward                                                                                                                                                                                   |                                                                                                       |
|                                                                                                       | MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.                                                                                                                                                                                                                  |                                                                                             | MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.<br>MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers. |                                                                                                       |
|  Essential Questions | <ul style="list-style-type: none"><li>How do you rewrite a subtraction-of-integers problem as addition of the additive inverses?</li><li>How do you subtract integers with values greater than 100?</li></ul>                                                                        |                                                                                             |                                                                                                                                                                                                  |                                                                                                       |
|  Lesson Narrative    | In this lesson, students develop procedural fluency with subtracting integers greater than 100. Students explore how subtraction can be rewritten as adding the additive inverse. Students write and solve subtraction of integers problems based on real-world elevation scenarios. |                                                                                             |                                                                                                                                                                                                  |                                                                                                       |
|  Component Summary | 6.8.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                              | Reflect on mastery of skills related to subtracting integers ( <i>SE p. 270</i> )           | 6.8.3 Try It (10-15 min.)                                                                                                                                                                        | Practice subtracting integers in the context of elevation ( <i>SE p. 271</i> )                        |
|                                                                                                       | 6.8.2 Guided Practice (15 min.)                                                                                                                                                                                                                                                      | Explore the equivalence of subtracting and adding the additive inverse ( <i>SE p. 270</i> ) | 6.8.4 Your Turn! (10 min.)                                                                                                                                                                       | Additional practice subtracting integers ( <i>SE p. 272</i> )                                         |
|                                                                                                       | 6.8.5 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                              | Demonstrate understanding of lesson learning target ( <i>SE p. 273</i> )                    |                                                                                                                                                                                                  |                                                                                                       |
|  Resources         | Glossary                                                                                                                                                                                                                                                                             |                                                                                             | Materials                                                                                                                                                                                        |                                                                                                       |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Existence of additive inverse</li></ul><br><u>Additional Terms:</u> N/A                                                                                                                                                      |                                                                                             | <ul style="list-style-type: none"><li>Chart paper &amp; markers</li></ul>                                                                                                                        |                                                                                                       |
|                                                                                                       |                                                                                                                                                                                                                                                                                      |                                                                                             | Additional Preparation                                                                                                                                                                           |                                                                                                       |
|                                                                                                       |                                                                                                                                                                                                                                                                                      |                                                                                             | <ul style="list-style-type: none"><li>Prepare an anchor chart with visual models, instructions, and examples.</li></ul>                                                                          |                                                                                                       |

| <br>Teacher Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                         |  | Things to Consider                                                                                                                                                                                                                                                                                           |  |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                                                                                                                | <ul style="list-style-type: none"><li>Students may struggle to represent a negative change as a subtraction expression.</li><li>Students may struggle justifying why expressions are equivalent.</li><li>Students may struggle representing subtraction equations as addition equations using the additive inverse.</li></ul> |  | <ul style="list-style-type: none"><li>Consider providing sentence starters for the written responses for 6.8.2 Guided Practice.</li><li>Show students how to create open number lines, because of the size of the integers, to model the problems in 6.8.2 Guided Practice.</li></ul>                        |  |
|                                                                                                                | Literacy and Language Routines                                                                                                                                                                                                                                                                                                |  | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                           |  |
|                                                                                                                | <b>Think-Pair-Share</b><br>6.8.2 Guided Practice provides opportunities for students to initially try the problems independently and then discuss their work with a partner. The discussions will help students gain confidence and understanding.                                                                            |  | <b>MA.K12.MTR.4.1: Error Analysis</b><br>6.8.2 Guided Practice provides an opportunity for students to analyze samples of student work. Students discuss the errors and how to correct them. These discussions help clear up common misconceptions and deepen student understanding of subtracting integers. |  |
| <br>Differentiate Instruction | 6.8.3 Try It and 6.8.4 Your Turn! both provide practice for what serves as the culmination of three lessons on subtracting integers before students transition to multiplication. This practice is an opportunity for students to engage in a variety of tools and resources that provide differentiated support.             |  |                                                                                                                                                                                                                                                                                                              |  |
|                                                                                                                | Consider utilizing On-Ramp to 6 <sup>th</sup> Grade Learning Tool to provide scaffolding for students in need of additional support subtracting integers. <ul style="list-style-type: none"><li>Addition and Subtraction Fluency</li><li>Number Lines: Whole-Number Sums and Differences</li></ul>                            |  |                                                                                                                                                                                                                                                                                                              |  |
| Lesson Notes:                                                                                                  |                                                                                                                                                                                                                                                                                                                               |  |                                                                                                                                                                                                                                                                                                              |  |

## 6.8.1 Warm-Up: Self-Reflection

**Component Overview:** Students reflect on their mastery of skills related to subtracting integers. Consider having students look back at the Wrap-Up formative assessments from previous lessons to see examples of each statement. Seeing the examples will help students make more valid and reliable ratings.



### 6.8.1 Warm-Up: Self-Reflection

- For each statement, rate yourself on a scale of 1-5. A rating of 5 means “absolutely yes” and 1 means “absolutely not.”

| Statement                                                             | Rating |
|-----------------------------------------------------------------------|--------|
| I can subtract integers using integer chips.                          |        |
| I understand how to show subtraction of integers on the number line.  |        |
| I can subtract integers without using integer chips or a number line. |        |
| I can explain to someone else how to subtract integers.               |        |

*Answers will vary.*



Consider utilizing these ratings or reflections to inform your differentiation in 6.8.3 Try It or 6.8.4 Your Turn!



## 6.8.2 Guided Practice: Subtracting Integers

**Component Overview:** With a partner, students determine if a subtraction and addition expression are equivalent. Students write an addition and a subtraction expression to represent changes in scores. Students analyze two sample students' answers, identifying and correcting errors. Finally, students consolidate their understanding that a subtraction expression can be rewritten as an equivalent addition expression by changing the operation from subtraction to addition and changing the second value to its opposite sign.



### 6.8.2 Guided Practice: Subtracting Integers

1. Two expressions are shown.

| Expression A | Expression B   |
|--------------|----------------|
| $-191 + 77$  | $-191 - (-77)$ |

With your partner, determine if the expressions are equivalent. Summarize your determination below.

*Answers will vary. The expressions are equivalent.*

*If thinking with integer chips in mind:*

*Expression A: In this expression, there is  $-191$  and  $+77$ . Eliminate the 77 zero pairs formed when combining these values, and there will be  $-114$  remaining.*

*Expression B: In this expression, there is  $-191$ , and  $-77$  is being removed. There will be  $-114$  remaining.*

OR

*The expressions are equivalent.*

*If picturing a number line:*

*Expression A: If using a number line, I would start at 0 and move 191 units to the left to represent  $-191$ . From there, I would move 77 units to the right. The final location will be at 114 on the negative side of the number line.*

*Expression B: If using a number line, I would start at 0 and move 191 units to the left to represent  $-191$ . I would turn around from  $-191$  because of the double negative and move 77 units to the right. The final location will be at 114 on the negative side of the number line.*

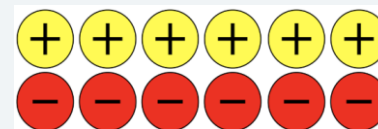


**MA.K12.MTR.3.1: Fluency**

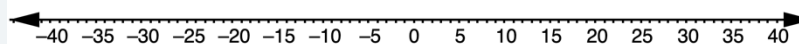
Monitor for students who share different approaches (as seen in the answer key) and elicit those responses whole group to highlight the flexibility of fluency in problem-solving.



The first sample response in the answer key would be for students thinking about integer chips to explain their answer.



The second sample response in the answer key would be for students thinking about a number line to explain their answer.



## 6.8.2 Guided Practice: Subtracting Integers (continued)

2. Annalise is on a trivia game show. Her score after the first round is  $-500$  points. She misses the first question in the second round, losing an additional 125 points.

Kiersten said she could represent Annalise's score using subtraction, while Dylan said it could only be represented with addition.

- a. Complete the table with the expression each student was thinking of.

| Kiersten's Subtraction Expression | Dylan's Addition Expression |
|-----------------------------------|-----------------------------|
| $-500 - 125$                      | $-500 + (-125)$             |

- b. Kiersten said Annalise's score was  $-375$  after missing the first question in the second round, but Dylan said it was  $-625$ . Explain which student is correct.

*Answers will vary. In this scenario, Annalise **lost** 500 points in the first round. Then she **lost** 125 points in the second round.*

*Dylan is correct. Because Annalise lost points in two consecutive rounds, she should have more negative points after the second round. Dylan added two negative integers to indicate the loss of points. The result is a negative number less than the values of the points lost in either of the two rounds.*

*Kiersten's expression,  $-500 - 125$  indicates that 125 positive integers are to be subtracted from 500 negative integers.*

*Her answer, however, indicates that she subtracted 125 negative integers from the 500 negative integers. Her expression is correct, but her answer is not.*



Consider having students who finish early explain why it is possible to represent one scenario as both a subtraction and addition expression and why they will be equivalent. Students should mention concepts of multiple representations (MA.K12.MTR.2.1) or provide examples.



If students struggle to start writing a response, provide them with sentence starters or a scaffolded fill-in-the-blank paragraph. Alternatively, prompt students to participate in the Stronger and Clearer language routine to structure responses.

## 6.8.2 Guided Practice: Subtracting Integers (continued)

3. Rewrite each subtraction equation as an equivalent addition equation.

|                             |                      |                     |
|-----------------------------|----------------------|---------------------|
| <b>Subtraction Equation</b> | $513 - 382 = 131$    | $957 - 136 = 1,093$ |
| <b>Addition Equation</b>    | $513 + (-382) = 131$ | $957 + 136 = 1,093$ |

Addition and subtraction are **inverse operations**, which means they are operations that have the opposite effect of each other.

Understanding this relationship and the existence of additive inverses can be helpful when subtracting integers.



### Existence of Additive Inverses

For every  $a$  there exists  $(-a)$  so that  $a + (-a) = 0$  and  $(-a) + a = 0$ .

4. Complete the statements below using this information.

A subtraction expression can be rewritten as an equivalent addition expression by changing the operation from subtraction to **addition** and changing the second value to its **additive inverse**.



Activate prior knowledge by referencing the integer chips table of representations.



- What happens when you add additive inverses?
- How do you know they will always equal zero?
- How can additive inverses or zero pairs help us when subtracting integers?

## 6.8.3 Try It: Subtracting Integers

**Component Overview:** Students write and solve expressions to represent the changes in elevation. Students also explain the meaning of negative elevations. Finally, students practice subtracting integers.



### 6.8.3 Try It: Subtracting Integers

- The table shows the beginning and ending elevations, in feet (ft), of six different hiking trails. The change in elevation along the trail can be found by subtracting the initial elevation from the final elevation.

$(\text{final elevation}) - (\text{initial elevation}) = (\text{change in elevation})$

- Write an expression that represents the change in elevation along each hiking trail. Then, write the value that represents the change in elevation. The first row has been completed as an example.

| Initial Elevation (ft) | Final Elevation (ft) | Difference Between Final and Beginning | Change in Elevation (ft) |
|------------------------|----------------------|----------------------------------------|--------------------------|
| 400                    | 900                  | $900 - 400$                            | 500                      |
| 115                    | 68                   | $68 - 115$                             | -47                      |
| 294                    | -121                 | $-121 - 294$                           | -415                     |
| -200                   | 610                  | $610 - (-200)$                         | 810                      |
| -163                   | -57                  | $-57 - (-163)$                         | 106                      |
| -17                    | -87                  | $-87 - (-17)$                          | -70                      |

- Explain what a negative change in elevation means in this context.

*Answers will vary. A negative change in elevation means that the hiker's final elevation is less than the initial elevation. It also means that a hiker moved down from, or below his or her initial elevation.*

- Find each difference.

- $-664 - 231$   
-895

- $109 - (-115)$   
224



To activate prior knowledge, consider reviewing how to represent positive and negative elevations. Prompt students to explain or represent concepts like sea level or mountains using vertical number lines.



How did you find the change in elevation?  
If a hiker is moving down the hiking trail, will there be a positive change in elevation or a negative change in elevation? Why?



Is it possible for a hiker to be at a negative elevation?  
Is that a reasonable (MA.K12.MTR.6.1) solution? Why or why not?

## 6.8.4 Your Turn!

**Component Overview:** Students have additional practice subtracting three-digit integers.



### 6.8.4 Your Turn!

1. Evaluate each expression.

a.  $740 - (-171)$

911

b.  $569 - 143$

426

c.  $-738 - (-212)$

-526

d.  $333 - (-204)$

537

e.  $-877 - 814$

-1691

f.  $-691 - 293$

-984

g.  $157 - (-120)$

277

h.  $341 - (-504)$

845



Consider methods for differentiated support during this activity based on your students' performances and your informal observations during Lessons 6-8.

**Station Rotations**

- **Station 1: Teacher-led small group**  
(for additional support)

- **Station 2: On-Ramp\* video and practice**  
(for additional support)

- **Station 3: Your Turn! independent practice**  
(for on-level benchmark practice)

- **Station 4: Desmos\* Activity**  
(for application and extension)

Students can either opt-in to each station as they need it, opting in to On-Ramp if they want more support with subtraction fluency from a Study Expert or trying some practice problems first before checking in to a teacher-led small group. Alternatively, students can be placed in three groups based on their responses from the 6.8.1 Warm-Up in this lesson or the Wrap-Up from the previous lesson.

\*On-Ramp and Desmos are available in multiple languages for added scaffolding and can be found in the Teacher Area.

## 6.8.5 Wrap-Up: Ticket Out the Door

**Component Overview:** Students evaluate three-digit integer subtraction expressions. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should not use a calculator to complete the Wrap-Up.



### 6.8.5 Wrap-Up: Ticket Out the Door

1. Evaluate each expression.

a.  $447 - (-702)$

1149

b.  $-834 - (-610)$

-224



Provide blank number-line models for students to utilize if necessary.

c.  $-309 - 525$

-834



Challenge students to choose one question and create a math story similar to question 1b from 6.8.3 Try It.



## 6.9 Multiplying Integers Using Models

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                  |                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.                                                                                                                                                                                                                                                                                                                                                |                                                                                                  |  Learning Target                                                                                                                | <ul style="list-style-type: none"><li>I can multiply integers using integer chips, number lines, and patterns.</li></ul>                                                                                                                                                                                        |
|  Benchmark Coherence | Building On<br>MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                  | Working Toward<br>MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.<br>MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers. |                                                                                                                                                                                                                                                                                                                 |
|  Essential Questions | <ul style="list-style-type: none"><li>How do you use integer chips and number lines to model multiplication of integers?</li><li>What pattern can you use to determine the sign of the product of integers?</li></ul>                                                                                                                                                                                                                                                                          |                                                                                                  |                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |
|  Lesson Narrative    | In this lesson, students use integer chips and number lines to model integer multiplication. Students work with integers that have absolute values less than or equal to 12. The lesson progresses from representing multiplication using integer chips to multiplying integers with opposite signs before closing with multiplying two negative integers. The focus is on students observing a pattern in the signs of the products while making sense of algebraic expressions using models. |                                                                                                  |                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |
|  Component Summary | 6.9.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Which One Doesn't Belong? – Number-Line Models ( <i>SE p. 274</i> )                              | 6.9.4 Guided Instruction (10 min.)                                                                                                                                                                                 | Discover how to use number lines to model multiplying integers ( <i>SE p. 276</i> )                                                                                                                                                                                                                             |
|                                                                                                       | 6.9.2 Exploration (10 min.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Use integer chips to model multiplication of integers ( <i>SE p. 274</i> )                       | 6.9.5 Guided Practice (10 min.)                                                                                                                                                                                    | Multiply integers using integer chips and number lines ( <i>SE p. 276</i> )                                                                                                                                                                                                                                     |
|                                                                                                       | 6.9.3 Exploration (5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Explore how zero pairs are used to multiply integers that are both negative ( <i>SE p. 275</i> ) | 6.9.6 Your Turn! (10 min.)                                                                                                                                                                                         | Practice multiplying integers using your preferred method ( <i>SE p. 277</i> )                                                                                                                                                                                                                                  |
|                                                                                                       | 6.9.7 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Determine sign of products of integers ( <i>SE p. 277</i> )                                      |                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                  | Materials                                                                                                                                                                                                          | Additional Preparation                                                                                                                                                                                                                                                                                          |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Commutative property of multiplication</li></ul> <u>Additional Terms:</u> N/A                                                                                                                                                                                                                                                                                                                                                          |                                                                                                  | <ul style="list-style-type: none"><li>Number lines</li><li>Integer chips</li><li>(Optional) Colored pencils</li></ul>                                                                                              | <ul style="list-style-type: none"><li>If you do not have integer chips, you can print and cut them out from a free online template, have students draw pictorial representations, or use virtual manipulatives.</li><li>For 6.9.6 Your Turn! consider having prelabeled blank number lines available.</li></ul> |



|                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher<br>Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                      | Things to Consider                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                                                                   | <ul style="list-style-type: none"><li>Students may struggle multiplying integers if they are not fluent multiplying whole numbers.</li><li>Students may not recognize a pattern in the sign of the products when multiplying integers.</li><li>Students may struggle to understand the concept of multiplying two negative integers.</li><li>Students may not recall the commutative property of multiplication.</li></ul> | <ul style="list-style-type: none"><li>Monitor for students who are correctly modeling the problems with the integer chips and have them share their work with the class.</li><li>If you do not have integer chips, students can draw pictorial representations or use virtual integer chips.</li><li>Model the precise language of multiplication and how to model it on the number lines accurately in the 6.9.4 Guided Instruction.</li><li>If time is an issue, consider assigning 6.9.6 Your Turn! for homework.</li></ul> |
|                                                                                                                   | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                             | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                                                                   | <b>Which One Doesn't Belong</b><br>The 6.9.1 Warm-Up offers a low-floor, high-ceiling opportunity to engage with number lines. This instructional routine allows for a low barrier of entry, as any informal observations (mathematical or not) contribute to the discussion, while students can extend their thinking with a higher ceiling of possible mathematical answers for all four choices.                        | <b>MA.K12.MTR.3.1: Mathematical Fluency</b><br>In 6.9.5 Guided Practice, students represent integer multiplication using both integer chips and number lines. Students progress from representing integer multiplication concretely to abstractly as they recognize and internalize the patterns with the signs of the products.                                                                                                                                                                                               |
| <br>Differentiate<br>Instruction | If students have delays or deficiencies in their fine motor skills, they can work with a peer tutor or use virtual manipulatives instead of concrete integer chips in the 6.9.6 Your Turn! activities. If students finish 6.9.3 Exploration activities early, ask them to create example multiplication problems with their partners, swap with other pairs, and solve each other's problems.                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Lesson Notes:                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

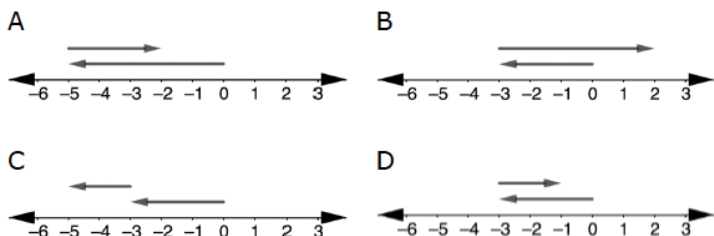
## 6.9.1 Warm-Up: Which One Doesn't Belong?

**Component Overview:** Students revisit number-line models for adding integers to prepare for multiplying integers on number lines in the lesson. After examining each model, students determine which one they think doesn't belong with the others and justify with mathematics.



### 6.9.1 Warm-Up: Which One Doesn't Belong?

1. Four number lines modeling addition are shown.



Explain which one doesn't belong.

*Answers will vary. C doesn't belong because both arrows are pointing to the left. Every other number line has one arrow starting at 0 and going left, showing a negative number, and one arrow going right, showing adding a positive number. C shows adding two negative numbers.*



#### Which One Doesn't Belong

This instructional routine provides a low-floor, high-ceiling approach to engaging with number lines. Encourage students to choose more than one answer. After they have selected their answer choice and written an explanation, prompt them to find another answer with an explanation of why that one does not belong.



If students struggle, consider providing students with sentence stems such as "I think the \_\_\_\_\_ doesn't belong because..."

## 6.9.2 Exploration: Using Integer Chips to Represent Multiplication

**Component Overview:** Students relate prior knowledge of how to represent multiplication as repeated addition to modeling integer multiplication with integer chips. Students work with integers with the same and with different signs. The commutative property of multiplication is also revisited.



### 6.9.2 Exploration: Using Integer Chips to Represent Multiplication

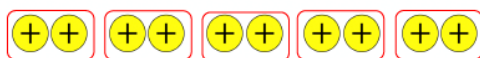
- The addition expression  $7 + 7 + 7$  can be written as  $3 \cdot 7$ . This is because repeated addition of integers can be

expressed as

- subtraction.
- **multiplication.**
- division.

- The expression  $5 \times 2$  can be interpreted as adding 5 groups of 2 or adding 2 groups of 5.
- Use integer chips to represent one of these interpretations.

Recall that  $\ominus = -1$  and  $\oplus = +1$ .  
Draw a picture of your representation.  
**Adding 5 groups of 2**



**OR Adding 2 groups of 5**



- Describe how the integer chips represent the value of the expression  $5 \times 2$ .  
**Answers will vary. In the representation,  $5 \times 2$  is the sum of the 5 groups of 2 positive integer chips. The sum is 10. Likewise,  $2 \times 5$  is the sum of the 2 groups of 5 positive integer chips. The sum is also 10.**
- The expression  $5 \times (-2)$  can be interpreted as adding 5 groups of  $-2$  or adding  $-2$  groups of 5.
- Explain which interpretation of  $5 \times (-2)$  you think would be easier to represent using integer chips.  
**Answers will vary. "Adding 5 groups of  $-2$ " is easier to represent using integer chips because this expression means adding 5 groups of 2 negative integer chips. The other interpretation, "adding  $-2$  groups of 5", may be interpreted as the opposite of 2 groups of 5 or  $-(2 \times 5)$ , which is more difficult to represent using integer chips.**



What do you notice about adding 5 groups of 2 and adding 2 groups of 5? Why might this be important?

## 6.9.2 Exploration: Using Integer Chips to Represent Multiplication (continued)

7. Discuss your choice with your partner. Then, work together to use integer chips to represent one of the interpretations. Draw a picture of the representation.

Answers will vary. 5 groups of  $-2$



OR

$-2$  groups of 5



8. Describe how the integer chips represent the value of the expression  $5 \times (-2)$ .

$5 \times (-2)$  means adding 5 groups of negative 2.



The representation shows 10 negative integer chips.

Therefore, the integer chips represent their sum.

OR The other interpretation, "adding  $-2$  groups of 5", may be interpreted as the opposite of 2 groups of 5 or  $-(2 \times 5)$ . First,  $(2 \times 5)$  can be represented as



The opposite of this would then be



9. What is the value of the expression  $5 \times (-2)$ ?

$-10$



### Commutative Property of Multiplication

$$a \times b = b \times a$$

The **commutative property of multiplication** states that the order in which multiplication is performed does not change the value of the expression.

10. Using the commutative property of multiplication, what is the value of the expression  $-2 \times 5$ ?

$-10$



Encourage mathematical discourse throughout the Exploration, as students have an opportunity to both verbally process and independently or collaboratively write explanations or descriptions in questions 4, 6, 7, and 8. Monitor for student responses that demonstrate understanding or possible misconceptions while students are exploring. Elicit these responses when facilitating a short debrief after the activity.



What do you notice about the value of both expressions?

## 6.9.3 Exploration: Multiplying a Negative Integer by a Negative Integer

**Component Overview:** Students explore how zero pairs are used to multiply integers that are both negative. Students learn to think of a negative integer multiplied by a negative integer as removing negative integers from the zero pairs that represent the expression. They generalize about whether the sign will always be positive.



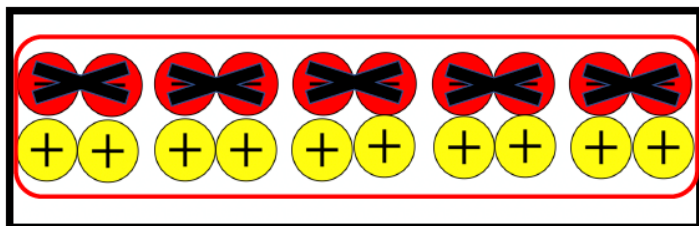
### 6.9.3 Exploration: Multiplying a Negative Integer by a Negative Integer

1. Complete the statement.

The expression,  $(-5)(-2)$ , indicates that 5 groups of  $-2$  should be removed.

2. Use the steps below to discover how to represent  $(-5)(-2)$  using integer chips.

**Step 1:** First, use your integer chips to create the set of zero pairs that will be needed to find  $(-5)(-2)$ . Draw the groups of zero pairs in the box below.



**Step 2:** Remove 5 groups of 2 negative integer chips.

**Step 3:** Remove any remaining zero pairs, if needed.

3. Complete the statement.

The value of expression  $(-5)(-2)$  is 10.

4. With your partner, discuss whether or not you think the product when multiplying two negative values will always be positive. Explain your thinking.

Answers will vary. I think the product will always be positive because when we create the zero pairs, we are removing groups of negative integers, which only leaves us with positive integers left.



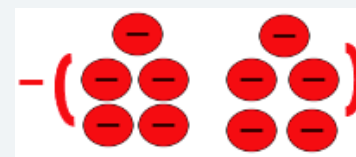
Based on the previous activities, what do you predict the sign will be when multiplying a negative integer by another negative integer?



Challenge students to find more than one way to represent multiplying two negative integers with integer chips accompanied by a different verbal expression, building on the use of the word “opposite” from previous activities.

Option 1: As seen and described in the answer key.

Option 2: “The opposite of two groups of negative five.”



Consider polling the class on question 4. Then ask a few students to justify their responses. After the discussion, repoll the class to see if any students changed their minds.



## 6.9.4 Guided Instruction: Multiplying Integers Using the Number Line

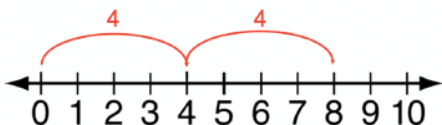
**Component Overview:** Students are guided through how to model multiplying integers on a number line. Consider having students compare the use of number lines for multiplying integers to using integer chips.



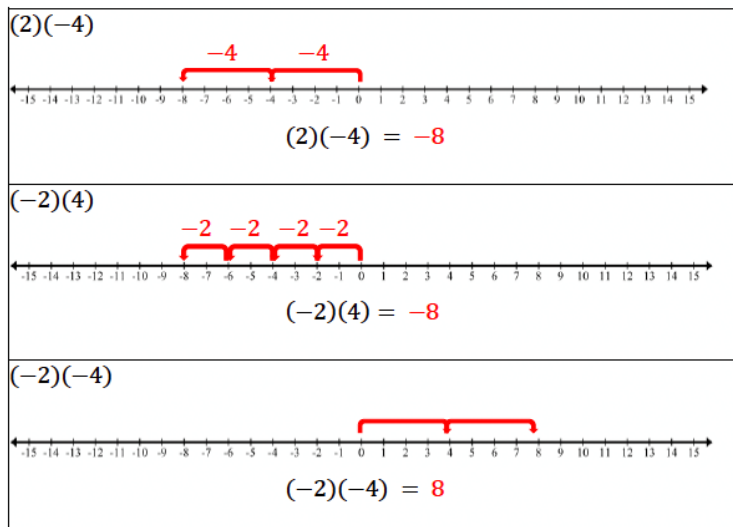
### 6.9.4 Guided Instruction: Multiplying Integers Using the Number Line

Recall that  $(2)(4)$  means 2 groups with 4 units in each group. The product is the total number of objects.

This can be represented on the number line as shown.



1. Represent each multiplication expression on the number line. Then, use it to complete the equation.



Consider displaying the examples, instructions, and visual models from this activity on an anchor chart.



Consider thinking aloud through each of the examples as you demonstrate how to model each expression on the number lines. This provides the opportunity to model precise language and ensure that each expression is modeled correctly on the number line. Consider using the following language for each of the expressions:

$(2)(-4)$  “Two groups of negative 4.”

$(-2)(4)$  “The opposite of two groups of 4.”

$(-2)(-4)$  “The opposite of two groups of negative 4.”

This should be guided instruction, as it will be released to the students in the next component.

## 6.9.5 Guided Practice: Multiplying Integers Using Models

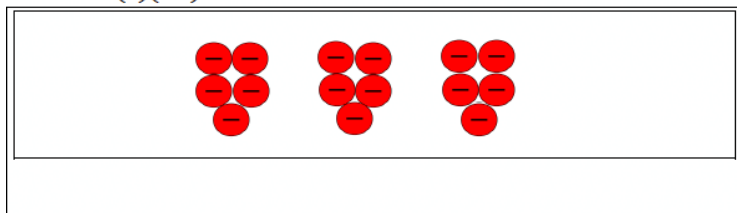
**Component Overview:** Students model integer multiplication with integer chips and number lines. Students reflect on which representation they prefer to use and why.



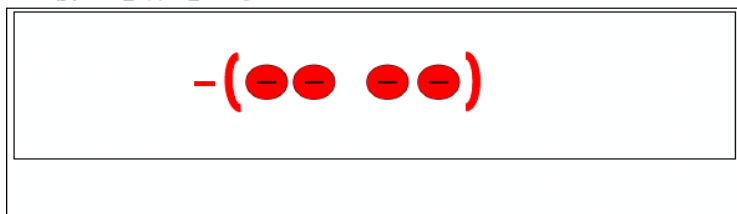
### 6.9.5 Guided Practice: Multiplying Integers Using Models

- Use integer chips to model and evaluate each expression.

a.  $(3)(-5) = -15$



b.  $-2 \times -2 = 4$

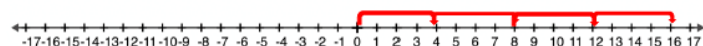


- Use a number line model to evaluate each expression.

a.  $(8)(-1) = -8$



b.  $(-4)(-4) = 16$



- Which model for multiplication of integers do you prefer: multiplication chips or the number line?

*Answers will vary. Number line is faster.*



Notice both representations of using integer chips for multiplying integers, with the second representation displaying how students used integer chips in 6.9.2. Prompt students to think creatively and include a different way to represent using integer chips.  
Representation 1: “The opposite of two groups of negative two.”  
Representation 2: “Two groups of negative two should be removed.”



## 6.9.6 Your Turn!

**Component Overview:** Students practice multiplying integers. Instructions include students “using your preferred method” to purposefully highlight and build fluency.



### 6.9.6 Your Turn!

1. Find the value of each expression using your preferred method.

a.  $2 \times -8$   
-16

b.  $-6 \cdot (-3)$   
18

c.  $(-7)(-6)$   
42

d.  $5(-3)$   
-15

e.  $-9 \times -4$   
36

f.  $7 \cdot 5$   
35



Students develop agency over their learning by having the opportunity to choose which representation they prefer to use.



- How did you solve the problem?
- Why did you choose this method?
- Can you solve it using another representation?

## 6.9.7 Wrap-Up: Cool Down

**Component Overview:** Students determine the signs of products of integers. Consider using data from this formative assessment to determine which students need remediation before the next lesson.



### 6.9.7 Wrap-Up: Cool Down

1. Complete each rule by drawing the sign of the product as an integer chip.

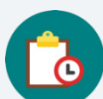
| Visual Model of Multiplication | Sign of Product |
|--------------------------------|-----------------|
| ×                              |                 |
| ×                              |                 |
| ×                              |                 |
| ×                              |                 |



Prompt students to provide means of proving their answer. Students can choose to reference an example from the lesson and writing the number next to the assigned row, or students could produce their own examples using integer chips or write a numeric expression.

## 6.10 Dividing Integers Using Integer Chips

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                    |                                                                                                   |                                                                                                                                                                                                  |                                                                                                                               |
|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.                                                                                                                                                                                                    |                                                                                                   |  Learning Target                                                                                              | <ul style="list-style-type: none"><li>I can use integer chips and patterns to discover rules for dividing integers.</li></ul> |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                                                                        |                                                                                                   | Working Toward                                                                                                                                                                                   |                                                                                                                               |
|                                                                                                       | MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.                                                                                                                                                                                                                                                                                |                                                                                                   | MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.<br>MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers. |                                                                                                                               |
|  Essential Questions | <ul style="list-style-type: none"><li>How can you use integer chips to divide integers?</li><li>How are multiplication and division of integers related?</li><li>What patterns are there with the sign of the quotient of integers?</li></ul>                                                                                                      |                                                                                                   |                                                                                                                                                                                                  |                                                                                                                               |
|  Lesson Narrative    | In this lesson, students learn how to divide integers. Students relate division to repeated subtraction and model division with integer chips. Students then relate the generalizations they can make about dividing integers to their previous learning multiplying integers. Students discover patterns with the signs of quotients of integers. |                                                                                                   |                                                                                                                                                                                                  |                                                                                                                               |
|  Component Summary  | 6.10.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                           | Notice and wonder about an image of a double negative ( <i>SE p. 278</i> )                        | 6.10.5 Collaboration (10 min.)                                                                                                                                                                   | Determine which division expressions do not belong with a partner ( <i>SE p. 281</i> )                                        |
|                                                                                                       | 6.10.2 Guided Instruction (10 min.)                                                                                                                                                                                                                                                                                                                | Use integer chips to represent division ( <i>SE p. 279</i> )                                      | 6.10.6 Guided Practice (5 min.)                                                                                                                                                                  | Practice dividing integers ( <i>SE p. 282</i> )                                                                               |
|                                                                                                       | 6.10.3 Collaboration (5 min.)                                                                                                                                                                                                                                                                                                                      | Collaboratively consider how integer multiplication and division are related ( <i>SE p. 280</i> ) | 6.10.7 Your Turn! (5 min.)                                                                                                                                                                       | Collaboratively write several division expressions that all result in a given quotient ( <i>SE p. 282</i> )                   |
|                                                                                                       | 6.10.4 Bring It Together (5 min.)                                                                                                                                                                                                                                                                                                                  | Consolidate how multiplication and division are related ( <i>SE p. 281</i> )                      | 6.10.8 Wrap-Up (~5 min.)                                                                                                                                                                         | Rate level of confidence with understanding and using rules for multiplying and dividing integers ( <i>SE p. 283</i> )        |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                           | Materials                                                                                         | Additional Preparation                                                                                                                                                                           |                                                                                                                               |
|                                                                                                       | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> <ul style="list-style-type: none"><li>Quotient</li><li>Divisor</li><li>Dividend</li></ul>                                                                                                                                                                                                    | <ul style="list-style-type: none"><li>Integer chips</li></ul>                                     | <ul style="list-style-type: none"><li>Create an anchor chart with instructions, examples, and visual models.</li></ul>                                                                           |                                                                                                                               |

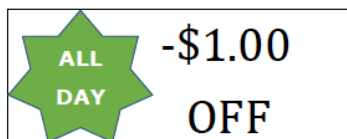
|                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher Tips | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>Students may have trouble understanding why the divisor must be a positive integer.</li><li>Students may struggle to represent division expressions using integer chips.</li><li>Students may struggle to relating division equations to multiplication equations.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Things to Consider</b> <ul style="list-style-type: none"><li>Consider creating an anchor chart to illustrate how multiplication and division of signed integers are related for students to reference.</li></ul>                                                                                                                                                                                                                         |
|                                                                                                   | <b>Literacy and Language Routines</b><br><b>Notice and Wonder</b><br>6.10.3 Collaboration provides an opportunity for students to discuss connections they are noticing and wondering about between the rules for dividing integers and the rules for multiplying integers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Mathematical Thinking and Reasoning (MTR) Standard</b><br><b>MA.K12.MTR.5.1: Patterns and Structure</b><br>6.10.3 Collaboration and 6.10.4 Bring It Together provide opportunities for students to engage in mathematical discussions comparing and connecting the rules for multiplication and division of integers. Students' arguments are based on the division and multiplication problems they have previously modeled and solved. |
|                                                                                                   | <br>Differentiate Instruction<br><p>Students are exploring the relationship between multiplication and division throughout this lesson. Calculation is not the focus of this lesson, but some students may need additional support when dividing integers if they are not confident in their math facts.</p> <p>Consider utilizing On-Ramp to 6<sup>th</sup> Grade learning tool to provide scaffolding for students in need of additional support, particularly students who learn well with Study Expert videos and practice (available in English and Spanish).</p> <ul style="list-style-type: none"><li>A Fraction Is a Quantity Formed by Its Parts</li><li>Evaluate Expressions</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Lesson Notes:                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## 6.10.1 Warm-Up: Notice and Wonder

**Component Overview:** Students write what they notice and wonder about a sale sign displaying a double negative. If students do not recognize the issue with the image, have them model the problem on a number line or with integer chips.



### 6.10.1 Warm-Up: Notice and Wonder



Does the image represent an increase or a decrease in the price? How do you know?

- Write down one thing you notice about the picture.  
*Answers will vary. I notice there is a negative sign next to the \$1.00.*
- Write down one thing you wonder about the picture.  
*Answers will vary. I wonder what "\$1.00 off" means.*
- Isaac described the image as a math fail. Why do you think he described it that way?  
*Answers will vary. I think he described it as a math fail because -\$1.00 off doesn't make sense, it would actually be a double negative which would mean that the price is actually \$1.00 more!*

## 6.10.2 Guided Instruction: Using Integer Chips to Represent Division

**Component Overview:** Students will receive guided instruction on how to use integer chips to represent division. Students will also learn how to write equivalent division expressions in order to use integer chips. Lastly, students will use what they have learned to create rules for the sign of a quotient.



### 6.10.2 Guided Instruction: Using Integer Chips to Represent Division

Recall that division is repeated subtraction.

For example,  $24 \div 8$  can be expressed as  $24 - 8 - 8 - 8$  because 3 groups of 8 can be subtracted from 24.

This shows that 24 is made up of 3 groups of 8.

The expression  $24 \div 8$  can also be interpreted as dividing 24 into 8 equal-sized groups. In the representation shown, 24 positive integer chips are divided into 8 equal-sized groups.

|   |   |   |
|---|---|---|
| + | + | + |
| + | + | + |
| + | + | + |
| + | + | + |
| + | + | + |
| + | + | + |
| + | + | + |
| + | + | + |

The quotient is the result of the division, or the value of the integer chips in each group.

$$24 \div 8 = 3$$

- Complete the table to describe the division, then draw the visual model using integer chips, and determine the quotient.

| Division Expression | Written Description                 | Visual Model of Division | Quotient |
|---------------------|-------------------------------------|--------------------------|----------|
| $\frac{15}{5}$      | Divide 15 into 5 equal-sized groups |                          | 3        |
| $-6 \div 2$         | Divide -6 into 2 equal-sized groups |                          | -3       |



### Turn and Talk

Utilize an instructional routine to leverage student-centered instruction for this activity. Students have used integer chips for multiplication, so a Turn and Talk provides them an opportunity to make meaningful connections from multiplication to division.

- How is this the same as multiplication?
- How is it different?

## 6.10.2 Guided Instruction: Using Integer Chips to Represent Division (continued)

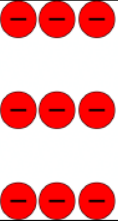
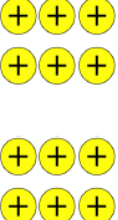
When modeling division with integer chips, the divisor must be positive, as this indicates the number of groups needed. When the divisor is negative, the expression can be rewritten using opposites.

For example,  $\frac{8}{-2}$  can be rewritten as  $\frac{-8}{2}$  because  $\frac{8}{-2} = \frac{-8}{2}$ .



*Consider providing a kinesthetic entry point by having students who are struggling model the problems with concrete integer chips.*

2.

| Division Expression | Equivalent Division Expression | Written Description                     | Visual Model of Division                                                          | Quotient |
|---------------------|--------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------|----------|
| $\frac{9}{-3}$      | $\frac{-9}{3}$                 | Divide $-9$ into $3$ equal-sized groups |  | $-3$     |
| $-12 \div -2$       | $\left(\frac{12}{2}\right)$    | Divide $12$ into $2$ equal-sized groups |  | $6$      |



## 6.10.2 Guided Instruction: Using Integer Chips to Represent Division (continued)

3. Use the pattern of the division in the tables to complete the rules by drawing the sign of the quotient as an integer chip.

| Visual Model of Division              | Sign of Quotient |
|---------------------------------------|------------------|
| $\begin{array}{r} + \\ + \end{array}$ | $+$              |
| $\begin{array}{r} + \\ - \end{array}$ | $-$              |
| $\begin{array}{r} - \\ - \end{array}$ | $-$              |
| $\begin{array}{r} - \\ + \end{array}$ | $+$              |



- How do the signs of the dividend and divisor relate to the sign of the quotient?
- For what other operation is this true?

## 6.10.3 Collaboration: How Are Integer Multiplication and Division Related?

**Component Overview:** Have the students read and discuss the problems with a partner. Students create integer division equations and relate them to integer multiplication equations.



### 6.10.3 Collaboration: How Are Integer Multiplication and Division Related?

Jeannie and Solimar were discussing the relationship between  $5 \times -2 = -10$  and  $(-10) \div (-2) = 5$ .

Solimar said, "The multiplication equation can be interpreted as 5 groups of  $-2$  and the division equation can be interpreted as how many groups of  $-2$  are in  $-10$ . Isn't that basically the same thing?"

Jeannie replied, "I agree. I also think it is interesting to see how the rules seem to be related. In the multiplication equation, we are multiplying a positive by a negative and the answer is a negative. In the division equation, we do the reverse of the multiplication problem and find that a negative divided by a negative is a positive."

Solimar said, "Yeah, that's a cool connection. Do you think the rules are the same for multiplication and division?"

Jeannie said, "I think they might be related to each other."

1. Discuss Solimar and Jeannie's conversation with your partner.

*Answers will vary.*

2. Create two integer division equations that can be rewritten with the related multiplication equation.

*Answers will vary. Examples shown in table below.*

| Division Equation    | Related Multiplication Equation |
|----------------------|---------------------------------|
| $\frac{-14}{7} = -2$ | $7 \cdot (-2) = -14$            |
| $\frac{45}{3} = 15$  | $3 \times 15 = 45$              |



#### Notice and Wonder

Have students discuss what they notice and wonder about the relationship between the equations.



Circulate while students are collaborating and listen for moments to ask questions to extend student thinking.

- What do you notice about the two equations? What do you wonder?
- Why does Jeannie think the rules of multiplication and division are related? Give an example to illustrate this point.
- How can you show that multiplication and division are connected?
- What do you notice about the rules of both multiplication and division?

## 6.10.4 Bring It Together: How Are Integer Multiplication and Division Related?

**Component Overview:** Students consolidate their understanding of how multiplication and division are related.



### 6.10.4 Bring It Together: How Are Integer Multiplication and Division Related?

- Write the missing factor in each given equation. Then, write the related division or multiplication equation.

| Given Equation                    | Can be Thought of as... | Related Equation    |
|-----------------------------------|-------------------------|---------------------|
| $3 \times 7 = 21$                 | $\longleftrightarrow$   | $21 \div 3 = 7$     |
| $\frac{-48}{4} = -12$             | $\longleftrightarrow$   | $(-12)(4) = -48$    |
| $-8 \times -7 = 56$               | $\longleftrightarrow$   | $56 \div (-8) = -7$ |
| $20 \div -4 = -5$                 | $\longleftrightarrow$   | $(-5)(-4) = 20$     |
| $-\left(\frac{36}{4}\right) = -9$ | $\longleftrightarrow$   | $-9 \times 4 = -36$ |



Consider utilizing inquiry to position students for synthesis.

- What do you notice about the factors in each related equation? What do you wonder?
- What do you notice when one factor is negative? What about when both factors are negative?
- How does this connect to what we already know about multiplying integers?

- How are rules for multiplication of integers related to the rules for division of integers?

*Answers will vary. The rules for multiplication of integers and the rules for division of integers are the same.*

## 6.10.5 Collaboration: Which One Doesn't Belong?

**Component Overview:** Students first work independently to determine and explain which division of integers expression does not belong. Then students discuss their work with a partner to verify their work or make corrections.



### 6.10.5 Collaboration: Which One Doesn't Belong?

1. Circle the expression that does not belong with the other three.

$$\frac{10}{-5} \quad \frac{-10}{5} \quad \frac{-10}{-5} \quad -\frac{10}{5}$$

2. Explain why the expression does not belong.

*Answers will vary. Expression 3 does not belong because it is dividing a negative number by a negative number. The quotient of integers with opposite signs is negative. The quotient of integers with the same sign is positive. The value of this expression is positive while the values of the other three expressions are negative.*

3. Verify your answers with your partner and, if necessary, resolve any differences.

Quotients can have negative signs in different places.

4. The values  $\frac{12}{-2}$ ,  $\frac{-12}{2}$ , and  $-\left(\frac{12}{2}\right)$  are equivalent. Explain why they are equivalent using your understanding of dividing integers.

*Answers will vary. Since the divisor should be positive to indicate the number of groups,  $\frac{12}{-2}$  can be rewritten as  $\frac{-12}{2}$ . Therefore, the two expressions are equivalent. The expression  $-\left(\frac{12}{2}\right)$  reads, "the opposite of positive 12 divided into 2 equal-sized groups." The quotient would be equal to 6, and the opposite of 6 is  $-6$ . All three expressions are equivalent, as they are all equal to  $-6$ .*

5. Discuss your explanation with your partner. If necessary, edit your explanation based on your discussion.



#### Which One Doesn't Belong?

Monitor for students with different responses and elicit their responses whole group to orient students to each other and the concept of equivalence.



- Why does the expression you selected not belong?
- How do you know what the sign of the quotient will be?
- Are any of the expressions equivalent?



Consider providing sentence starters for problems requiring written responses.

## 6.10.5 Collaboration: Which One Doesn't Belong? (continued)

6. Create two equivalent expressions for  $21 \div (-7)$ .

$$-\left(\frac{21}{7}\right) \text{ or } \left(\frac{-21}{7}\right)$$

7.  $\frac{-40}{-8}$  is not equivalent to  $-\left(\frac{40}{8}\right)$ . Explain why not.

*Answers will vary. The resulting sign of the quotient  $\frac{-40}{-8}$  is positive because when you divide a negative integer by a negative integer, it always results in a positive integer. The resulting sign of  $-\left(\frac{40}{8}\right)$  is negative because it indicates the quotient is the opposite of  $\left(\frac{40}{8}\right)$ , which is equal to  $-5$ .*



- What do you know about the quotient when the dividend is positive, and the divisor is negative? Are these expressions equivalent?
- How do you know?
- How does this compare to multiplying integers with the same sign?

## 6.10.6 Guided Practice: Dividing Integers

**Component Overview:** Students practice dividing integers. Consider providing integer chips, if needed.



### 6.10.6 Guided Practice: Dividing Integers

1. Find each quotient.

a.  $12 \div -4 = -3$

b.  $\frac{-63}{-9} = 7$

c.  $-\left(\frac{28}{4}\right) = -7$

d.  $60 \div 5 = 12$

e.  $-\left(\frac{-30}{-5}\right) = -6$



For students who finish early, consider having them create the related multiplication equations for each division equation. To add complexity, consider replacing some of the integers in each division equation with more complex integers.



How do you know the expressions are equivalent using quotient rules?

2. Create two expressions equivalent to  $\frac{-20}{10}$ .

$\frac{20}{-10}$  and  $-\left(\frac{20}{10}\right)$

3. What is the value of the expressions above?

$-2$

## 6.10.7 Your Turn!

**Component Overview:** Students work independently to create three different division expressions to represent a given quotient. Consider modeling an example before having students work through the table on their own. After students are finished working, have them work in pairs to verify their responses before sharing responses whole group.



### 6.10.7 Your Turn!

- In each column below, create three different division expressions that would result in the given quotient.  
*Answers will vary.*

| -4                              | 10                               | -7                                |
|---------------------------------|----------------------------------|-----------------------------------|
| 1. $\frac{-40}{10}$             | 1. $\left(\frac{100}{10}\right)$ | 1. $\left(\frac{70}{-10}\right)$  |
| 2. $-\left(\frac{12}{3}\right)$ | 2. $\left(\frac{-20}{-2}\right)$ | 2. $\left(\frac{-84}{12}\right)$  |
| 3. $\frac{16}{-4}$              | 3. $-\left(\frac{-50}{5}\right)$ | 3. $-\left(\frac{-35}{-5}\right)$ |



For students who may finish early, consider challenging them to create more division expressions that would result in the given quotients for each column.



Display an anchor chart for students who are struggling. When determining the equivalent expressions, prompt students to determine the sign of the quotients first and then divide the integers.



## 6.10.8 Wrap-Up: Reflection

**Component Overview:** Students complete a reflection indicating their confidence in using rules to multiply and divide integers.



### 6.10.8 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with understanding and using the rules for multiplying and dividing integers.

*Answers will vary.*



0 1 2 3 4 5 6 7 8 9 10



*If time allows, ask students to justify their emoji rating. They could write an explanation summarizing the rules. They can also write and solve a few sample problems to demonstrate their knowledge and skills.*

## 6.11 Multiplying and Dividing Integers

### Lesson Introduction

| Lesson Introduction                                                                                   |                                                                                                                                                                                                                                                                                                                                      |                                                                                                  |                                                                                                                                                                                                  |                                                                                                             |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.                                                                                                                                                                                      |                                                                                                  |  Learning Target                                                                                              | <ul style="list-style-type: none"><li>I can multiply and divide integers with procedural fluency.</li></ul> |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                                                          |                                                                                                  | Working Toward                                                                                                                                                                                   |                                                                                                             |
|                                                                                                       | MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.                                                                                                                                                                                                                                                                  |                                                                                                  | MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.<br>MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers. |                                                                                                             |
|  Essential Questions | <ul style="list-style-type: none"><li>How can you use the signs of the factors to determine the sign of the product?</li><li>How can you use the signs of the dividend and divisor to determine the sign of the quotient?</li><li>How do you multiply integers? How do you divide integers? What is similar and different?</li></ul> |                                                                                                  |                                                                                                                                                                                                  |                                                                                                             |
|  Lesson Narrative    | In this lesson, students develop procedural fluency with multiplying and dividing integers. Students multiply and divide integers and determine factor pairs for given products and the dividends and divisors of given quotients.                                                                                                   |                                                                                                  |                                                                                                                                                                                                  |                                                                                                             |
|  Component Summary | 6.11.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                             | Analyze a temperature graph and determine a headline to describe it ( <i>SE p. 284</i> )         | 6.11.4 Guided Practice (5 min.)                                                                                                                                                                  | Determine the missing factors, products, and quotients in equations ( <i>SE p. 286</i> )                    |
|                                                                                                       | 6.11.2 Refresher (10 min.)                                                                                                                                                                                                                                                                                                           | Write and solve multiplication and division problems based on visual models ( <i>SE p. 284</i> ) | 6.11.5 Your Turn! (5-10 min.)                                                                                                                                                                    | Determine the products and quotients of integers ( <i>SE p. 286</i> )                                       |
|                                                                                                       | 6.11.3 Collaboration (10-15 min.)                                                                                                                                                                                                                                                                                                    | Given products and quotients, determine factors, dividends, and divisors ( <i>SE p. 285</i> )    | 6.11.6 Wrap-Up (~5 min.)                                                                                                                                                                         | Reflection on learning from lesson ( <i>SE p. 287</i> )                                                     |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                             |                                                                                                  | Additional Preparation                                                                                                                                                                           |                                                                                                             |
|                                                                                                       | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> <ul style="list-style-type: none"><li>Product</li><li>Factor</li><li>Quotient</li><li>Dividend</li><li>Divisor</li></ul>                                                                                                                                                       |                                                                                                  | <ul style="list-style-type: none"><li>Consider preparing sentence stems for 6.11.3 Collaboration.</li><li>Consider preparing answer choices for 6.11.4 Guided Practice.</li></ul>                |                                                                                                             |
|                                                                                                       |                                                                                                                                                                                                                                                                                                                                      | Materials                                                                                        |                                                                                                                                                                                                  |                                                                                                             |
|                                                                                                       |                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"><li>Anchor charts</li></ul>                                    |                                                                                                                                                                                                  |                                                                                                             |

|                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher<br>Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                        | Things to Consider                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                   | <ul style="list-style-type: none"><li>Students may struggle to find the factors, given the product, and the dividends and divisors, given the quotient, if they are not confident in their math facts.</li></ul>                                                                                                                                                                                                                             | <ul style="list-style-type: none"><li>Consider providing sentence starters for the written responses (questions 2, 3, 4, 5, and 6) in 6.11.3 Collaboration.</li><li>It may be helpful to provide answer choices for question 2 in 6.11.4 Guided Practice.</li><li>In 6.11.3 Collaboration, consider initially providing one of the factors or a dividend or divisors to help students who struggle getting started.</li></ul> |
|                                                                                                                   | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                               | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                   | <b>Turn and Talk</b><br>In 6.11.3 Collaboration, students are given products and quotients and asked to determine the factors and dividends and divisors. They are also asked to explain their work. Due to the complexity of these problems, students will benefit from pausing to check their work and answers with a partner.                                                                                                             | <b>MA.K12.MTR.4.1: Discussion and Reflection</b><br>6.11.4 Guided Practice provides an opportunity to analyze samples of student work for errors. By discussing and correcting the errors, students will be less likely to make the same mistakes themselves.                                                                                                                                                                 |
| <br>Differentiate<br>Instruction | Utilize resources from previous lessons like manipulatives and Study Expert videos for additional scaffolding. <ul style="list-style-type: none"><li>Consider providing one of the factors, a dividend, or a divisor initially in 6.11.3 Collaboration.</li><li>The 6.11.6 Wrap-Up is a reflection with open-ended questions. Consider allowing students to use an audio recorder or speech-to-text tool to share their responses.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Lesson Notes:                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                               |

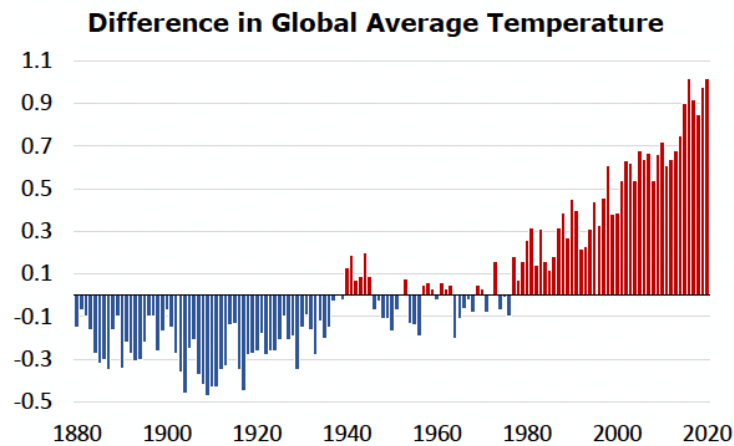
## 6.11.1 Warm-Up: What's Going On in This Graph?

**Component Overview:** Students analyze a graph of the difference in global average temperatures from 1880 to 2020 and the middle of the 19th century. Years with averages above the average are represented with red bars. Years with averages below the average are represented with blue bars.



### 6.11.1 Warm-Up: What's Going on in This Graph?

The graph shows the average global temperatures between 1880 and 2020 compared to the middle of the 20<sup>th</sup> century. All temperatures are in degrees Celsius ( $^{\circ}\text{C}$ ).



#### Notice and Wonder

Ask students to share what they notice and wonder about the graph. This will give students the opportunity to formalize their ideas before completing their written responses.



- What do you notice about the temperatures in the 1800s and 1900s?
- What trend do you notice in the temperatures over time?

1. What are you curious about after examining the graph?  
*Answers will vary.*
2. Imagine this graph is included in a newspaper article. Write a catchy headline that captures what's going on in the graph.  
*Answers will vary.*

## 6.11.2 Refresher: Multiplying and Dividing Integers

**Component Overview:** Given visual models of the signs of the factors, products, dividends, divisors, and quotients, students work with a partner to create an example equation that could be represented by the visual model. This component is meant to provide students an opportunity to review the rules they have developed for multiplying and dividing integers.



### 6.11.2 Refresher: Multiplying and Dividing Integers

1. Work with your partner to complete the tables. The last row of each table is missing both the rule and an example equation.

*Answers will vary.*

| Visual Model of Multiplication | Sign of Product | Example Equation       |
|--------------------------------|-----------------|------------------------|
| $\times$                       |                 | $3 \times (-14) = -42$ |
| $\times$                       |                 | $9 \times 5 = 45$      |
| $\times$                       |                 | $-14 \cdot 4 = -56$    |
| $\times$                       |                 | $(-7) \cdot (-8) = 56$ |

| Visual Model of Division | Sign of Quotient | Example Equation       |
|--------------------------|------------------|------------------------|
| $\div$                   |                  | $\frac{-144}{-3} = 48$ |
| $\div$                   |                  | $22 \div (-2) = -11$   |
| $\div$                   |                  | $\frac{24}{4} = 6$     |
| $\div$                   |                  | $-28 \div 4 = -7$      |



- How can you use the signs of the factors to predict the sign of the product? How can you use the signs of the dividend and divisor to predict the sign of the quotient?
- How can you use this to check the reasonableness of your answers when multiplying and dividing integers?

## 6.11.3 Collaboration: Multiplying and Dividing Integers

**Component Overview:** Students are given two products and two quotients. They are asked to work with a partner to determine five possible factor pairs and dividends and divisors that result in each product or quotient. Next, students explain the strategies they used.



### 6.11.3 Collaboration: Multiplying and Dividing Integers

- The table shows two different products. Work with your partner to find five different factor pairs that, when multiplied, will result in the product. The factors cannot include 1 or  $-1$ .

*Answers will vary.*

| Product | Factor | Factor |
|---------|--------|--------|
| (-90)   | -2     | 45     |
|         | 3      | -30    |
|         | -5     | 18     |
|         | -6     | 15     |
|         | 10     | -9     |
| 42      | -2     | -21    |
|         | 3      | 14     |
|         | -6     | -7     |
|         | 21     | 2      |
|         | 7      | 6      |

- Explain a strategy or an approach that each partner suggested.

*Answers will vary. Paying attention to the sign of the product and following the rules:*

*The sign of the product of integers with opposite signs is negative.*

*The sign of the product of integers with the same sign is positive.*

- Explain which of the products,  $(-90)$  or  $42$ , was the easiest to find factor pairs for and why.

*Answers will vary. I think 90 was easier because I knew more factors of 90 and then just had to make one factor negative and one factor positive in each pair.*



*It may be helpful to initially provide students with one of the factors and ask them to find the other missing factor.*



- How did you determine the factor pairs?
- What do you know about the signs of the factor pairs if the product is negative?
- What do you know about the signs of the factor pairs if the product is positive?



#### Turn and Talk

*Have students discuss their responses to questions 2 and 3. These discussions will help students gain confidence and understanding with multiplying integers.*

### 6.11.3 Collaboration: Multiplying and Dividing Integers (continued)

4. The table shows two different quotients. Work with your partner to find four different pairs of dividends and divisors that, when divided, result in the quotient. The numbers 1 or  $-1$  cannot be used.

*Answers will vary.*

$\text{dividend} \div \text{divisor} = \text{quotient}$

| Dividend | Divisor | Quotient |
|----------|---------|----------|
| $-16$    | $2$     | $(-8)$   |
| $32$     | $-4$    |          |
| $-48$    | $6$     |          |
| $40$     | $-5$    |          |
| $-26$    | $-2$    |          |
| $143$    | $11$    | $13$     |
| $-65$    | $-5$    |          |
| $-52$    | $-4$    |          |



*It may be helpful to initially provide either a missing dividend or a missing divisor for students who struggle getting started. Allowing students to use a multiplication table may also be helpful.*

5. Explain a strategy or an approach that each partner suggested.

*Answers will vary. Paying attention to the sign of the quotient and following the rules:*

*The sign of the quotient of integers with opposite signs is negative.*

*The sign of the quotient of integers with the same sign is positive.*

6. Explain which of the quotients,  $(-8)$  or  $13$ , was the easiest to find dividends and divisors for and why.  
*Answers will vary. I think  $-8$  was easier because it was easier for me to think of multiples of  $-8$  to get the dividend, and then I just had to make the dividend and divisor such that one was negative and one was positive.*



*How are the dividends and divisors for the quotient of  $(-8)$  the same or different than the dividends and divisors for the quotient of  $13$ ?*



## 6.11.4 Guided Practice: Finding Missing Factors, Products, or Quotients

**Component Overview:** Students find the missing values in integer multiplication and division equations. Students apply rules of integer multiplication and division to analyze a given equation and determine which sample of student work is correct.



### 6.11.4 Guided Practice: Finding Missing Factors, Products, or Quotients

- Fill in the missing values to make each equation true.

a.  $(-7) \cdot 2 = -14$

b.  $(-35) \div (-5) = 7$

c.  $\frac{-40}{8} = -5$

d.  $(-7) \times (-9) = 63$

- Consider the equation shown. Phillip said  $-12$  is the missing factor while Alicia claimed it is  $12$ .

$$(-1) \cdot (-2) \cdot \underline{\quad} = 24$$

Explain which student is correct using rules for multiplying integers.

*Alicia is correct.  $(-1) \cdot (-2) = 2$  because of the rule,*

$$\textcircled{+} \times \textcircled{+} = \textcircled{+}$$

*Since the product 24 is positive, and applying the same rule, the missing integer is 12.*



Consider providing sentence starters for students who may struggle explaining their work in writing. For example, “\_\_\_\_\_ is correct because of the rule \_\_\_\_\_. The correct product is \_\_\_\_\_, and the missing integer is \_\_\_\_\_.”



Consider discussing errors during a whole class debrief if you observe similar errors throughout the class. If students are aware of common mistakes, they will be less likely to make them in their work in the future. If most students are not making errors, then have them move on to the 6.11.5 Your Turn! independently.

## 6.11.5 Your Turn!

**Component Overview:** Students determine the products and quotients of integer expressions. These problems will help students build procedural fluency.



### 6.11.5 Your Turn!

1. Find each product or quotient.

|                         |                           |                          |                           |
|-------------------------|---------------------------|--------------------------|---------------------------|
| $-11 \times 9$<br>-99   | $-4 \cdot (-12)$<br>48    | $-100 \div 25$<br>-4     | $\frac{-60}{-5}$<br>12    |
| $96 \div (-6)$<br>-16   | $(-5) \times (-13)$<br>65 | $\frac{84}{-12}$<br>-7   | $(-7)(-7)$<br>49          |
| $110 \div (-11)$<br>-10 | $(-6)(-24)$<br>144        | $9 \times (-15)$<br>-135 | $-18 \times 96$<br>-1,728 |
| $26 \div (-13)$<br>-2   | $150 \div (-3)$<br>-50    | $-39 \times (-4)$<br>156 | $\frac{-88}{-4}$<br>22    |



Consider your classroom when planning for the way in which you want your students engaging with these questions, if needed.

- Allow students to choose a certain number of questions to answer to increase agency and provide choice for entry points.
- Chunk all 16 questions into four groups of four by difficulty, and students must choose two of the four questions from each group.

## 6.11.6 Wrap-Up: Reflection

**Component Overview:** Students reflect on their learning from the lesson. Use information from this reflection along with students' performances in 6.11.5 Your Turn! to determine which students need remediation before the next lesson.



### 6.11.6 Wrap-Up: Reflection

1. Complete the learning period based on today's lesson.  
*Answers will vary.*

|                                                |                                        |
|------------------------------------------------|----------------------------------------|
| One question I would like to have answered ... |                                        |
| One mistake I learned from today ...           | One thing I am not sure about ... YET! |
| One thing I am confident I know is ...         |                                        |



Consider allowing students to use an audio recorder or speech-to-text tool to share their responses.

## 6.12 Multiplying and Dividing Integers With Procedural Fluency

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                   |                                                                                                                       |                                                                                                                                                                                                                                                                                                            |                                                                                                             |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.                                                                                                                                                   |                                                                                                                       |  Learning Target                                                                                                                                                                                                        | <ul style="list-style-type: none"><li>I can multiply and divide integers with procedural fluency.</li></ul> |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                       |                                                                                                                       | Working Toward                                                                                                                                                                                                                                                                                             |                                                                                                             |
|                                                                                                       | MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.                                                                                                                                                                                                                               |                                                                                                                       | MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.<br>MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.                                                                                                           |                                                                                                             |
|  Essential Questions | <ul style="list-style-type: none"><li>How are the rules for multiplying more than two factors similar to the rules for multiplying two factors?</li><li>How are multiplication and division related?</li></ul>                                                                                    |                                                                                                                       |                                                                                                                                                                                                                                                                                                            |                                                                                                             |
|  Lesson Narrative    | In this lesson, students discover that when multiplying or dividing integers, the sign will be positive if there is an even number of negative factors and negative if there is an odd number of negative factors. Students develop procedural fluency through working several practice problems. |                                                                                                                       |                                                                                                                                                                                                                                                                                                            |                                                                                                             |
|  Component Summary | 6.12.1 Warm-Up (~5-10 min.)                                                                                                                                                                                                                                                                       | Students select negative expressions from a given number line with numbers and variables plotted ( <i>SE p. 288</i> ) | 6.12.4 Guided Practice (5-10 min.)                                                                                                                                                                                                                                                                         | Practice finding products and quotients of integers ( <i>SE p. 289</i> )                                    |
|                                                                                                       | 6.12.2 Collaboration (10 min.)                                                                                                                                                                                                                                                                    | Collaboratively multiply multiple integers and identify patterns with signs ( <i>SE p. 288</i> )                      | 6.12.5 Your Turn! (10 min.)                                                                                                                                                                                                                                                                                | Additional practice finding products and quotients of integers ( <i>SE p. 290</i> )                         |
|                                                                                                       | 6.12.3 Activity (10 min.)                                                                                                                                                                                                                                                                         | Practice multiplying integers by playing Integer War ( <i>SE p. 289</i> )                                             | 6.12.6 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                                                   | Demonstrate understanding of lesson learning goals ( <i>SE p. 290</i> )                                     |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                          | Materials                                                                                                             | Additional Preparation                                                                                                                                                                                                                                                                                     |                                                                                                             |
|                                                                                                       | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> <ul style="list-style-type: none"><li>Integer</li><li>Expression</li></ul>                                                                                                                                                                  | <ul style="list-style-type: none"><li>One deck of cards per student pair</li></ul>                                    | <ul style="list-style-type: none"><li>(Optional) Prepare and print sheets for students to record their work for the card game.</li><li>Prepare decks of cards by removing face cards or ask students to do this before beginning the game.</li><li>Prepare student grouping for 6.12.3 Activity.</li></ul> |                                                                                                             |

| <br>Teacher<br>Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Things to Consider                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                   | <ul style="list-style-type: none"><li>Students may struggle to reason about the absolute value of <math>a</math> and <math>b</math> on the number line in the 6.12.1 Warm-Up.</li><li>Students may struggle multiplying more than two integer factors.</li><li>Students may struggle accurately multiplying and dividing two- and three-digit numbers.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"><li>Consider creating and displaying an anchor chart with the rules for multiplying multiple factors.</li></ul>                                                                                                                                                                                                                                         |
|                                                                                                                   | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                                        |
|                                                                                                                   | <b>Think-Pair-Share</b><br>6.12.2 Collaboration provides students the opportunity to discuss their work and answers with a partner. These discussions will help students gain confidence and understanding.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>MA.K12.MTR.3.1: Mathematical Fluency</b><br>6.12.2 Collaboration and 6.12.5 Your Turn! provide opportunities for students to apply what they have learned about multiplying integers to expressions with more than two factors. Encourage students to make connections from what they learned about multiplying two integers to determine the sign of the product of multiple numbers. |
| <br>Differentiate<br>Instruction | <b>Integer War</b><br>6.12.3 Activity provides a unique, engaging, and collaborative opportunity to increase fluency in integer operations. Consider your students when planning groups, using data and informal observations from the previous lessons to inform pairings as well as the method in which students are engaging. <ul style="list-style-type: none"><li>Pair students by their current understanding or fluency with multiplying integers for “even” pairings.</li><li>Alternatively, rather than focusing on speed, one student in each pair is negative and one is positive, or one student is less than 40 and one student is greater than 40. The product determines who wins the cards rather than the speed of who finishes first.</li><li>Groups of three can be formed to increase the difficulty for some groups, if appropriate.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                           |
| Lesson Notes:                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                           |

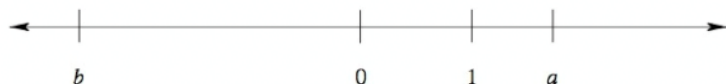
## 6.12.1 Warm-Up: Thinking Algebraically

**Component Overview:** Students work independently to use variables and numbers plotted on a number line to identify expressions that will have negative values. After explaining their thinking independently, they can work with a partner to compare expressions and explanations, providing an opportunity to refine or revise their thinking.



### 6.12.1 Warm-Up: Thinking Algebraically

A number line is shown. The numbers 0 and 1 are marked on the line, along with two other numbers,  $a$  and  $b$ .



1. Circle one expression from those shown below that will have a negative value.

|         |         |         |         |          |
|---------|---------|---------|---------|----------|
| $a - 1$ | $a - 2$ | $a + b$ | $a - b$ | $ab + 1$ |
|---------|---------|---------|---------|----------|

2. Explain how you know the expression selected will have a negative value based on the number line.

*Answers will vary. Since the distance between 0 and 1 is greater than the distance between 1 and  $a$ , I know that  $a < 2$ , so  $a - 2 < 0$ . I see that  $b$  is farther away from 0 than  $a$  is. Therefore,  $b$  has a greater absolute value than  $a$ , so  $a + b$  will be negative. Since the absolute values of  $a$  and  $b$  are both greater than 1,  $ab$  will be less than  $-1$ , so  $ab + 1$  will still be negative.*

3. Find a classmate who chose a different expression that will have a negative value.

Record their expression and explanation for why they chose it in the table. Then, have them initial that you recorded it correctly. *You are the only person who should write in the first two columns of the table.*

*Answers will vary.*

| Classmate's Selected Expression | Classmate's Explanation for Why the Value is Negative | Initials |
|---------------------------------|-------------------------------------------------------|----------|
|                                 |                                                       |          |



Students may benefit from sentence stems, such as “I know the expression \_\_\_\_\_ will have a negative value, because \_\_\_\_\_.”

## 6.12.2 Collaboration: Multiplying Multiple Integers

**Component Overview:** Students work with a partner to discover patterns in the signs when multiplying multiple integers. Students discover that the sign will be positive if there are an even number of negative factors and negative if there are an odd number of negative factors. Debrief with a whole-class discussion to consolidate student understanding.



### 6.12.2 Collaboration: Multiplying Multiple Integers

1. Work with your partner to find the product of each expression.

|                                                                       |                                    |                           |
|-----------------------------------------------------------------------|------------------------------------|---------------------------|
| a. $5(-9)$<br>-45                                                     | b. $-6(-4)$<br>24                  | c. $0(-3)$<br>0           |
| d. $(-1)(-1)(3)$<br>3                                                 | e. $4 \cdot (-2) \cdot (5)$<br>-40 | f. $-2(-2)(-2)(-2)$<br>16 |
| g. $(-1) \times (-2) \times 3 \times (-2) \times (-1) \times 2$<br>24 |                                    |                           |

2. In expressions d-g, more than two factors are being multiplied.

Discuss with your partner how the rules you developed previously for multiplying two integers could be extended for multiplying more than two integers.

Write at least one rule that would apply in this situation.

*Answers will vary.*

$$\begin{array}{c} \text{Red circle with } - \times \text{Red circle with } - = \text{Yellow circle with } + \\ \text{Red circle with } - \times \text{Yellow circle with } + = \text{Red circle with } - \end{array}$$

*Multiplying an odd number of negative factors will result in a negative product.*

*Multiplying an even number of negative factors will result in a positive product.*



Consider providing sentence starters for question 2. For example, “The sign will be positive if...” and “The sign will be negative if...”



## 6.12.3 Activity: Integer War – Multiplication Edition

**Component Overview:** Students use a deck of playing cards to play Integer War in pairs. Both students flip over a card, and the first student to state the product wins the round and collects both cards. Before beginning, model a few rounds for students. Consider providing sheets for students to work their products and record their answers. Circulate while students work to monitor student progress and resolve any disagreements that arise.



### 6.12.3 Activity: Integer War – Multiplication Edition

#### Set Up:

- Remove all jokers and face cards (Kings, Queens, and Jacks) from the deck.
- With your partner, determine who is the dealer. The dealer deals the entire deck of cards to each player, face down.

#### Rules:

- The black cards (Spades ♠ and Clubs ♣) represent positive values.
- The red cards (Diamonds ♦ and Hearts ♥) represent negative values.
- Aces represent either 1 or  $-1$ , while the other cards represent the positive or negative value indicated on the card.

#### How to Play:

- To begin, each player will flip over one card from their deck and place it in the center.
- The first player who figures out the product of the two integers keeps both cards and adds them to the bottom of their stack.
  - NOTE: If a player says an incorrect product, the other player responds for a chance to win the hand before the first player can change their answer.
- Repeat this process until time is called or until one player has all of the cards in the deck.



Consider your students when planning partners and groups. Produce partnerships that focus on fluency over speed and comprehension over calculation.



Teachers can also increase difficulty by forming some groups of three to play.

## 6.12.4 Guided Practice: Finding Products or Quotients

**Component Overview:** Students find the products and quotients of integers. Consider having students check their work with a partner after completing the expressions. Then debrief with a whole-class discussion to clarify any miscalculations and misconceptions, if needed.



### 6.12.4 Guided Practice: Finding Products or Quotients

1. Find each product or quotient.

a.  $10 \times (-65)$

**-650**

b.  $-19 \times 99$

**-1,881**

c.  $-812 \div (-7)$

**116**

d.  $\frac{208}{-16}$

**-13**

e.  $-125 \div (-5)$

**25**

f.  $(-21)(-16)$

**336**



Consider allowing students to choose which questions they answer and in what order they answer the questions.



Prompt students to first make sense of what the question is asking before calculating an answer. Provide an example for question 1.

- “The opposite of 65 groups of 10 is what number?”
- “The sign of the product will be negative.”
- “65 groups of 10 would be 650. So, in other words, 650 divided by 65 is 10.”

Students can also decide on the sign of the answer before performing the calculation to use as a way to check the reasonableness of their answers.

## 6.12.5 Your Turn!

**Component Overview:** Students work independently to find the products and quotients of integers.



### 6.12.5 Your Turn!

1. Find each product or quotient.

a.  $-17 \times 29 = -493$

b.  $41 \cdot (-19) = -779$

c.  $-184 \div 23 = -8$

d.  $\frac{-175}{-5} = 35$

e.  $96 \div (-6) = -16$

f.  $(-51) \times (-31) = 1,581$

g.  $-256 \div (-16) = 16$

h.  $(-17) \cdot 704 \cdot (-1) = 11,968$



*How did you determine the sign of the product or quotient?  
How can you check your answer to ensure it is correct?*

## 6.12.6 Wrap-Up: Ticket Out the Door

**Component Overview:** Students demonstrate understanding of the lesson learning target by filling in missing factors, divisors, dividends, and quotients in equations. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should not use a calculator to complete the Wrap-Up.



### 6.12.6 Wrap-Up: Ticket Out the Door

- Complete the table by writing a value in each blank that completes each related equation.

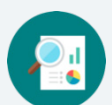
| Multiplication Equation | ...related to...      | Division Equation      |
|-------------------------|-----------------------|------------------------|
| $(-12)(-42) = 504$      | $\longleftrightarrow$ | $504 \div (-42) = -12$ |
| $-7 \times 18 = -126$   | $\longleftrightarrow$ | $-126 \div 18 = -7$    |
| $-50 \cdot 13 = -650$   | $\longleftrightarrow$ | $-650 \div 13 = -50$   |



Notice the focus of this Wrap-Up is centered on the relationship between multiplication and division rather than the calculation. Consider this when planning your instruction throughout the lesson.

## 6.13 Evaluating Exponents

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                          |                                                                                                                                                                                                 |                                                                                                                                                                             |                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.6.NSO.3.3 Evaluate positive rational numbers and integers with natural number exponents.                                                                                                                                                                                                                                                                                                                                                |                                                                                                          |  Learning Targets                                                                                            | <ul style="list-style-type: none"><li>I can evaluate integers with natural number exponents.</li><li>I can evaluate a positive rational number raised to a power.</li></ul> |                                                                                                                                                                                                              |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                          | Working Toward                                                                                                                                                                                  |                                                                                                                                                                             |                                                                                                                                                                                                              |
|                                                                                                       | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                          | MA.7.NSO.1.1 Know and apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases. |                                                                                                                                                                             |                                                                                                                                                                                                              |
|  Essential Questions | <ul style="list-style-type: none"><li>How do you represent repeated multiplication using exponents?</li><li>How can you represent exponential expressions as repeated multiplication?</li><li>How do you determine whether an exponential expression will have a positive or negative value?</li></ul>                                                                                                                                     |                                                                                                          |                                                                                                                                                                                                 |                                                                                                                                                                             |                                                                                                                                                                                                              |
|  Lesson Narrative    | In this lesson, students extend and apply their knowledge of multiplying integers and of exponential expressions to evaluate exponential expressions with integer bases. Students explore how to determine whether an exponential expression will have a positive or negative value and practice rewriting exponential expressions, including those with positive decimal and fractional bases, as repeated multiplication and vice versa. |                                                                                                          |                                                                                                                                                                                                 |                                                                                                                                                                             |                                                                                                                                                                                                              |
|  Component Summary  | 6.13.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                   | Describe what they want to get better at after watching a motivational video ( <i>SE p. 291</i> )        | 6.13.3 Bring It Together (10 min.)                                                                                                                                                              | Discover when the signs of exponential expressions will be positive and negative ( <i>SE p. 293</i> )                                                                       |                                                                                                                                                                                                              |
|                                                                                                       | 6.13.2 Exploration (10 min.)                                                                                                                                                                                                                                                                                                                                                                                                               | Explore how to evaluate exponential expressions with positive and negative bases ( <i>SE p. 292</i> )    | 6.13.4 Your Turn! (10 min.)                                                                                                                                                                     | Additional practice raising rational numbers to powers ( <i>SE p. 294</i> )                                                                                                 |                                                                                                                                                                                                              |
|                                                                                                       | 6.13.5 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                   | Self-assessment rating of levels of understanding of learning targets from the unit ( <i>SE p. 294</i> ) |                                                                                                                                                                                                 |                                                                                                                                                                             |                                                                                                                                                                                                              |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                          | Materials                                                                                                                                                                                       |                                                                                                                                                                             | Additional Preparation                                                                                                                                                                                       |
|                                                                                                       | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> <ul style="list-style-type: none"><li>Base</li><li>Exponent/Power</li><li>Factors</li><li>Value</li></ul>                                                                                                                                                                                                                                                                            |                                                                                                          | <ul style="list-style-type: none"><li>(Optional) Chart paper</li><li>(Optional) Markers</li><li>(Optional) Note cards</li></ul>                                                                 |                                                                                                                                                                             | <ul style="list-style-type: none"><li>Warm-Up 6.13.1 contains a video to accompany the question prompts.</li><li>Consider creating an anchor chart with instructions, examples, and visual models.</li></ul> |

|                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher<br>Tips | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>Students may not correctly evaluate exponential expressions when there is a negative sign inside the parentheses. For example, <math>(-4)^2 \neq -16</math>.</li><li>Students may think that an exponent of 1 means to multiply the number by itself once. For example, <math>16^1 = 16 \times 16</math>.</li></ul>                                                                                                                                                                                                                                                                                        | <b>Things to Consider</b> <ul style="list-style-type: none"><li>Consider displaying and adding to the anchor chart created with Lesson 1 with instructions, examples, and visual models of evaluating exponential expression in 6.13.2 Exploration.</li></ul>                                                                                                               |
|                                                                                                      | <b>Literacy and Language Routines</b><br><b>Think-Pair-Share</b><br>In 6.13.3 Bring It Together, have students discuss their work and answers after first working individually. The discussions will help students gain confidence and understanding.                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Mathematical Thinking and Reasoning (MTR) Standard</b><br><b>MA.K12.MTR.4.1: Discussion and Reflection</b><br>6.13.4 Guided Practice question 3 provides an opportunity for students to analyze errors in samples of student work. Students identify and correct the errors. Students will be less likely to make the mistake themselves after completing this analysis. |
|                                                                                                      | <br>Differentiate<br>Instruction<br><p>6.13.4 Your Turn! includes fractions raised to powers for students to evaluate. Notice benchmark MA.6.NSO.3.3 includes “rational numbers” in the benchmark language, as fluency with fractions and decimals is an expectation by the end of Grade 6. This is spiraled from Lesson 1 in this unit.</p> <p>Consider utilizing the On-Ramp to 6<sup>th</sup> Grade learning tool to provide scaffolding for students in need of additional support when evaluating expressions.</p> <ul style="list-style-type: none"><li>Evaluate Expressions</li></ul> |                                                                                                                                                                                                                                                                                                                                                                             |
| Lesson Notes:                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                             |

## 6.13.1 Warm-Up: Consider the YET

**Component Overview:** Play motivational video “The Power of Yet” for students. Then have students reflect by answering the following questions.



### 6.13.1 Warm-Up: Consider the YET

1. Consider the belief that “you can get better at whatever you want to get better at.” Explain whether you agree or disagree.  
*Answers will vary.*
2. Consider the quote, “I’ve never heard of someone who took up running and didn’t get faster the more they did it.” That person just may not have been a fast runner ... YET.

Complete the statement to describe something you want to get better at.

*Answers will vary.*

I \_\_\_\_\_  
\_\_\_\_\_...YET!



*Consider providing an auditory entry point by having students discuss question 1. Write sentence stems on the board to encourage conversation, such as “I agree with the statement because...” or “I disagree with the statement because....”*



*Prompt students to reference a specific moment in this unit in which they exhibited growth mindset or heard their partner or classmate exhibit growth mindset.*



## 6.13.2 Exploration: Building on the Understanding of Exponents

**Component Overview:** Review the vocabulary as a class before students complete questions 1 and 2 independently. Students then work with a partner to explore how to evaluate exponential expressions with positive and negative bases.



### 6.13.2 Exploration: Building on the Understanding of Exponents

Recall the vocabulary of exponential expressions.

$$\begin{array}{c} \text{exponent (or power)} \\ \nearrow \\ \text{base} \end{array} 3^4 = \underbrace{3 \cdot 3 \cdot 3 \cdot 3}_{\text{factors}} = \begin{array}{c} \text{value} \\ \text{(of the expression)} \end{array} 81$$

1. Think of a number that, when raised to a power less than 6, will result in a value of 16.

2. Write the base and its exponent.

*Answers will vary.*



*Answers could also include  $(-2)^4$ ,  $(4)^2$ , or  $(-4)^2$ .*

3. Compare your expression with your partner and check for accuracy.

4. If your partner found a different base and exponent that meets the description, write it below. If your partner wrote the same base and exponent as you, work together to find a second exponential expression that also works.

*Answers will vary. See examples from question 2.*



#### Turn and Talk

Prompt students to discuss vocabulary terms, their expressions, and possible values.



What are some factors of 16 that you could use as a base? Is there more than one exponent and base combination that will equal 16? How do you know?



Remind students who may be struggling how to find the prime factorization of a number. Suggest using the prime factorization of 16 to determine possible bases and exponents to use.

## 6.13.2 Exploration: Building on the Understanding of Exponents (continued)

5. With your partner, choose one of the exponential expressions used and rewrite it making the base negative.

*Answers will vary.*

$$(-4)^{\boxed{2}}$$

6. Explain whether the negative base changes the value of the expression.

*Answers will vary. The exponent is even, so the value is the same whether the base is negative or positive. The negative base does not change the value of the expression because when you multiply an even number of negative factors, the sign of the product is positive. Therefore, the value is still 16.*

7. Discuss with your partner how you could change the exponent to result in a negative answer. Write a brief summary of your discussion below.

*Answers will vary. You can change the exponent to an odd number to result in a negative answer because when you multiply an odd number of negative factors, the product will be negative.*



*How does a negative base change the value of the expression when the exponent is even? How does it change the value when the exponent is odd? Why do you think that is? (See extension in question 6 for 6.13.3.)*



*If students struggle expressing themselves in writing, consider allowing students to respond to the questions verbally first, taking turns with a partner. Students can then respond to questions 6 and 7 as a synthesis rather than a starting point.*



*The next component will solidify understanding for questions 6 and 7, so it is not necessary that students are able to answer these accurately at this point. The goal is for them to discuss their thinking related to what they have learned already in the unit. Listen as students discuss to bring their thinking into the next component.*

## 6.13.3 Bring It Together: Building on the Understanding of Exponents

**Component Overview:** Students build their understanding of evaluating exponents and how the rules of multiplication with integers apply when the signs of the bases are positive and negative.



### 6.13.3 Bring It Together: Building on the Understanding of Exponents

When evaluating exponents with a negative base, the same rules of multiplication with integers apply.

1. In the expression  $(-4)^2$ , the base is  $-4$  and the exponent is  $2$ , resulting in the repeated multiplication sentence of  $(-4)(-4)$ .
2. Why do the results of  $(-4)^2$  result in a positive number?  
*Answers will vary. The product of two negative integers is positive.*
3. In the expression  $(-4)^3$ , the base is  $-4$  and the exponent is  $3$ , resulting in the repeated multiplication sentence of  $(-4)(-4)(-4)$ .
4. Why do the results of  $(-4)^3$  result in a negative number?  
*A positive times a positive = a positive.*  
 $(-4)(-4) = 16$   
*Then, a positive times a negative = a negative.*  
 $(16)(-4) = -64$



Utilize inquiry to position students for synthesis.

- What impact does the negative sign have on the value of the outcome?
- How does that change if the negative sign was the coefficient on the outside of the parenthesis? Provide an example.



If students struggle expressing themselves in writing, consider providing sentence starters for question 4, such as “A negative times a negative equals a \_\_\_\_\_,” “A positive times a negative equals a \_\_\_\_\_,” or “Therefore, the answer must result in an \_\_\_\_\_ answer.”

### 6.13.3 Bring It Together: Building on the Understanding of Exponents (continued)

5. For each expression, determine whether its value is positive or negative.

| Expression  | Positive or Negative Value? |
|-------------|-----------------------------|
| $(-2)^2$    | <i>Positive</i>             |
| $(-2)^3$    | <i>Negative</i>             |
| $(-2)^4$    | <i>Positive</i>             |
| $(-2)^5$    | <i>Negative</i>             |
| $(-2)^{19}$ | <i>Negative</i>             |



#### Notice and Wonder

To encourage the connection between the value of the exponent and sign of the base, consider asking students what they notice and wonder about the sign of the values of each exponential expression.

6. When evaluating exponents with a negative base, the value of the expression will be ☐ positive ☐ negative if the exponent is an ☐ odd ☐ even number, and the value of the expression will be ☐ positive ☐ negative if the exponent is an ☐ odd ☐ even number.



Note there are two correct answers for question 6. If students complete the first blank with “negative,” then the remaining blanks would be “odd,” “positive,” and “even,” respectively.

## 6.13.4 Your Turn!

**Component Overview:** Students independently evaluate exponential expressions and rewrite repeated multiplication expressions in exponential form. Students also assess whether two expressions are equivalent and show their work to justify their responses.



### 6.13.4 Your Turn!

1. Evaluate each expression.

a.  $(-3)^4$   
81

b.  $\left(\frac{15}{17}\right)^2$   
 $\frac{225}{289}$

c.  $(-1)^5$   
-1

2. Rewrite each repeated multiplication sentence as an exponential expression. Then, find the value of the expression.

a.  $(-12)(-12)(-12)$   
 $(-12)^3$   
= -1,728

b.  $3.4 \times 3.4$   
 $(3.4)^2$   
= 11.56

3. Determine if each equation in the table is true or false. Show your work to support your answer.

| Equation                      | True or False? | Work                                                                                                                                  |
|-------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| $16^1 = (-8)^2$               | False          | $16^1 = 16$<br>$(-8)^2 = (-8)(-8)$<br>$= 64$                                                                                          |
| $(-5)^3 = -5 \cdot 5 \cdot 5$ | True           | $(-5)^3 = -5 \cdot -5 \cdot -5$<br>$-5 \cdot -5 \cdot -5 = 25 \cdot -5$<br>$= -125$<br>$-5 \cdot 5 \cdot 5 = -25 \cdot 5$<br>$= -125$ |



Consider displaying the anchor chart created in Lesson 1 with instructions, examples, and visual models for students to reference.

Alternatively, offer students a station to opt-in for support using On-Ramp to 6<sup>th</sup> Grade videos to learn alongside Study Experts.



Prompt students to use the word “equivalent” in their verbal explanations of their work or findings.

## 6.13.5 Wrap-Up: Reflection

**Component Overview:** Students complete a self-assessment, rating their level of understanding of the topics covered throughout the unit.



### 6.13.5 Wrap-Up: Reflection

1. In this unit, you have learned multiple concepts about operations with integers. Rate yourself on the following skills based on your level of understanding.

*Answers will vary.*

| Statement                                                                | Rating                                                                                                 |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
|                                                                          | 1 – I struggle with this.<br>2 – I have some skills.<br>3 – I am almost there.<br>4 – I have got this! |
| I can evaluate positive rational numbers and integers raised to a power. |                                                                                                        |
| I can express composite whole numbers as a product of prime factors.     |                                                                                                        |
| I can add integers.                                                      |                                                                                                        |
| I can subtract integers.                                                 |                                                                                                        |
| I can multiply integers.                                                 |                                                                                                        |
| I can divide integers.                                                   |                                                                                                        |



*Consider having students create and solve examples of each learning target they rate as a 4 or 5. Ask students to consider a plan for filling in gaps in their learning for targets they rated as a 1, 2, or 3.*